

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

(In the Name of Allah, the Most Compassionate, the Most Merciful)

MATHEMATICS



**PUNJAB CURRICULUM AND
TEXTBOOK BOARD, LAHORE**

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Unit 1

Real Numbers

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Explain, with examples, that civilizations throughout history have systematically studied living things [e.g., the history of numbers from Sumerians and its development to the present Arabic system]
- Describe the set of real numbers as a combination of rational and irrational numbers
- Demonstrate and verify the properties of equality and inequality of real numbers
- Apply laws of indices to simplify radical expressions
- Apply concepts of real numbers to real-world problems (such as temperature, banking, measures of gain and loss, sources of income and expenditure)

1.1 Introduction to Real Numbers

The history of numbers comprises thousands of years, from ancient civilization to the modern Arabic system. Here is a brief overview:

Sumerians (1900 – 4500 BCE) used a sexagesimal (base 60) system for counting. The Sumerians used a small cone, bead, large cone, large perforated cone, sphere and perforated sphere, corresponding to 1, 10, 60, 600, 3600 and 36000.

1	𐎶	11	𐎶𐎵	100	𐎶𐎵𐎶
2	𐎶𐎶	12	𐎶𐎵𐎶𐎵	200	𐎶𐎵𐎶𐎵𐎶
3	𐎶𐎶𐎶	20	𐎶𐎵𐎶𐎵𐎶	300	𐎶𐎵𐎶𐎵𐎶𐎶
4	𐎶𐎶𐎶𐎶	30	𐎶𐎵𐎶𐎵𐎶𐎶	400	𐎶𐎵𐎶𐎵𐎶𐎶𐎶
5	𐎶𐎶𐎶𐎶𐎶	40	𐎶𐎵𐎶𐎵𐎶𐎶𐎶	500	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶
6	𐎶𐎶𐎶𐎶𐎶𐎶	50	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶	600	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶
7	𐎶𐎶𐎶𐎶𐎶𐎶𐎶	60	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶	700	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶
8	𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶	70	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶	800	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶𐎶
9	𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶	80	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶𐎶	900	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶
10	𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶	90	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶	1000	𐎶𐎵𐎶𐎵𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶

Egyptians (2000 – 3000 BCE) used a decimal (base 10) system for counting.

Here are some of the symbols used by the Egyptians, as shown in the figure below:

The Egyptians usually wrote numbers left to right, starting with the highest place value. For example, 2525 would be written with 2000 first, then 500, 20, and 5.

						
1	10	100	1,000	10,000	100,000	1,000,000

Romans (500BCE-500CE) used the Roman numerals system for counting.

Roman numerals represent a number system that was widely used throughout Europe as the standard writing system until the late Middle Ages. The ancient Romans explained that when a number reaches 10 it is not easy to count on one's fingers. Therefore, there was a need to create a proper number system that could be used for trade and communications. Roman numerals use 7 letters to represent different numbers. These are I, V, X, L, C, D, and M which represent the numbers 1, 5, 10, 50, 100, 500 and 1000 respectively.

Indians (500 – 1200 CE) developed the concept of zero (0) and made a significant contribution to the decimal (base 10) system.

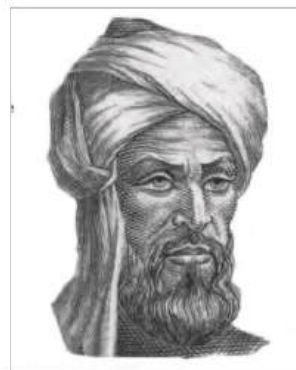
Ancient Indian mathematicians have contributed immensely to the field of mathematics. The invention of zero is attributed to Indians, and this contribution outweighs all others made by any other nation since it is the basis of the decimal number system, without which no

—	=	≡	𑀓	𑀔	𑀕	𑀖	𑀗	𑀘
1	2	3	4	5	6	7	8	9
𑀠	𑀡	𑀢	𑀣	𑀤	𑀥	𑀦	𑀧	𑀨
10	20	30	40	50	60	70	80	90
𑀩	𑀪	𑀫	𑀬	𑀭	𑀮	𑀯	𑀰	𑀱
100	200	500	1,000	4,000	70,000			

advancement in mathematics would have been possible. The number system used today was invented by Indians, and it is still called Indo-Arabic numerals because Indians invented them and the Arab merchants took them to the Western world.

Arabs (800 – 1500 CE) introduced Arabic numerals (0 – 9) to Europe. The Islamic world underwent significant developments in mathematics. Muhammad ibn Musa al-Khwārizmī played a key role in this transformation, introducing algebra as a distinct field in the 9th century. Al-Khwārizmī's approach, departing from earlier arithmetical traditions, laid the groundwork for the arithmetization of algebra, influencing mathematical thought for an extended period. Successors like Al-Karaji expanded on his work, contributing to advancements in various mathematical domains.

The practicality and broad applicability of these mathematical methods facilitated the dissemination of Arabic mathematics to the West, contributing substantially to the evolution of Western mathematics.



Modern era (1700 – present): Developed modern number systems e.g., binary system (base - 2) and hexadecimal system (base - 16).

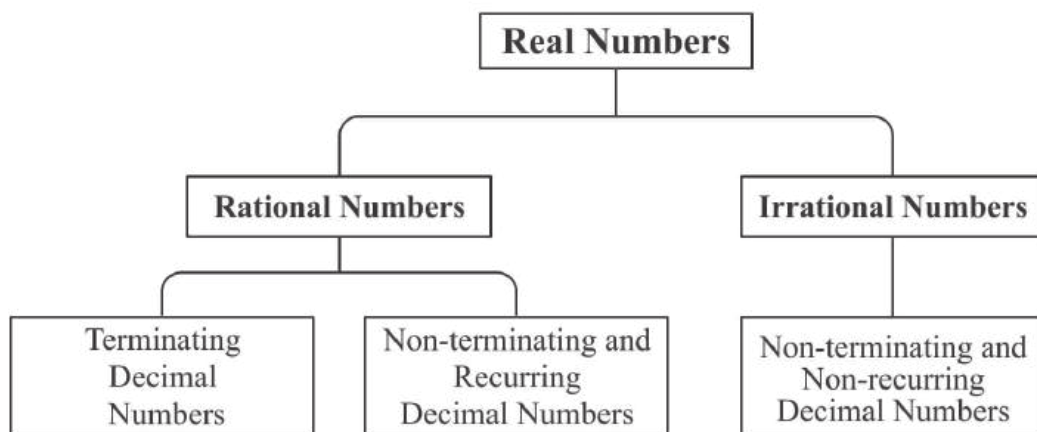
The Arabic system is the basis for modern decimal system used globally today. Its development and refinement comprise thousands of years from ancient Sumerians to modern mathematicians.

In the modern era, the set $\{1, 2, 3, \dots\}$ was adopted as the counting set. This counting set represents the set of natural numbers was extended to set of real numbers which is used most frequently in everyday life.

1.1.1 Combination of Rational and Irrational Numbers

We know that the set of rational numbers is defined as $Q = \left\{ \frac{p}{q}; p, q \in Z \wedge q \neq 0 \right\}$

and set of irrational numbers (Q') contains those elements which cannot be expressed as quotient of integers. The set of Real numbers is the union of the set of rational numbers and irrational numbers i.e., $R = Q \cup Q'$



1.1.2 Decimal Representation of Rational Numbers

(i) Terminating Decimal Numbers

A decimal number with a finite number of digits after the decimal point is called a terminating decimal number.

For example, $\frac{1}{4} = 0.25$, $\frac{8}{25} = 0.32$, $\frac{3}{8} = 0.375$, $\frac{4}{5} = 0.8$ are all terminating decimal numbers.

(ii) Non-Terminating and Recurring Decimal Numbers

The decimal numbers with an infinitely repeating pattern of digits after the decimal point are called non-terminating and recurring decimal numbers.

Here are some examples.

$$\frac{1}{3} = 0.333... = 0.\overline{3} \text{ (3 repeats infinitely)}$$

$$\frac{1}{6} = 0.1666... = 0.1\overline{6} \text{ (6 repeats infinitely)}$$

$$\frac{22}{7} = 3.142857142857... = 3.\overline{142857} \text{ (the pattern 142857 repeats infinitely)}$$

$$\frac{4}{9} = 0.44444... = 0.\overline{4} \text{ (4 repeats infinitely)}$$

Non-terminating and recurring decimal numbers are also rational numbers.

1.1.3 Decimal Representation of Irrational Numbers

Decimal numbers that do not repeat a pattern of digits after the decimal point continue indefinitely without terminating.

Non-terminating and non-recurring decimal numbers are known as irrational numbers.

For examples,

- $\pi = 3.1415926535897932...$
- $e = 2.71828182845904...$
- $\sqrt{2} = 1.41421356237309...$

Remember!

$e = 2.7182...$ is called Euler's Number.

Example 1: Identify the following decimal numbers as rational or irrational numbers:

- (i) 0.35 (ii) 0.444... (iii) $3.\overline{5}$
 (iv) 3.36788542... (v) 1.709975947...

Solution: (i) 0.35 is a terminating decimal number, therefore it is a rational number.
 (ii) 0.444... is a non-terminating and recurring decimal number, therefore it is a rational number.
 (iii) $3.\overline{5} = 3.5555...$ is a non-terminating and recurring decimal number, therefore it is a rational number.
 (iv) 3.36788542... is a non-terminating and non-recurring decimal number, therefore it is an irrational number.

- (v) 1.709975947... is a non-terminating and non-recurring decimal number, therefore it is an irrational number.

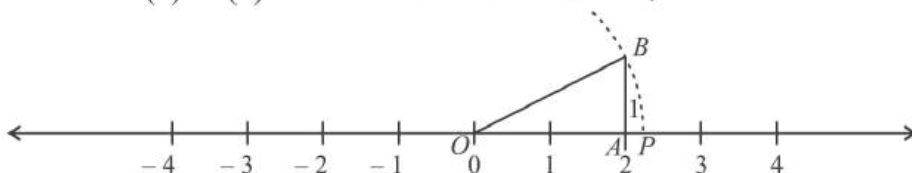
1.1.4 Representation of Rational and Irrational Numbers on Number Line

In previous grades, we have learnt to represent rational numbers on a number line. Now, we move to the next step and learn how to represent irrational numbers on a number line.

Example 2: Represent $\sqrt{5}$ on a number line.

Solution: $\sqrt{5}$ can be located on the number line by geometric construction. As, $\sqrt{5} = 2.236...$ which is near to 2. Draw a line of $m\overline{AB} = 1$ unit at point A, where $m\overline{OA} = 2$ units, and we have a right-angled triangle OAB. By using Pythagoras theorem

$$\begin{aligned}(m\overline{OB})^2 &= (m\overline{OA})^2 + (m\overline{AB})^2 \\ &= (2)^2 + (1)^2 = 4 + 1 = 5 \Rightarrow m\overline{OB} = \sqrt{5}\end{aligned}$$



Draw an arc of radius $m\overline{OB} = \sqrt{5}$ taking O as centre, we got point “P” representing $\sqrt{5}$ on the number line. So, $|OP| = \sqrt{5}$

Remember!

- (i) Rational no. + Irrational no. = Irrational no.
- (ii) Rational no. ($\neq 0$) $\times$ Irrational no. = Irrational no.

Example 3: Express the following recurring decimals as the rational number $\frac{p}{q}$, where p and q are integers.

- (i) $0.\overline{5}$ (ii) $0.\overline{93}$

Solution: (i) $0.\overline{5}$

$$\begin{aligned}\text{Let } x &= 0.\overline{5} \\ x &= 0.55555... \quad \dots(i)\end{aligned}$$

Multiply both sides by 10

$$\begin{aligned}10x &= 10(0.55555...) \\ 10x &= 5.55555... \quad \dots(ii)\end{aligned}$$

Subtracting (i) from (ii)

$$10x - x = (5.55555...) - (0.55555...)$$

$$9x = 5$$

$$\Rightarrow x = \frac{5}{9}$$

Which shows the decimal number in the form of $\frac{p}{q}$.

(ii) Let $x = 0.\overline{93}$

$$x = 0.939393... \quad \dots(i)$$

Multiply by 100 on both sides

$$100x = 100 (0.939393...)$$

$$100x = 93.939393... \quad \dots(ii)$$

Subtracting (i) from (ii)

$$100x - x = 93.939393... - 0.939393...$$

$$99x = 93$$

$$x = \frac{93}{99} \text{ which is in the form of } \frac{p}{q}.$$

Example 4 : Find two rational numbers between 2 and 3.

Solution: There are infinite rational numbers between 2 and 3.

We have to find any two of them.

For this, find the average of 2 and 3 as $\frac{2+3}{2} = \frac{5}{2}$

So, $\frac{5}{2}$ is a rational number between 2 and 3, to find another rational number between

2 and 3 we will again find average of $\frac{5}{2}$ and 3

$$\text{i.e., } \frac{\frac{5}{2} + 3}{2} = \frac{\frac{5+6}{2}}{2} = \frac{\frac{11}{2}}{2} = \frac{11}{4}$$

Try Yourself!

What will be the product of two irrational numbers?

Hence, two rational numbers between 2 and 3 are $\frac{5}{2}$ and $\frac{11}{4}$.

1.1.5 Properties of Real Numbers

All calculations involving addition, subtraction, multiplication, and division of real numbers are based on their properties. In this section, we shall discuss these properties.

Additive properties

Name of the property	$\forall a, b, c \in R$	Examples
Closure	$a + b \in R$	$2 + 3 = 5 \in R$
Commutative	$a + b = b + a$	$2 + 5 = 5 + 2$ $7 = 7$
Associative	$a + (b + c) = (a + b) + c$	$2 + (3 + 5) = (2 + 3) + 5$ $2 + 8 = 5 + 5$ $10 = 10$
Identity	$a + 0 = a = 0 + a$	$5 + 0 = 5 = 0 + 5$
Inverse	$a + (-a) = -a + a = 0$	$6 + (-6) = (-6) + 6 = 0$

Multiplicative properties

Name of the property	$\forall a, b, c \in R$	Examples
Closure	$ab \in R$	$2 \times 5 = 10 \in R$
Commutative	$ab = ba$	$2 \times 3 = 3 \times 2 = 6 \in R$
Associative	$a(bc) = (ab)c$	$2 \times (3 \times 5) = (2 \times 3) \times 5$ $2 \times 15 = 6 \times 5$ $30 = 30$
Identity	$a \times 1 = 1 \times a = a$	$5 \times 1 = 1 \times 5 = 5$
Inverse	$a \times \frac{1}{a} = \frac{1}{a} \times a = 1$	$7 \times \frac{1}{7} = \frac{1}{7} \times 7 = 1$

Distributive Properties

For all real numbers a, b, c

- $a(b + c) = ab + ac$ is called left distributive property of multiplication over addition.
- $a(b - c) = ab - ac$ is called left distributive property of multiplication over subtraction.
- $(a + b)c = ac + bc$ is called right distributive property of multiplication over addition.
- $(a - b)c = ac - bc$ is called right distributive property of multiplication over subtraction.

Do you know?

0 and 1 are the additive and multiplicative identities of real numbers respectively.

Remember!

$0 \in R$ has no multiplicative inverse.

Properties of Equality of Real Numbers

i	Reflexive property	$\forall a \in R, a = a$
ii	Symmetric property	$\forall a, b \in R, a = b \Rightarrow b = a$
iii	Transitive property	$\forall a, b, c \in R, a = b \wedge b = c \Rightarrow a = c$
iv	Additive property	$\forall a, b, c \in R, a = b \Rightarrow a + c = b + c$
v	Multiplicative property	$\forall a, b, c \in R, a = b \Rightarrow ac = bc$
vi	Cancellation property w.r.t addition	$\forall a, b, c \in R, a + c = b + c \Rightarrow a = b$
vii	Cancellation property w.r.t multiplication	$\forall a, b, c \in R \text{ and } c \neq 0, ac = bc \Rightarrow a = b$

Order Properties

i	Trichotomy property	$\forall a, b \in R, \text{either } a = b \text{ or } a > b \text{ or } a < b$
ii	Transitive Property	$\forall a, b, c \in R$ $\bullet a > b \wedge b > c \Rightarrow a > c$ $\bullet a < b \wedge b < c \Rightarrow a < c$
iii	Additive property	$\forall a, b, c \in R$ $\bullet a > b \Rightarrow a + c > b + c$ $\bullet a < b \Rightarrow a + c < b + c$
iv	Multiplicative property	$\forall a, b, c \in R$ $\bullet a > b \Rightarrow ac > bc$ if $c > 0$ $\bullet a < b \Rightarrow ac < bc$ if $c > 0$ $\bullet a > b \Rightarrow ac < bc$ if $c < 0$ $\bullet a < b \Rightarrow ac > bc$ if $c < 0$ $\bullet a > b \wedge c > d \Rightarrow ac > bd$ $\bullet a < b \wedge c < d \Rightarrow ac < bd$
v	Division property	$\forall a, b, c \in R$ $\bullet a < b \Rightarrow \frac{a}{c} < \frac{b}{c}$ if $c > 0$ $\bullet a < b \Rightarrow \frac{a}{c} > \frac{b}{c}$ if $c < 0$ $\bullet a > b \Rightarrow \frac{a}{c} > \frac{b}{c}$ if $c > 0$ $\bullet a > b \Rightarrow \frac{a}{c} < \frac{b}{c}$ if $c < 0$

vi	Reciprocal property	$\forall a, b \in R$ and have same sign $\bullet \quad a < b \Rightarrow \frac{1}{a} > \frac{1}{b}$ $\bullet \quad a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$
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Example 5 : If $a = \frac{2}{3}$, $b = \frac{3}{2}$, $c = \frac{5}{3}$ then verify the distributive properties over addition.

Solution: (i) Left distributive property

$$\begin{array}{lcl}
 & a(b+c) = ab+ac & \\
 \text{LHS} = a(b+c) & | & \text{RHS} = ab+ac \\
 = \frac{2}{3}\left(\frac{3}{2} + \frac{5}{3}\right) = \frac{2}{3}\left(\frac{9+10}{6}\right) & & = \left(\frac{2}{3}\right)\left(\frac{3}{2}\right) + \left(\frac{2}{3}\right)\left(\frac{5}{3}\right) = 1 + \frac{10}{9} \\
 = \frac{2}{3}\left(\frac{19}{6}\right) = \frac{19}{9} & & = \frac{9+10}{9} = \frac{19}{9} \\
 & \text{LHS} = \text{RHS} &
 \end{array}$$

Hence, it is verified that $a(b+c) = ab+ac$

(ii) Right distributive property

$$\begin{array}{lcl}
 & (a+b)c = ac+bc & \\
 \text{LHS} = (a+b)c & | & \text{RHS} = ac+bc \\
 = \left(\frac{2}{3} + \frac{3}{2}\right)\frac{5}{3} = \left(\frac{4+9}{6}\right)\frac{5}{3} & & = \left(\frac{2}{3}\right)\left(\frac{5}{3}\right) + \left(\frac{3}{2}\right)\left(\frac{5}{3}\right) = \frac{10}{9} + \frac{15}{6} \\
 = \left(\frac{13}{6}\right)\left(\frac{5}{3}\right) = \frac{65}{18} & & = \frac{20+45}{18} = \frac{65}{18} \\
 & \text{LHS} = \text{RHS} &
 \end{array}$$

Hence, it is verified that $(a+b)c = ac+bc$

Example 6: Identify the property that justifies the statement

- If $a > 13$ then $a + 2 > 15$
- If $3 < 9$ and $6 < 12$ then $9 < 21$
- If $7 > 4$ and $5 > 3$ then $35 > 12$
- If $-5 < -4 \Rightarrow 20 > 16$

Solution:

- (i) $a > 13$
 Add 2 on both sides
 $a + 2 > 13 + 2$
 $a + 2 > 15$ (order property w.r.t addition)
 $a + 2 > 13 + 2$
 $a + 2 > 15$
- (ii) As $3 < 9$ and $6 < 12$
 $\Rightarrow 3 + 6 < 9 + 12$
 $9 < 21$ (order property w.r.t addition)
- (iii) $7 > 4$ and $5 > 3$
 $\Rightarrow 7 \times 5 > 4 \times 3$
 $\Rightarrow 35 > 12$ (order property w.r.t multiplication)
- (iv) As $-5 < -4$
 Multiply on both sides by -4
 $(-5) \times (-4) > (-4) \times (-4)$
 $\Rightarrow 20 > 16$ (order property w.r.t multiplication)

EXERCISE 1.1

1. Identify each of the following as a rational or irrational number:

- (i) 2.353535 (ii) $0.\bar{6}$ (iii) 2.236067... (iv) $\sqrt{7}$
 (v) e (vi) π (vii) $5 + \sqrt{11}$ (viii) $\sqrt{3} + \sqrt{13}$
 (ix) $\frac{15}{4}$ (x) $(2 - \sqrt{2})(2 + \sqrt{2})$

2. Represent the following numbers on number line:

- (i) $\sqrt{2}$ (ii) $\sqrt{3}$ (iii) $4\frac{1}{3}$
 (iv) $-2\frac{1}{7}$ (v) $\frac{5}{8}$ (vi) $2\frac{3}{4}$

3. Express the following as a rational number $\frac{p}{q}$ where p and q are integers and $q \neq 0$:

- (i) $0.\bar{4}$ (ii) $0.\bar{37}$ (iii) $0.\bar{21}$

4. Name the property used in the following:

(i) $(a + 4) + b = a + (4 + b)$

(ii) $\sqrt{2} + \sqrt{3} = \sqrt{3} + \sqrt{2}$

(iii) $x - x = 0$

(iv) $a(b + c) = ab + ac$

(v) $16 + 0 = 16$

(vi) $100 \times 1 = 100$

(vii) $4 \times (5 \times 8) = (4 \times 5) \times 8$

(viii) $ab = ba$

5. Name the property used in the following:

(i) $-3 < -1 \Rightarrow 0 < 2$

(ii) If $a < b$ then $\frac{1}{a} > \frac{1}{b}$

(iii) If $a < b$ then $a + c < b + c$

(iv) If $ac < bc$ and $c > 0$ then $a < b$

(v) If $ac < bc$ and $c < 0$ then $a > b$

(vi) Either $a > b$ or $a = b$ or $a < b$

6. Find two rational numbers between:

(i) $\frac{1}{3}$ and $\frac{1}{4}$

(ii) 3 and 4

(iii) $\frac{3}{5}$ and $\frac{4}{5}$

1.2 Radical Expressions

If n is a positive integer greater than 1 and a is a real number, then any real number x such that $x = \sqrt[n]{a}$ is called n^{th} root of a .

Here, $\sqrt{\quad}$ is called radical and n is the index of radical. A real number under the radical sign is called a radicand. $\sqrt[3]{5}, \sqrt[5]{7}$ are the examples of radical form.

Exponential form of $x = \sqrt[n]{a}$ is $x = (a)^{\frac{1}{n}}$.

1.2.1 Laws of Radicals and Indices

Laws of Radical	Laws of Indices
(i) $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$	(i) $a^m \cdot a^n = a^{m+n}$
(ii) $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$	(ii) $(a^m)^n = a^{mn}$
(iii) $\sqrt[n]{a^m} = (\sqrt[n]{a})^m$	(iii) $(ab)^n = a^n b^n$
(iv) $(\sqrt[n]{a})^n = (a^{\frac{1}{n}})^n = a$	(iv) $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
	(v) $\frac{a^m}{a^n} = a^{m-n}$
	(vi) $a^0 = 1$

Example 7: Simplify the following:

(i) $\sqrt[4]{16x^4y^8}$

(ii) $\sqrt[3]{27x^6y^9z^3}$

(iii) $(64)^{-\frac{4}{3}}$

Solution: (i) $\sqrt[4]{16x^4y^8} = (16x^4y^8)^{\frac{1}{4}}$ $\therefore \sqrt[n]{a} = a^{\frac{1}{n}}$

$$= (16)^{\frac{1}{4}} (x^4)^{\frac{1}{4}} (y^8)^{\frac{1}{4}}$$

$$= 2^{4 \cdot \frac{1}{4}} \times x^{4 \cdot \frac{1}{4}} \times y^{8 \cdot \frac{1}{4}}$$

$$= 2xy^2$$

$\therefore (ab)^m = a^m b^m$

$\therefore (a^m)^n = a^{mn}$

(ii) $\sqrt[3]{27x^6y^9z^3} = (27x^6y^9z^3)^{\frac{1}{3}}$ $\therefore \sqrt[n]{a} = a^{\frac{1}{n}}$

$$= (27)^{\frac{1}{3}} (x^6)^{\frac{1}{3}} (y^9)^{\frac{1}{3}} (z^3)^{\frac{1}{3}}$$

$$= (3^3)^{\frac{1}{3}} (x^6)^{\frac{1}{3}} (y^9)^{\frac{1}{3}} (z^3)^{\frac{1}{3}}$$

$$= 3^{3 \times \frac{1}{3}} \cdot x^{6 \times \frac{1}{3}} \cdot y^{9 \times \frac{1}{3}} \cdot z^{3 \times \frac{1}{3}}$$

$$= 3x^2y^3z$$

$\therefore (ab)^m = a^m b^m$

$\therefore (a^m)^n = a^{mn}$

(iii) $(64)^{-\frac{4}{3}} = \frac{1}{(64)^{\frac{4}{3}}}$

$$= \frac{1}{4^{\frac{3 \times 4}{3}}} = \frac{1}{4^4}$$

$$= \frac{1}{256}$$

1.2.2 Surds and their Applications

An irrational radical with rational radicand is called a surd.

For example, if we take the n^{th} root of any rational number a then $\sqrt[n]{a}$ is a surd. $\sqrt{5}$ is a surd because the square root of 5 does not give a

whole number but $\sqrt{9}$ is not a surd because it simplifies to a whole number 3 and our result is not an irrational number. Therefore, the radical $\sqrt[n]{a}$ is irrational $\sqrt{7}, \sqrt{2}, \sqrt[3]{11}$ are surds but $\sqrt{\pi}, \sqrt{e}$ are not surds.

The different types of surds are as follow:

Remember!

Every surd is an irrational number but every irrational number is not a surd e.g., $\sqrt{\pi}$ is not a surd.

- (i) A surd that contains a single term is called a monomial surd e.g., $\sqrt{5}, \sqrt{7}$ etc.
- (ii) A surd that contains the sum of two monomial surds is called a binomial surd e.g., $\sqrt{3} + \sqrt{5}, \sqrt{2} + \sqrt{7}$ etc.
- (iii) $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ are called conjugate surds of each other.

Remember!

The product of two conjugate surds is a rational number.

1.2.3 Rationalization of Denominator

To rationalize a denominator of the form $a + b\sqrt{x}$ or $a - b\sqrt{x}$, we multiply both the numerator and denominator by the conjugate factor.

Example 8: Rationalize the denominator of:

(i) $\frac{3}{\sqrt{5} + \sqrt{2}}$

(ii) $\frac{3}{\sqrt{5} - \sqrt{3}}$

Solution (i):

$$\begin{aligned}\frac{3}{\sqrt{5} + \sqrt{2}} &= \frac{3}{\sqrt{5}} \times \frac{\sqrt{5} - \sqrt{2}}{\sqrt{5} - \sqrt{2}} \\ &= \frac{3(\sqrt{5} - \sqrt{2})}{(\sqrt{5})^2 - (\sqrt{2})^2} = \frac{3(\sqrt{5} - \sqrt{2})}{5 - 2} \\ &= \frac{3(\sqrt{5} - \sqrt{2})}{3} = \sqrt{5} - \sqrt{2}\end{aligned}$$

(ii)

$$\begin{aligned}\frac{3}{\sqrt{5} - \sqrt{3}} &= \frac{3}{\sqrt{5} - \sqrt{3}} \times \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} + \sqrt{3}} \\ &= \frac{3(\sqrt{5} + \sqrt{3})}{(\sqrt{5})^2 - (\sqrt{3})^2} = \frac{3(\sqrt{5} + \sqrt{3})}{5 - 3} \\ &= \frac{3(\sqrt{5} + \sqrt{3})}{2}\end{aligned}$$

EXERCISE 1.2

1. Rationalize the denominator of following:

(i) $\frac{13}{4 + \sqrt{3}}$

(ii) $\frac{\sqrt{2} + \sqrt{5}}{\sqrt{3}}$

(iii) $\frac{\sqrt{2} - 1}{\sqrt{5}}$

$$(iv) \frac{6-4\sqrt{2}}{6+4\sqrt{2}} \quad (v) \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \quad (vi) \frac{4\sqrt{3}}{\sqrt{7}+\sqrt{5}}$$

2. Simplify the following:

$$(i) \left(\frac{81}{16}\right)^{-\frac{3}{4}} \quad (ii) \left(\frac{3}{4}\right)^{-2} \div \left(\frac{4}{9}\right)^3 \times \frac{16}{27} \quad (iii) (0.027)^{-\frac{1}{3}}$$

$$(iv) \sqrt[7]{\frac{x^{14} \times y^{21} \times z^{35}}{y^{14} z^7}} \quad (v) \frac{5 \cdot (25)^{n+1} - 25 \cdot (5)^{2n}}{5 \cdot (5)^{2n+3} - (25)^{n+1}}$$

$$(vi) \frac{(16)^{x+1} + 20(4^{2x})}{2^{x-3} \times 8^{x+2}} \quad (vii) (64)^{-\frac{2}{3}} \div (9)^{-\frac{3}{2}}$$

$$(viii) \frac{3^n \times 9^{n+1}}{3^{n-1} \times 9^{n-1}} \quad (ix) \frac{5^{n+3} - 6.5^{n+1}}{9 \times 5^n - 4 \times 5^n}$$

3. If $x = 3 + \sqrt{8}$ then find the value of:

$$(i) x + \frac{1}{x} \quad (ii) x - \frac{1}{x} \quad (iii) x^2 + \frac{1}{x^2}$$

$$(iv) x^2 - \frac{1}{x^2} \quad (v) x^4 + \frac{1}{x^4} \quad (vi) \left(x - \frac{1}{x}\right)^2$$

4. Find the rational numbers p and q such that $\frac{8-3\sqrt{2}}{4+3\sqrt{2}} = p + q\sqrt{2}$

5. Simplify the following:

$$(i) \frac{(25)^{\frac{3}{2}} \times (243)^{\frac{5}{3}}}{(16)^{\frac{5}{4}} \times (8)^{\frac{4}{3}}} \quad (ii) \frac{54 \times \sqrt[3]{(27)^{2x}}}{9^{x+1} + 216(3^{2x-1})}$$

$$(iii) \sqrt{\frac{(216)^{\frac{2}{3}} \times (25)^{\frac{1}{2}}}{(0.04)^{\frac{-3}{2}}}} \quad (iv) \left(a^{\frac{1}{3}} + b^{\frac{2}{3}}\right) \times \left(a^{\frac{2}{3}} - a^{\frac{1}{3}}b^{\frac{2}{3}} + b^{\frac{4}{3}}\right)$$

1.3 Application of Real Numbers in Daily Life.

Real numbers are extremely useful in our daily life. That is probably one of the main reasons we learn how to count, add and subtract from a very young age. We cannot imagine life without numbers.

Real numbers are used in various fields including

- Science and engineering (physics, mechanical systems, electrical circuits)
- Medicine and Health
- Environmental science (climate modding, pollution monitoring etc.)
- Computer science (algorithm design, data compression, graphic rendering)
- Navigation and transportation (GPS, flight planning)
- Surveying and architecture
- Statistics

Example 9: The sum of two real numbers is 8, and their difference is 2. Find the numbers.

Solution: Let a and b be two real numbers then

$$a + b = 8 \quad \dots(i)$$

$$a - b = 2 \quad \dots(ii)$$

Add (i) and (ii)

$$2a = 10 \Rightarrow a = 5 \quad \text{put in (ii)}$$

$$\Rightarrow 5 - b = 2 \Rightarrow -b = 2 - 5 \Rightarrow -b = -3 \Rightarrow b = 3$$

So, 5 and 3 are the required real numbers.

1.3.1 Temperature Conversions

In the figure, three types of thermometers are shown.

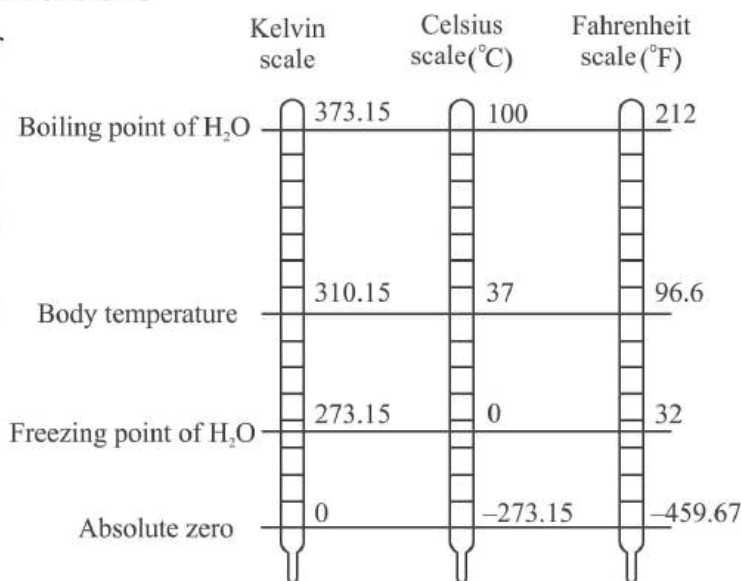
We can convert three temperature scales, Celsius, Fahrenheit, and Kelvin, with each other.

Conversion formulae are given below:

$$(i) \quad K = ^\circ C + 273$$

$$(ii) \quad ^\circ C = \frac{5}{9} (F - 32)^\circ$$

$$(iii) \quad ^\circ F = \frac{9^\circ C}{5} + 32$$



Where K , $^\circ C$, and $^\circ F$ show the Kelvin, Celsius, and Fahrenheit scales respectively.

Example 10: Normal human body temperature is $98.6^{\circ}F$. Convert it into Celsius and Kelvin scale.

Solution: Given that $^{\circ}F = 98.6$

So, to convert it into Celsius scale, we use

$$^{\circ}C = \frac{5}{9}(F - 32)^{\circ}$$

$$^{\circ}C = \frac{5}{9}(98.6 - 32)$$

$$= \frac{5}{9}(66.6)$$

$$= (0.55)(66.6)$$

$$^{\circ}C = 37^{\circ}$$

Hence, normal human body temperature at Celsius scale is 37° .

Now, we convert it into Kelvin scale.

$$K = C + 273^{\circ}$$

$$K = 37^{\circ} + 273^{\circ}$$

$$K = 310 \text{ kelvin}$$

1.3.2 Profit and Loss

The traders may earn profit or incur losses. Profit and loss are a part of business. Profit and loss can be calculated by the following formulae:

(i) Profit = selling Price – cost price

$$P = SP - CP$$

$$\text{Profit \%} = \left(\frac{\text{profit}}{CP} \times 100 \right) \%$$

(ii) Loss = cost price – selling price

$$\text{Loss} = CP - SP$$

$$\text{Loss \%} = \left(\frac{\text{loss}}{CP} \times 100 \right) \%$$

Example 11: Hamail purchased a bicycle for Rs. 6590 and sold it for Rs.6850. Find the profit percentage.

Solution:

$$\begin{aligned}\text{Cost Price} &= \text{CP} = \text{Rs. } 6590 \\ \text{Selling Price} &= \text{SP} = \text{Rs. } 6850 \\ \text{Profit} &= \text{SP} - \text{CP} \\ &= 6850 - 6590 \\ &= \text{Rs. } 260\end{aligned}$$

Now, we find the profit percentage.

$$\begin{aligned}\text{Profit \%} &= \left(\frac{\text{profit}}{\text{CP}} \times 100 \right) \% \\ &= \left(\frac{260 \times 100}{6590} \right) \% \\ &= 3.94\% \\ &\approx 4\%\end{aligned}$$

Example 12: Umair bought a book for Rs. 850 and sold it for Rs. 720. What was his loss percentage?

Solution:

$$\begin{aligned}\text{Cost price of book} &= \text{CP} = \text{Rs. } 850 \\ \text{Selling price of book} &= \text{SP} = \text{Rs. } 720 \\ \text{Loss} &= \text{CP} - \text{SP} \\ &= 850 - 720 \\ &= \text{Rs. } 130 \\ \text{Loss percentage} &= \left(\frac{\text{loss}}{\text{CP}} \times 100 \right) \% \\ &= \left(\frac{130}{850} \times 100 \right) \% \\ &= 15.29\%\end{aligned}$$

Example 13: Saleem, Nadeem, and Tanveer earned a profit of Rs. 4,50,000 from a business. If their investments in the business are in the ratio 4 : 7 : 14, find each person's profit.

Solution:

$$\begin{aligned}\text{Profit earned} &= \text{Rs. } 4,50,000 \\ \text{Given ratio} &= 4 : 7 : 14 \\ \text{Sum of ratios} &= 4 + 7 + 14 \\ &= 25\end{aligned}$$

$$\text{Saleem earned profit} = \frac{4}{25} \times 4,50,000 = \text{Rs. } 72,000$$

$$\text{Nadeem earned profit} = \frac{7}{25} \times 4,50,000 = \text{Rs. } 126,000$$

$$\text{Tanveer earned profit} = \frac{14}{25} \times 4,50,000 = \text{Rs. } 252,000$$

Hence, the profit of Saleem is Rs. 72,000, profit of Nadeem is Rs. 126,000 and profit of Tanveer is Rs. 252,000.

Example 14: If the simple profit on Rs. 6400 for 12 years is Rs. 3840. Find the rate of profit.

Solution: Principal = Rs. 6400
Simple profit = Rs. 3840
Time = 12 years

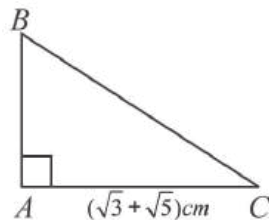
To find the rate we use the following formula:

$$\begin{aligned} \text{Rate} &= \frac{\text{amount of profit} \times 100}{\text{time} \times \text{principal}} \\ &= \frac{3840 \times 100}{12 \times 6400} = 5\% \end{aligned}$$

Thus, rate of profit is 5%.

EXERCISE 1.3

- The sum of three consecutive integers is forty-two, find the three integers.
- The diagram shows right angled $\triangle ABC$ in which the length of $\overline{AC}$ is $(\sqrt{3} + \sqrt{5})$ cm. The area of $\triangle ABC$ is $(1 + \sqrt{15})$ cm². Find the length $\overline{AB}$ in the form $(a\sqrt{3} + b\sqrt{5})$ cm, where a and b are integers.
- A rectangle has sides of length $2 + \sqrt{18}$ m and $\left(5 - \frac{4}{\sqrt{2}}\right)$ m. Express the area of the rectangle in the form $a + b\sqrt{2}$, where a and b are integers.
- Find two numbers whose sum is 68 and difference is 22.
- The weather in Lahore was unusually warm during the summer of 2024. The



TV news reported temperature as high as 48°C . By using the formula $(^{\circ}\text{F} = \frac{9}{5} ^{\circ}\text{C} + 32)$ find the temperature as Fahrenheit scale.

6. The sum of the ages of the father and son is 72 years. Six years ago, the father's age was 2 times the age of the son. What was son's age six years ago?
7. Mirha bought a toy for Rs. 1500 and sold for Rs. 1520. What was her profit percentage?
8. The annual income of Tayyab is Rs. 9,60,000, while the exempted amount is Rs. 1,30,000. How much tax would he have to pay at the rate of 0.75%?
9. Find the compound markup on Rs. 3,75,000 for one year at the rate of 14% compounded annually.

REVIEW EXERCISE 1

1. Four options are given against each statement. Encircle the correct option.
 - (i) $\sqrt{7}$ is:
 - (a) integer
 - (b) rational number
 - (c) irrational number
 - (d) natural number
 - (ii) π and e are:
 - (a) natural numbers
 - (b) integers
 - (c) rational numbers
 - (d) irrational numbers
 - (iii) If n is not a perfect square, then $\sqrt{n}$ is:
 - (a) rational number
 - (b) natural number
 - (c) integer
 - (d) irrational number
 - (iv) $\sqrt{3} + \sqrt{5}$ is:
 - (a) whole number
 - (b) integer
 - (c) rational number
 - (d) irrational number
 - (v) For all $x \in R$, $x = x$ is called:
 - (a) reflexive property
 - (b) transitive number
 - (c) symmetric property
 - (d) trichotomy property
 - (vi) Let $a, b, c \in R$, then $a > b$ and $b > c \Rightarrow a > c$ is called _____ property.
 - (a) trichotomy
 - (b) transitive
 - (c) additive
 - (d) multiplicative

(vii) $2^x \times 8^x = 64$ then $x =$

(a) $\frac{3}{2}$

(b) $\frac{3}{4}$

(c) $\frac{5}{6}$

(d) $\frac{2}{3}$

(viii) Let $a, b \in R$, then $a = b$ and $b = a$ is called _____ property.

(a) reflexive

(b) symmetric

(c) transitive

(d) additive

(ix) $\sqrt{75} + \sqrt{27} =$

(a) $\sqrt{102}$

(b) $9\sqrt{3}$

(c) $5\sqrt{3}$

(d) $8\sqrt{3}$

(x) The product of $(3 + \sqrt{5})(3 - \sqrt{5})$ is:

(a) prime number

(b) odd number

(c) irrational number

(d) rational number

2. If $a = \frac{3}{2}$, $b = \frac{5}{3}$ and $c = \frac{7}{5}$, then verify that

(i) $a(b + c) = ab + ac$

(ii) $(a + b)c = ac + bc$

3. If $a = \frac{4}{3}$, $b = \frac{5}{2}$, $c = \frac{7}{4}$, then verify the associative property of real numbers w.r.t addition and multiplication.

4. Is 0 a rational number? Explain.

5. State trichotomy property of real numbers.

6. Find two rational numbers between 4 and 5.

7. Simplify the following:

(i) $\sqrt[5]{\frac{x^{15}y^{35}}{z^{20}}}$

(ii) $\sqrt[3]{(27)^{2x}}$

(iii) $\frac{6(3)^{n+2}}{3^{n+1}-3^n}$

8. The sum of three consecutive odd integers is 51. Find the three integers.

9. Abdullah picked up 96 balls and placed them into two buckets. One bucket has twenty-eight more balls than the other bucket. How many balls were in each bucket?

10. Salma invested Rs. 3,50,000 in a bank, which paid simple profit at the rate of $7\frac{1}{4}\%$ per annum. After 2 years, the rate was increased to 8% per annum. Find the amount she had at the end of 7 years.

Unit 2

Logarithms

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Express a number in scientific notation and vice versa.
- Describe logarithm of a number
- Differentiate between common and natural logarithm

INTRODUCTION

Logarithms are powerful mathematical tools used to simplify complex calculations, particularly those involving exponential growth or decay. They are widely applicable across various fields, including banking, science, engineering, and information technology. In chemistry, the pH scale, which measures the acidity or alkalinity of a solution, is based on logarithms. They help in transforming non-linear data into linear form for analysis, solving exponential equations and managing calculations involving very large or small numbers efficiently.

2.1 Scientific Notation

A method used to express very large or very small numbers in a more manageable form is known as Scientific notation. It is commonly used in science, engineering and mathematics to simplify complex calculations.

A number in scientific notation is written as:

$$a \times 10^n, \text{ where } 1 \leq a < 10 \text{ and } n \in \mathbb{Z}$$

Here “ a ” is called the coefficient or base number.

Remember!

If the number is greater than 1 then n is positive and if the number is less than 1 then n is negative.

2.1.1 Conversion of Numbers from Ordinary Notation to Scientific Notation

Example 1: Convert 78,000,000 to scientific notation.

Solution: **Step 1:** Move the decimal to get a number between 1 and 10:

$$7.8$$

Step 2: Count the number of places you moved the decimal:

7 places

Step 3: Write in scientific notation:

$$78,000,000 = 7.8 \times 10^7$$

Since we moved the decimal to the **left**, the exponent is **positive**.

Example 2: Convert 0.0000000315 to scientific notation.

Solution:

Step 1: Move the decimal to get a number between 1 and 10:

$$3.15$$

Step 2: Count the number of places you moved the decimal:

8 places

Step 3: Write in scientific notation:

$$0.0000000315 = 3.15 \times 10^{-8}$$

Since we moved the decimal to the **right**, the exponent is **negative**.

Try Yourself!

Convert the following into scientific notation:

(i) 29,000,000

(ii) 0.000006

2.1.2 Conversion of Numbers from Scientific Notation to Ordinary Notation

Example 3: Convert 3.47×10^6 to ordinary notation.

Solution: **Step 1:** Identify the parts:

Coefficient: 3.47

Exponent: 10^6

Remember!

If exponent is positive then the decimal will move to the right.
If exponent is negative then the decimal will move to the left.

Step 2: Since the exponent is **positive** 6, move the decimal point 6 places to the right.

$$3.47 \times 10^6 = 3,470,000$$

Example 4: Convert 6.23×10^{-4} to ordinary notation.

Solution: **Step 1:** Identify the parts:

Coefficient: 6.23

Exponent: 10^{-4}

Try Yourself!

Convert the following into ordinary notation:

(i) 5.63×10^3

(ii) 6.6×10^{-5}

Step 2: Since the exponent is **negative** 4, move the decimal point 4 places to the **left**.

$$6.23 \times 10^{-4} = 0.000623$$

EXERCISE 2.1

1. Express the following numbers in scientific notation:

(i) 2000000

(ii) 48900

(iii) 0.0042

(iv) 0.0000009

(v) 73×10^3

(vi) 0.65×10^2

2. Express the following numbers in ordinary notation:

(i) 8.04×10^2

(ii) 3×10^5

(iii) 1.5×10^{-2}

(iv) 1.77×10^7

(v) 5.5×10^{-6}

(vi) 4×10^{-5}

- The speed of light is approximately 3×10^8 metres per second. Express it in standard form.
- The circumference of the Earth at the equator is about 40075000 metres. Express this number in scientific notation.
- The diameter of Mars is 6.779×10^3 km. Express this number in standard form.
- The diameter of Earth is about 1.2756×10^4 km. Express this number in standard form.

2.2 Logarithm

A logarithm is based on two Greek words: logos and arithmos which means ratio or proportion. John Napier, a Scottish mathematician, introduced the word logarithm. It is a way to simplify complex calculations, especially those involving multiplication and division of large numbers. Now a days logarithm remain fundamental in mathematics, with applications in science, finance and technology.

2.2.1 Logarithm of a Real Number

In simple words, the logarithm of a real number tells us how many times one number must be multiplied by itself to get another number.

The general form of a logarithm is: $\log_b(x) = y$

Where:

- b is the **base**,
- x is the **result** or the number whose logarithm is being taken,
- y is the **exponent** or the logarithm of x to the base b .

This means that:

$$b^y = x \quad \begin{array}{ccc} b^y & = & x \text{ (Exponential form)} \\ \downarrow & \swarrow \searrow & \\ \log_b x & = & y \text{ (Logarithmic form)} \end{array}$$

In words, "**the logarithm of x to the base b is y** ", means that when b is raised to the power y , it equals x .

The relationship between logarithmic form and exponential form is given below:

$$\log_b(x) = y \Leftrightarrow b^y = x \text{ where } b > 0, x > 0 \text{ and } b \neq 1$$

Example 5: Convert $\log_2 8 = 3$ to exponential form.

Solution: $\log_2 8 = 3$

Its exponential form is: $2^3 = 8$

Example 6: Convert $\log_{10} 100 = 2$ to exponential form.

Solution: $\log_{10} 100 = 2$

Its exponential form is: $10^2 = 100$

Example 7: Find the value of x in each case:

(i) $\log_5 25 = x$ (ii) $\log_2 x = 6$

Solution: (i) $\log_5 25 = x$

Its exponential form is:

$$5^x = 25$$

$$\Rightarrow 5^x = 5^2$$

$$\Rightarrow x = 2$$

(ii) $\log_2 x = 6$

Its exponential form is:

$$2^6 = x$$

$$\Rightarrow x = 64$$

Example 8: Convert the following in logarithmic form:

(i) $3^4 = 81$ (ii) $7^0 = 1$

Solution: (i) $3^4 = 81$

Its logarithmic form is:

$$\log_3 81 = 4$$

(ii) $7^0 = 1$

Its logarithmic form is:

$$\log_7 1 = 0$$

EXERCISE 2.2

1. Express each of the following in logarithmic form:

(i) $10^3 = 1000$ (ii) $2^8 = 256$ (iii) $3^{-3} = \frac{1}{27}$

(iv) $20^2 = 400$ (v) $16^{-\frac{1}{4}} = \frac{1}{2}$ (vi) $11^2 = 121$

(vii) $p = q^r$ (viii) $(32)^{\frac{-1}{5}} = \frac{1}{2}$

2. Express each of the following in exponential form:

(i) $\log_5 125 = 3$ (ii) $\log_2 16 = 4$ (iii) $\log_{23} 1 = 0$

(iv) $\log_5 5 = 1$ (v) $\log_2 \frac{1}{8} = -3$ (vi) $\frac{1}{2} = \log_9 3$

(vii) $5 = \log_{10} 100000$ (viii) $\log_4 \frac{1}{16} = -2$

3. Find the value of x in each of the following:

(i) $\log_x 64 = 3$

(ii) $\log_5 1 = x$

(iii) $\log_x 8 = 1$

(iv) $\log_{10} x = -3$

(v) $\log_4 x = \frac{3}{2}$

(vi) $\log_2 1024 = x$

2.3 Common Logarithm

The **common logarithm** is the logarithm with a base of 10. It is written as $\log_{10}$ or simply as $\log$ (when no base is mentioned, it is usually assumed to be base 10).

For example:

$$10^1 = 10 \Leftrightarrow \log 10 = 1$$

$$10^2 = 100 \Leftrightarrow \log 100 = 2$$

$$10^3 = 1000 \Leftrightarrow \log 1000 = 3 \text{ and so on.}$$

$$10^{-1} = \frac{1}{10} = 0.1 \Leftrightarrow \log 0.1 = -1$$

$$10^{-2} = \frac{1}{100} = 0.01 \Leftrightarrow \log 0.01 = -2$$

$$10^{-3} = \frac{1}{1000} = 0.001 \Leftrightarrow \log 0.001 = -3 \text{ and so on.}$$

History

English mathematician Henry Briggs extended Napier's work and developed the common logarithm. He also introduced logarithmic table.

2.3.1 Characteristic and Mantissa of Logarithms

The logarithm of a number consists of two parts: **the characteristic** and **the mantissa**. Here is a simple way to understand them:

(a) Characteristic

The characteristic is the integral part of the logarithm. It tells us how big or small the number is.

Rules for Finding the Characteristic

(i) For a number greater than 1:

Characteristic = number of digits to the left of the decimal point – 1

For example, in $\log 567$ the characteristic = $3 - 1 = 2$

(ii) For a number less than 1:

Characteristic = – (number of zeros between the decimal point and the first non-zero digit + 1)

For example, in $\log 0.0123$ the characteristic = $-(1 + 1) = -2$ or $\bar{2}$

Remember!

When the characteristic is negative, we write it with bar.

Example 9: Find characteristic of the followings:

(i) $\log 725$

(ii) $\log 9.87$

(iii) $\log 0.00045$

(iv) $\log 0.54$

Solution: (i) $\log 725$

Characteristic = $3 - 1 = 2$

(ii) $\log 9.87$

Characteristic = $1 - 1 = 0$

(iii) $\log 0.00045$

Characteristic = $-(3 + 1) = \bar{4}$

(iv) $\log 0.54$

Characteristic = $-(0 + 1) = \bar{1}$

Characteristic of the logarithm of numbers can also be found by expressing them in scientific notation. For example,

Number	Scientific Notation	Characteristic of the logarithm
725	7.25×10^2	2
9.87	9.87×10^0	0
0.00045	4.5×10^{-4}	-4
0.54	5.4×10^{-1}	-1

(b) Mantissa

The mantissa is the decimal part of the logarithm. It represents the "fractional" component and is always positive.

For example, in $\log 5000 = 3.698$ the mantissa is 0.698

2.3.2 Finding Common Logarithm of a Number

Suppose we want to find the common logarithm of 13.45. The step-by-step procedure to find the logarithm is given below:

Step 1: Separate the integral and decimal parts.

Integral part = 13

Decimal part = 45

Remember!

$$\log (\text{Number}) = \text{Characteristic} + \text{Mantissa}$$

Step 2: Find the characteristic of the number

Characteristic = number of digits to the left of the decimal point - 1

$$= 2 - 1 = 1$$

Step 3: In common logarithm table (Complete table is given at the end of the book), check the intersection of row number 13 and column number 4 which is 1271.

Step 4: Find mean difference: Check the intersection of row number 13 and column number 5 in the mean difference which is 16.

Logarithm Table																			
	0	1	2	3	4	5	6	7	8	9	Mean Difference								
											1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170	0212	0253	0294	0334	0374	4	8	12	17	21	25	29	33	37
11	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755	4	8	11	15	19	23	26	30	34
12	0792	0828	0864	0899	0934	0969	1004	1038	1072	1106	3	7	10	14	17	21	24	28	31
13	1139	1173	1206	1239	1271	1303	1335	1367	1399	1430	3	6	10	13	16	19	23	26	29
14	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732	3	6	9	12	15	18	21	24	27

Step 5: Add the numbers found in step 3 and step 4. i.e., $1271 + 16 = 1287$ so the mantissa of given number is 0.1287.

Step 6: Finally, combine the characteristic and mantissa parts found in step 2 and step 5 respectively. We get 1.1287
So, the value of $\log 13.45$ is 1.1287

Example 10: Find logarithm of the following numbers:

- (i) $\log 345$ (ii) $\log 5.678$ (iii) $\log 0.0036$ (iv) $\log 0.0478$

Solution: (i) $\log 345$

$$\text{Characteristic} = 3 - 1 = 2$$

$$\text{Mantissa} = 0.5378 \quad (\text{Look for 34 in the row and 5 in the column of the log table})$$

$$\text{So, } \log (345) = 2 + 0.5378 = 2.5378$$

- (ii) $\log 5.678$

$$\text{Characteristic} = 1 - 1 = 0$$

$$\text{Mantissa} = 0.7542 \quad (7536 + 6 = 7542)$$

$$\text{So, } \log (5.678) = 0 + 0.7542 = 0.7542$$

- (iii) $\log 0.0036$

$$\text{Characteristic} = -(2 + 1) = -3$$

$$\text{Mantissa} = 0.5563 \quad (\text{Look for 36 in the row and 0 in the column of the log table})$$

$$\text{So, } \log (0.0036) = -3 + 0.5563 = -2.4437$$

- (iv) $\log 0.0478$

$$\text{Characteristic} = -(1 + 1) = -2$$

$$\text{Mantissa} = 0.6794 \quad (\text{Look for 47 in the row and 8 in the column of the log table})$$

$$\text{So, } \log 0.0478 = -2 + 0.6794 = -1.3206$$

Do you know?

$$\log (0) = \text{undefined}$$

$$\log (1) = 0$$

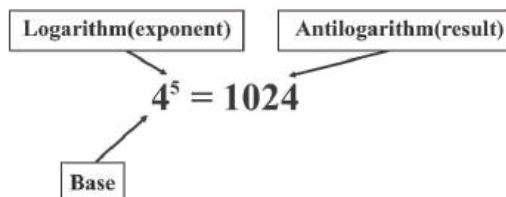
$$\log_a (a) = 1$$

2.3.3 Concept of Antilogarithm

An **antilogarithm** is the inverse operation of a logarithm. An antilogarithm helps to find a number whose logarithmic value is given.

In simple terms:

If $\log_b x = y \Leftrightarrow b^y = x$ then the process of finding x is called antilogarithm of y .



Finding Antilogarithm of a Number using Tables

Let us find the antilogarithm of 2.1245.

The step-by-step procedure to find the antilogarithm is given below:

Step 1: Separate the characteristic and mantissa parts:

Characteristic = 2

Mantissa = 0.1245

Step 2: Find corresponding value of mantissa from antilogarithm table (given at the end of the book):

Check the intersection of row number .12 and column number 4 which provides the number 1330.

Step 3: Find the mean difference:

Check the intersection of row number .12 and the column number 5 of the mean difference in the antilogarithm table which gives 2.

Remember!

The word antilogarithm is another word for the number or result. For example, in $4^3 = 64$, the result 64 is the antilogarithm.

Antilogarithm Table											Mean Difference								
	0	1	2	3	4	5	6	7	8	9									
											1	2	3	4	5	6	7	8	9
.11	1288	1291	1294	1297	1300	1303	1306	1309	1312	1315	0	1	1	1	2	2	2	2	3
.12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	0	1	1	1	2	2	2	2	3
.13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	0	1	1	1	2	2	2	3	3
.14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	0	1	1	1	2	2	2	3	3
.15	1413	1416	1419	1422	1426	1429	1432	1435	1439	1442	0	1	1	1	2	2	2	3	3

Step 4: Add the numbers found in the step 2 and step 3, we get $1330 + 2 = 1332$

Step 5: Insert the decimal point:

Since characteristic is 2, therefore the decimal point will be after 2 digits right from the reference position. So, we get 133.2.

Thus, the antilog (2.1245) = 133.2

Remember!

The place between the first non-zero digit from left and its next digit is called reference position. For example, in 1332, the reference position is between 1 and 3 (1.332).

Example 11: Find the value of x in the followings:

(i) $\log x = 0.2568$ (ii) $\log x = -1.4567$ (iii) $\log x = -2.1234$

Solution: (i) $\log x = 0.2568$

$$\text{Characteristic} = 0$$

$$\text{Mantissa} = 0.2568$$

$$\text{Table value of } 0.2568 = 1803 + 3 = 1806$$

$$\text{So, } x = \text{antilog } (0.2568) = 1.806 \quad (\text{Insert the decimal point at reference position because characteristic is 0.})$$

(ii) $\log x = -1.4567$

Since mantissa is negative, so we make it positive by adding and subtracting 2

$$\begin{aligned} \log x &= -2 + 2 - 1.4567 \\ &= -2 + 0.5433 = \bar{2}.5433 \end{aligned}$$

$$\text{Here characteristic} = \bar{2}, \text{ mantissa} = 0.5433$$

$$\text{Table value of } 0.5433 = 3491 + 2 = 3^{\wedge}493$$

$$\begin{aligned} \text{So, } x &= \text{antilog } (\bar{2}.5433) \\ &= 0.03493 \end{aligned}$$

Since characteristic is $\bar{2}$, therefore decimal point will be before 2 digits left from the reference position.

(iii) $\log x = -2.1234$

Since mantissa is negative, so we make it positive by adding and subtracting 3

$$\begin{aligned} \log x &= -3 + 3 - 2.1234 \\ &= -3 + 0.8766 = \bar{3}.8766 \end{aligned}$$

$$\text{Here characteristic} = \bar{3}, \text{ mantissa} = 0.8766$$

$$\text{Table value of } 0.8766 = 7516 + 10 = 7^{\wedge}526$$

$$\begin{aligned} \text{So, } x &= \text{antilog } (\bar{3}.8766) \\ &= 0.007526 \end{aligned}$$

History

Swiss mathematician and physicist Leonhard Euler introduced 'e' for the base of natural logarithm.

Since characteristic = $\bar{3}$, therefore decimal point will be before 3 digits left from the reference position.

2.3.4 Natural Logarithm

The natural logarithm is the logarithm with base e , where e is a mathematical constant approximately equal to 2.71828. It is denoted as $\ln$. The natural logarithm is commonly

used in mathematics, particularly in calculus, to describe exponential growth, decay and many other natural phenomena.

For example, $\ln e^2 = 2$ i.e., the logarithm of e^2 to the base e is 2.

Difference between Common and Natural Logarithms

Common Logarithm	Natural Logarithm
i. The base of a common logarithm is 10.	i. The base of a natural logarithm is e .
ii. It is written as $\log_{10}(x)$ or simply $\log(x)$ when no base is specified.	ii. It is written as $\ln(x)$
iii. Common logarithms are widely used in everyday calculations, especially in scientific and engineering applications.	iii. Natural logarithms are commonly used in higher level mathematics particularly calculus and applications involving growth/decay processes.

EXERCISE 2.3

1. Find characteristic of the following numbers:

(i) 5287

(ii) 59.28

(iii) 0.0567

(iv) 234.7

(v) 0.000049

(vi) 145000

2. Find logarithm of the following numbers:

(i) 43

(ii) 579

(iii) 1.982

(iv) 0.0876

(v) 0.047

(vi) 0.000354

3. If $\log 3.177 = 0.5019$, then find:

(i) $\log 3177$

(ii) $\log 31.77$

(iii) $\log 0.03177$

4. Find the value of x .

(i) $\log x = 0.0065$

(ii) $\log x = 1.192$

(iii) $\log x = -3.434$

(iv) $\log x = -1.5726$

(v) $\log x = 4.3561$

(vi) $\log x = -2.0184$

2.4 Laws of Logarithm

Laws of logarithm are also known as rules or properties of logarithm. These laws help to simplify logarithmic expressions and solve logarithmic equations.

1. Product Law

$$\log_b xy = \log_b x + \log_b y$$

The logarithm of the product is the sum of the logarithms of the factors.

Proof: Let $m = \log_b x$... (i)

and $n = \log_b y$... (ii)

Express (i) and (ii) in exponential form:

$$x = b^m \quad \text{and} \quad y = b^n$$

Multiply x and y , we get

$$x \cdot y = b^m \cdot b^n = b^{m+n}$$

Its logarithmic form is:

$$\log_b xy = m + n$$

$$\log_b xy = \log_b x + \log_b y \quad [\text{From (i) and (ii)}]$$

2. Quotient Law

$$\log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y$$

The logarithm of a quotient is the difference between the logarithms of the numerator and the denominator.

Proof:

Let $m = \log_b x$... (i)

and $n = \log_b y$... (ii)

Express (i) and (ii) in exponential form:

$$x = b^m \quad \text{and} \quad y = b^n$$

Divide x by y , we get

$$\frac{x}{y} = \frac{b^m}{b^n} = b^{m-n}$$

Its logarithmic form is:

$$\log_b \left(\frac{x}{y} \right) = m - n$$

$$\log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y$$

3. Power Law

$$\log_b x^n = n \cdot \log_b x$$

The logarithm of a number raised to a power is the product of the power and the logarithm of the base number.

Activity

- Divide the students into small groups.
- Distribute the logarithmic expression cards randomly among the groups.
- Each group will work together to identify which logarithmic law applies to each expression.
- After completing the task, each group will present its findings.

Proof:

Let $m = \log_b x \quad \dots(i)$

Its exponential form is:

$$x = b^m$$

Raise both sides to the power n

$$x^n = (b^m)^n = b^{nm}$$

Its logarithmic form is:

$$\log_b x^n = nm$$

$$\log_b x^n = n \cdot \log_b x \quad [\text{From (i)}]$$

4. Change of Base Law

$$\log_b x = \frac{\log_a x}{\log_a b}$$

This law allows to change the base of a logarithm from “ b ” to any other base “ a ”.

Proof: Let

$$m = \log_b x \quad \dots(i)$$

Its exponential form is:

$$b^m = x$$

Taking log with base “ a ” on both sides, we get

$$\log_a b^m = \log_a x$$

$$m \log_a b = \log_a x$$

$$m = \frac{\log_a x}{\log_a b}$$

$$\log_b x = \frac{\log_a x}{\log_a b} \quad [\text{From (i)}]$$

2.4.1 Applications of Logarithm

Logarithms have a wide range of applications in many fields. Some examples are given here about the applications of logarithms.

Example 12: Expand the following using laws of logarithms:

(i) $\log_3(20)$

(ii) $\log_2(9)^5$

(iii) $\log_{32} 27$

Solution: (i) $\log_3(20)$ $= \log_3(2 \times 2 \times 5)$ $= \log_3(2^2 \times 5)$ $= \log_3(2)^2 + \log_3 5$ $= 2 \log_3 2 + \log_3 5$	(ii) $\log_2(9)^5$ $= \log_2(3^2)^5$ $= \log_2(3)^{10}$ $= 10 \log_2 3$	(iii) $\log_{32} 27$ $= \frac{\log 27}{\log 32}$ $= \frac{\log 3^3}{\log 2^5}$ $= \frac{3 \log 3}{5 \log 2}$ $= \frac{3}{5} \log_2 3$
---	--	---

Example 13: Expand the following using laws of logarithms:

(i) $\log_2 \left(\frac{x-y}{z} \right)^3$ (ii) $\log_5 \left(\frac{xy}{z} \right)^8$

Solution: (i) $\log_2 \left(\frac{x-y}{z} \right)^3 = 3 \log_2 \left(\frac{x-y}{z} \right)$
 $= 3 [\log_2(x-y) - \log_2 z]$

(ii) $\log_5 \left(\frac{xy}{z} \right)^8 = 8 \log_5 \left(\frac{xy}{z} \right)$
 $= 8 [\log_5(xy) - \log_5 z]$
 $= 8 [\log_5 x + \log_5 y - \log_5 z]$

Example 14: Write the following as a single logarithm:

(i) $2 \log_3 10 - \log_3 4$ (ii) $6 \log_3 x + 2 \log_3 11$

Solution: (i) $2 \log_3 10 - \log_3 4$ $= \log_3(10)^2 - \log_3 4$ $= \log_3 100 - \log_3 4$ $= \log_3 \left(\frac{100}{4} \right)$ $= \log_3 25$	(ii) $6 \log_3 x + 2 \log_3 11$ $= \log_3 x^6 + \log_3 (11)^2$ $= \log_3 x^6 + \log_3 (121)$ $= \log_3 (121x^6)$
---	---

Example 15: The decibel scale measures sound intensity using the formula

$L = 40 \log_{10} \left(\frac{I}{I_0} \right)$. If a sound has an intensity (I) of 10^6 times the reference intensity

(I_o). What is the sound level in decibels?

Solution: $L = 40 \log_{10} \left(\frac{I}{I_o} \right)$

Put $I = 10^6 I_o$, we get

$$L = 40 \log_{10} \left(\frac{10^6 I_o}{I_o} \right)$$

$$L = 40 \log_{10} (10)^6$$

$$L = 40 \times 6 \log_{10} 10$$

$$L = 40 \times 6 \quad (\because \log_{10} 10 = 1)$$

$$L = 240 \text{ decibels}$$

Do you know?

$\ln(0)$ = undefined

$\ln(1) = 0$

$\ln(e) = 1$

EXERCISE 2.4

1. Without using calculator, evaluate the following:

(i) $\log_2 18 - \log_2 9$ (ii) $\log_2 64 + \log_2 2$ (iii) $\frac{1}{3} \log_3 8 - \log_3 18$

(iv) $2 \log 2 + \log 25$ (v) $\frac{1}{3} \log_4 64 + 2 \log_5 25$ (vi) $\log_3 12 + \log_3 0.25$

2. Write the following as a single logarithm:

(i) $\frac{1}{2} \log 25 + 2 \log 3$ (ii) $\log 9 - \log \frac{1}{3}$

(iii) $\log_5 b^2 \cdot \log_a 5^3$ (iv) $2 \log_3 x + \log_3 y$

(v) $4 \log_5 x - \log_5 y + \log_5 z$ (vi) $2 \ln a + 3 \ln b - 4 \ln c$

3. Expand the following using laws of logarithms:

(i) $\log \left(\frac{11}{5} \right)$ (ii) $\log_5 \sqrt{8a^6}$ (iii) $\ln \left(\frac{a^2 b}{c} \right)$

(iv) $\log \left(\frac{xy}{z} \right)^{\frac{1}{9}}$ (v) $\ln \sqrt[3]{16x^3}$ (vi) $\log_2 \left(\frac{1-a}{b} \right)^5$

4. Find the value of x in the following equations:

(i) $\log 2 + \log x = 1$ (ii) $\log_2 x + \log_2 8 = 5$

(iii) $(81)^x = (243)^{x+2}$ (iv) $\left(\frac{1}{27} \right)^{x-6} = 27$

(v) $\log(5x-10)=2$

(vi) $\log_2(x+1)-\log_2(x-4)=2$

5. Find the values of the following with the help of logarithm table:

(i) $\frac{3.68 \times 4.21}{5.234}$

(ii) $4.67 \times 2.11 \times 2.397$

(iii) $\frac{(20.46)^2 \times (2.4122)}{754.3}$

(iv) $\frac{\sqrt[3]{9.364} \times 21.64}{3.21}$

6. The formula to measure the magnitude of earthquakes is given by

$$M = \log_{10} \left(\frac{A}{A_0} \right)$$
 If amplitude (A) is 10,000 and reference amplitude (A_0) is 10.

What is the magnitude of the earthquake?

7. Abdullah invested Rs. 100,000 in a saving scheme and gains interest at the rate of 5% per annum so that the total value of this investment after
- t
- years is Rs
- y
- . This is modelled by an equation
- $y = 100,000 (1.05)^t$
- ,
- $t \geq 0$
- . Find after how many years the investment will be double.

8. Huria is hiking up a mountain where the temperature (
- T
-) decreases by 3% (or a factor of 0.97) for every 100 metres gained in altitude. The initial temperature (
- T_i
-) at sea level is
- 20°C
- . Using the formula
- $T = T_i \times 0.97^{\frac{h}{100}}$
- , calculate the temperature at an altitude (
- h
-) of 500 metres.

REVIEW EXERCISE 2

1. Four options are given against each statement. Encircle the correct option.

- (i) The standard form of
- 5.2×10^6
- is:

(a) 52,000

(b) 520,000

(c) 5,200,000

(d) 52,000,000

- (ii) Scientific notation of 0.00034 is:

(a) 3.4×10^3

(b) 3.4×10^{-4}

(c) 3.4×10^4

(d) 3.4×10^{-3}

- (iii) The base of common logarithm is:

(a) 2

(b) 10

(c) 5

(d) e

- (iv)
- $\log_2 2^3 =$
- _____.

(a) 1

(b) 2

(c) 5

(d) 3

- (v)
- $\log 100 =$
- _____.

(a) 2

(b) 3

(c) 10

(d) 1

- (vi) If
- $\log 2 = 0.3010$
- , then
- $\log 200$
- is:

(a) 1.3010

(b) 0.6010

(c) 2.3010

(d) 2.6010

(vii) $\log(0) = \underline{\hspace{2cm}}$.

- (a) positive (b) negative (c) zero (d) undefined

(viii) $\log 10,000 =$

- (a) 2 (b) 3 (c) 4 (d) 5

(ix) $\log 5 + \log 3 = \underline{\hspace{2cm}}$.

- (a)
- $\log 0$
- (b)
- $\log 2$
- (c)
- $\log\left(\frac{5}{3}\right)$
- (d)
- $\log 15$

(x) $3^4 = 81$ in logarithmic form is:

- (a)
- $\log_3 4 = 81$
- (b)
- $\log_4 3 = 81$
-
- (c)
- $\log_3 81 = 4$
- (d)
- $\log_4 81 = 3$

2. Express the following numbers in scientific notation:

- (i) 0.000567 (ii) 734 (iii)
- 0.33×10^3

3. Express the following numbers in ordinary notation:

- (i)
- 2.6×10^3
- (ii)
- 8.794×10^{-4}
- (iii)
- 6×10^{-6}

4. Express each of the following in logarithmic form:

- (i)
- $3^7 = 2187$
- (ii)
- $a^b = c$
- (iii)
- $(12)^2 = 144$

5. Express each of the following in exponential form:

- (i)
- $\log_4 8 = x$
- (ii)
- $\log_9 729 = 3$
- (iii)
- $\log_4 1024 = 5$

6. Find value of x in the following:

- (i)
- $\log_9 x = 0.5$
- (ii)
- $\left(\frac{1}{9}\right)^{3x} = 27$
- (iii)
- $\left(\frac{1}{32}\right)^{2x} = 64$

7. Write the following as a single logarithm:

- (i)
- $7 \log x - 3 \log y^2$
- (ii)
- $3 \log 4 - \log 32$

(iii) $\frac{1}{3}(\log_5 8 + \log_5 27) - \log_5 3$

8. Expand the following using laws of logarithms:

- (i)
- $\log(xy^6)$
- (ii)
- $\log_3 \sqrt[6]{m^5 n^3}$
- (iii)
- $\log \sqrt{8x^3}$

9. Find the values of the following with the help of logarithm table:

- (i)
- $\sqrt[3]{68.24}$
- (ii)
- 319.8×3.543
- (iii)
- $\frac{36.12 \times 750.9}{113.2 \times 9.98}$

10. In the year 2016, the population of a city was 22 millions and was growing at a rate of 2.5% per year. The function $p(t) = 22(1.025)^t$ gives the population in millions, t years after 2016. Use the model to determine in which year the population will reach 35 millions. Round the answer to the nearest year.

Unit 3

Sets and Functions

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Recall:
 - Describe mathematics as the study of patterns, structure, and relationships.
 - Identify sets and apply operations on three sets (Subsets, overlapping sets and disjoint sets), using Venn diagrams.
- Solve problems on classification and cataloguing by using Venn diagrams for scenarios involving two sets and three sets. Further application of sets.
- Verify and apply properties/laws of union and intersection of three sets through analytical and Venn diagram methods.
- Apply concepts from set theory to real-world problems (such as in demographic classification, categorizing products in shopping malls)
- Explain product, binary relations and its domain and range.
- Recognize that a relation can be represented by a table, ordered pair and graphs.
- Recognize notation and determine the value of a function and its domain and range.
- Identify types of functions (into, onto, one-to-one, injective, surjective and bijective) by using Venn diagrams.

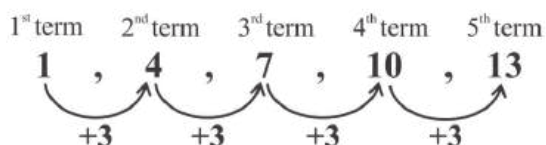
INTRODUCTION

In this unit, we will revise some basic concepts of set theory and functions, beginning with mathematics as an essential study of patterns, structure, and relationships. Students will learn to identify different types of sets, the laws of union and intersection for two and three sets, and their representation using Venn diagrams. Additionally, they will apply set theory to real-world problems to enhance their understanding of demographic classification and product categorization. Classification develops an understanding of the relationship between various sets. Students will also explore binary relations and functions and their representation in various forms including tables, ordered pairs, and graphs.

3.1 Mathematics as the Study of Patterns, Structures and Relationships

Mathematics is the science of patterns, structures, and relationships, comprising various branches that explore and analyze our world's logical and quantitative aspects. The strength of mathematics is based upon relations that enhance the understanding

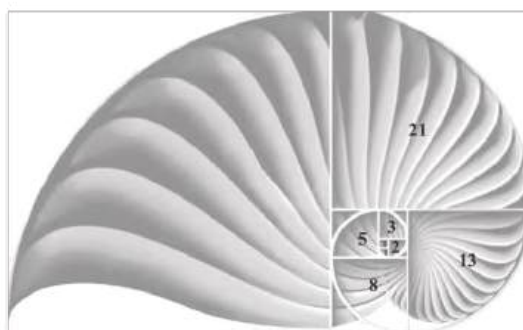
between the patterns and structure and their generalizations. A mathematical pattern is a predictable arrangement of numbers, shapes, or symbols that follows a specific rule or relationship. Virtually, patterns are the key to learning structural knowledge involving numerical and geometrical relationships. For example, look at the following numerical pattern of the numbers



In the above pattern, every term is obtained by adding 3 in the preceding term. This predictable rule or pattern extends continuously, making it a sequence where each term increases at a constant rate.

Consider another example of a famous sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, ..., known as the Fibonacci sequence. This sequence starts with two terms, 0 and 1. Each term of the sequence is obtained by adding the previous two terms. The formula for the Fibonacci sequence is

$F_n = F_{n-1} + F_{n-2}$, where $F_0 = 0$ and $F_1 = 1$ are the first and second terms respectively. This recursive pattern occurs more frequently in nature.



The study of mathematical structure is essential for mathematical competence. A mathematical structure is typically a rule of a numerical, geometric and logical relationship that holds consistency within a specific domain. A structure is a collection of items or objects, along with particular relationships defined among them. Consider a triangle made up of smaller triangles, as illustrated in Figure (iii).

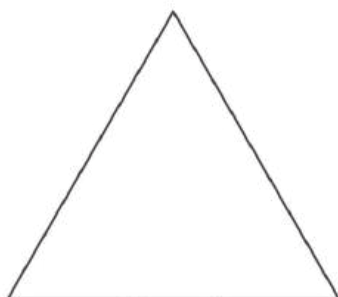


Figure (i)

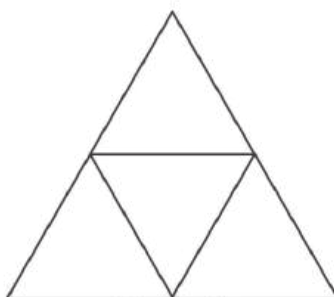


Figure (ii)

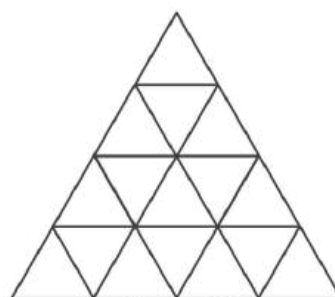


Figure (iii)

The pattern of arranging smaller triangles to form a larger triangle is clear. We can easily recognize the implicit structure: the larger triangle can be seen as consisting of several rows, where each row contains a decreasing number of smaller triangles (e.g., 7 triangles in the first row, 5 in the second, 3 in the third, and 1 at the top).

The repetition of the rows and the spatial relationships between the smaller triangles are critical structural features. The alignment of the smaller triangles creates a sense of congruence as each row is made up of triangles of the same size. At the same time, the arrangement illustrates parallel and perpendicular relationships when viewed in relation to the base of the larger triangle, as shown in Figure (iv). We can develop logical reasoning by understanding these patterns and structures and preparing them for more complex geometric concepts in various fields of mathematics. Similarly, we can establish a relationship between two sets when there is a correspondence between the numbers of these sets.

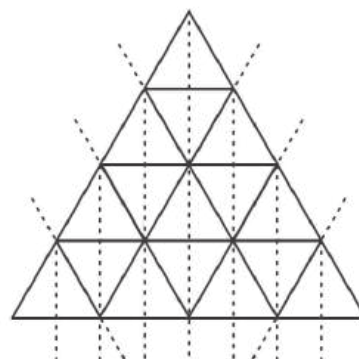


Figure (iv)

3.1.1 Basic Definitions

We are familiar with the notion of a **set** since the word is frequently used in everyday speech, for instance, water set, tea set and sofa set. It is a wonder that mathematicians have developed this ordinary word into a mathematical concept as much as it has become a language that is employed in most branches of modern mathematics. The study of sets helps in understanding the concept of relations, functions and especially in statistics we use sets to understand probability and other important ideas.

A **set** is described as a well-defined collection of distinct objects, numbers or elements, so that we may be able to decide whether the object belongs to the collection or not.

Capital letters A, B, C, X, Y, Z etc., are generally used as names of sets and small letters a, b, c, x, y, z etc., are used as members or elements of sets.

Georg Cantor (1845-1918) was a German mathematician

who significantly contributed to the development of set theory, a key area in mathematics. He showed how to compare two sets



by matching their members one-to-one. Cantor defined different types of infinite sets and proved that there are more real numbers than natural numbers. His proof revealed that there are many sizes of infinity. Additionally, he introduced the concepts of cardinal and ordinal numbers, along with their arithmetic operations.

https://en.wikipedia.org/wiki/Georg_Cantor

There are three different ways of describing a set.

- (i) **The Descriptive form:** A set may be described in words. For instance, the set of all vowels of the English alphabet.
- (ii) **The Tabular form:** A set may be described by listing its elements within brackets. If A is the set mentioned above, then we may write:

$$A = \{a, e, i, o, u\}$$

The tabular form is also known as the Roster form.

- (iii) **Set-builder method:** It is sometimes more convenient or useful to employ the method of set-builder notation in specifying sets. This is done by using a symbol or letter for an arbitrary set member and stating the property common to all the members. Thus, the above set may be written as:

$$A = \{x \mid x \text{ is a vowel of the English alphabets}\}$$

This is read as A is the set of all x such that x is a vowel of the English alphabets.

The symbol used for membership of a set is $\in$. Thus, $a \in A$ means a is an element of A or a belongs to A . $c \notin A$ means c does not belong to A or c is not a member of A . Elements of a set can be anything: people, countries, rivers, objects of our thought. In algebra, we usually deal with sets of numbers. Such sets, along with their names are given below: -

N = The set of natural numbers	$= \{1, 2, 3, \dots\}$
W = The set of whole numbers	$= \{0, 1, 2, \dots\}$
Z = The set of integers	$= \{0, \pm 1, \pm 2, \dots\}$
O = The set of odd integers	$= \{\pm 1, \pm 3, \pm 5, \dots\}$
E = The set of even integers	$= \{0, \pm 2, \pm 4, \dots\}$
P = The set of prime numbers	$= \{2, 3, 5, 7, 11, 13, 17, \dots\}$
Q = The set of all rational numbers	$= \left\{x \mid x = \frac{p}{q} \text{ where } p, q \in Z \text{ and } q \neq 0\right\}$
Q' = The set of all irrational numbers	$= \left\{x \mid x \neq \frac{p}{q}, \text{ where } p, q \in Z \text{ and } q \neq 0\right\}$
R = The set of all real numbers	$= Q \cup Q'$

A set with only one element is called a **singleton set**.

For example, $\{3\}$, $\{a\}$, and $\{\text{Saturday}\}$ are singleton sets. The set with no elements (zero number of elements) is called an **empty set**, **null set**, or **Void set**.

The empty set is denoted by the symbol $\emptyset$ or $\{\}$.

Remember!

The set $\{0\}$ is a singleton set having zero as its only element, and not the empty set.

Equal sets: Two sets A and B are equal if they have exactly the same elements or if every element of set A is an element of set B . If two sets A and B are equal, we write $A=B$. Thus, the sets $\{1, 2, 3\}$ and $\{2, 1, 3\}$ are equal.

Equivalent sets: Two sets A and B are equivalent if they have the same number of elements. For example, if $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4, 5\}$, then A and B are equivalent sets. The symbol $\sim$ is used to represent equivalent sets. Thus, we can write $A \sim B$.

Subset: If every element of a set A is an element of set B , then A is a subset of B . Symbolically this is written as $A \subseteq B$ (A is a subset of B).

In such a case, we say B is a superset of A . Symbolically this is written as:

$$B \supseteq A \text{ (} B \text{ is a superset of } A \text{)}.$$

Remember!

The subset of a set can also be stated as follows:
 $A \subseteq B$ iff $\forall x \in A \Rightarrow x \in B$

Proper subset: If A is a subset of B and B contains at least one element that is not an element of A , then A is said to be a proper subset of B . In such a case, we write:

$$A \subset B \text{ (} A \text{ is a proper subset of } B \text{)}.$$

Improper subset: If A is a subset of B and $A = B$, then we say that A is an improper subset of B . From this definition, it also follows that every set A is a subset of itself and is called an improper subset.

For example, let $A = \{a, b, c\}$, $B = \{c, a, b\}$ and $C = \{a, b, c, d\}$, then clearly

$$A \subset C, B \subset C \text{ but } A = B.$$

Notice that each of sets A and B is an improper subset of the other because $A = B$.

Universal set: The set that contains all objects or elements under consideration is called the universal set or the universe of discourse. It is denoted by U .

Power set: The power set of a set S denoted by $P(S)$ is the set containing all the possible subsets of S . For Example:

(i) If $C = \{a, b, c, d\}$, then

$$P(C) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \\ \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b, c, d\}\}.$$

(ii) If $D = \{a\}$, then $P(D) = \{\emptyset, \{a\}\}$

If S is a finite set with $n(S) = m$ representing the number of elements of the set S , then $n\{P(S)\} = 2^m$ is the number of the elements of the power set.

Remember!

When we do not want to distinguish between proper and improper subsets, we may use the symbol $\subseteq$ for the relationship. It is easy to see that:

$$N \subset W \subset Z \subset Q \subset R$$

EXERCISE 3.1

- Write the following sets in set builder notation:
 - $\{1, 4, 9, 16, 25, 36, \dots, 484\}$
 - $\{2, 4, 8, 16, \dots, 256\}$
 - $\{0, \pm 1, \pm 2, \dots, \pm 1000\}$
 - $\{6, 12, 18, \dots, 120\}$
 - $\{100, 102, 104, \dots, 400\}$
 - $\{1, 3, 9, 27, 81, \dots\}$
 - $\{1, 2, 4, 5, 10, 20, 25, 50, 100\}$
 - $\{5, 10, 15, \dots, 100\}$
 - The set of all integers between -100 and 1000
- Write each of the following sets in tabular forms:
 - $\{x | x \text{ is a multiple of } 3 \wedge x \leq 36\}$
 - $\{x | x \in R \wedge 2x + 1 = 0\}$
 - $\{x | x \in P \wedge x < 12\}$
 - $\{x | x \text{ is a divisor of } 128\}$
 - $\{x | x = 2^n, n \in N \wedge n < 8\}$
 - $\{x | x \in N \wedge x + 4 = 0\}$
 - $\{x | x \in N \wedge x = x\}$
 - $\{x | x \in Z \wedge 3x + 1 = 0\}$
- Write two proper subsets of each of the following sets:
 - $\{a, b, c\}$
 - $\{0, 1\}$
 - N
 - Z
 - Q
 - R
 - $\{x | x \in Q \wedge 0 < x \leq 2\}$
- Is there any set which has no proper subset? If so, name that set.
- What is the difference between $\{a, b\}$ and $\{\{a, b\}\}$?
- What is the number of elements of the power set of each of the following sets?
 - $\{\}$
 - $\{0, 1\}$
 - $\{1, 2, 3, 4, 5, 6, 7\}$
 - $\{0, 1, 2, 3, 4, 5, 6, 7\}$
 - $\{a, \{b, c\}\}$
 - $\{\{a, b\}, \{b, c\}, \{d, e\}\}$
- Write down the power set of each of the following sets:
 - $\{9, 11\}$
 - $\{+, -, \times, \div\}$
 - $\{\emptyset\}$
 - $\{a, \{b, c\}\}$

3.2 Operations on Sets

Just as operations of addition, subtraction etc., are performed on numbers, the operations of union, intersection etc., are performed on sets. We are already familiar with them. A review of the main rules is given below:

Union of Two Sets

The union of two sets A and B , denoted by $A \cup B$, is the set of all elements which belong to A or B . Symbolically;

$$A \cup B = \{x | x \in A \vee x \in B\}.$$

Thus if $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5\}$, then $A \cup B = \{1, 2, 3, 4, 5\}$

Intersection of Two Sets

The intersection of two sets A and B , denoted by $A \cap B$, is the set of all elements that belong to both A and B . Symbolically:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Thus, for the above sets A and B , $A \cap B = \{2, 3\}$.

Remember!

The symbol $\vee$ means or.
The symbol $\wedge$ means and.

Disjoint Sets

If the intersection of two sets is the empty set, the sets are said to be disjoint. For example, if

S_1 = The set of odd natural numbers and S_2 = The set of even natural numbers, then S_1 and S_2 are disjoint sets. Similarly, the set of arts students and the set of science students of a college are disjoint sets.

Overlapping Sets

If the intersection of two sets is non-empty but neither is a subset of the other, the sets are called overlapping sets, e.g., if

$L = \{2, 3, 4, 5, 6\}$ and $M = \{5, 6, 7, 8, 9, 10\}$, then L and M are overlapping sets.

Difference of Two Sets

The difference between the sets A and B denoted by $A - B$, consists of all the elements that belong to A but do not belong to B .

Symbolically, $A - B = \{x \mid x \in A \wedge x \notin B\}$ and $B - A = \{x \mid x \in B \wedge x \notin A\}$

For example, if $A = \{1, 2, 3, 4, 5\}$ and

$$B = \{4, 5, 6, 7, 8, 9, 10\},$$

then $A - B = \{1, 2, 3\}$ and $B - A = \{6, 7, 8, 9, 10\}$.

Notice that: $A - B \neq B - A$.

Complement of a Set

The complement of a set A , denoted by A' or A^c relative to the universal set U is the set of all elements of U , which do not belong to A . Symbolically:

$$A' = \{x \mid x \in U \wedge x \notin A\}$$

For example, if $U = Z$, then $E' = O$ and $O' = E$

For example, If U = Set of alphabets of English language, C = Set of consonants,

W = Set of vowels, then $C' = W$ and $W' = C$.

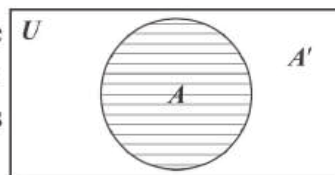
Note:

In view of the definition of complement and difference sets it is evident that for any set A , $A' = U - A$

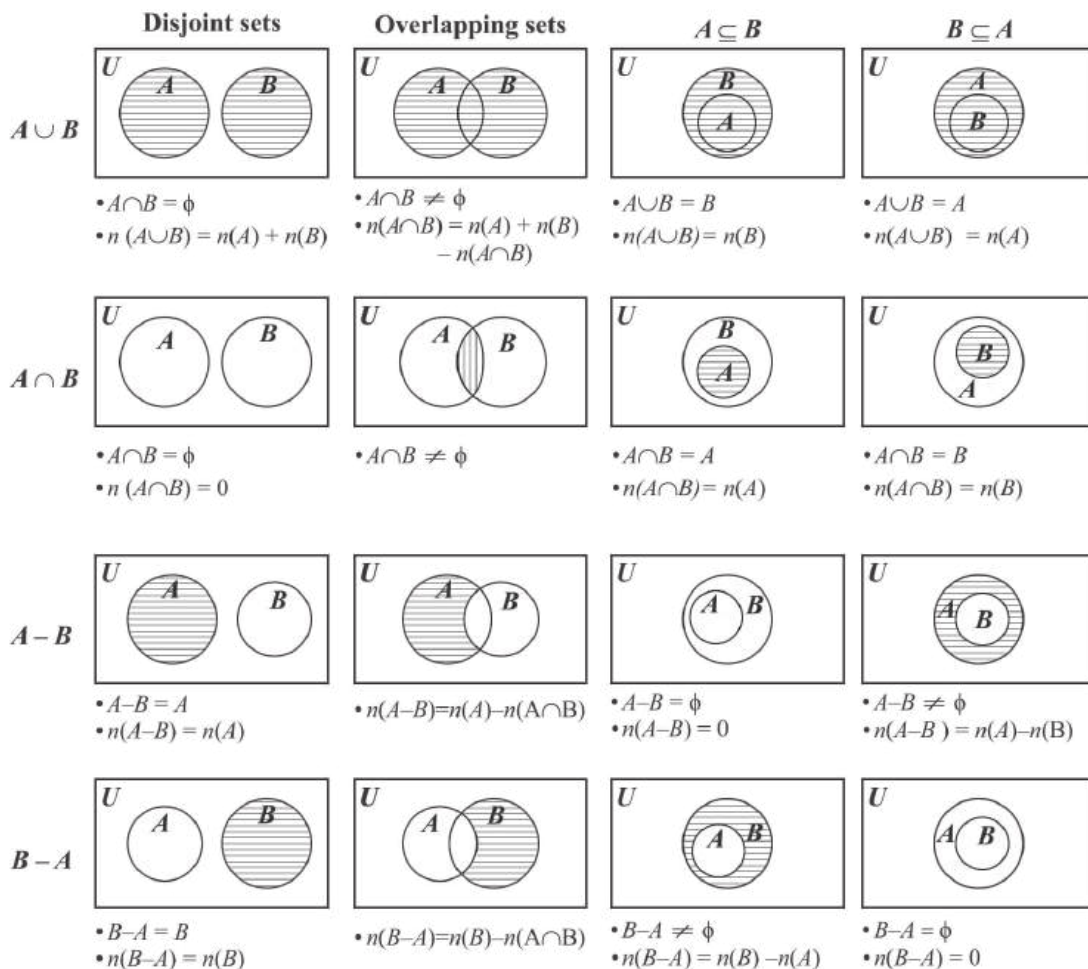
3.2.1 Identification of Sets Using Venn Diagram

Venn diagrams are very useful in depicting visually the basic concepts of sets and relationships between sets. These diagrams were first used by an English logician and mathematician John Venn (1834 to 1883 A.D).

In the adjoining figures, the rectangle represents the universal set U and the lined circular region represents a set A and the remaining portion of the rectangle represents the A' or $U - A$.



Below are given some more diagrams illustrating basic operations on two sets in different cases (the lined region represents the result of the relevant operation in each case shown below).



3.2.2 Operations on Three Sets

If A , B and C are three given sets, operations of union and intersection can be performed on them in the following ways:

- | | | |
|---------------------------|----------------------------|-----------------------------------|
| (i) $A \cup (B \cup C)$ | (ii) $(A \cup B) \cup C$ | (iii) $A \cap (B \cup C)$ |
| (iv) $(A \cap B) \cap C$ | (v) $A \cup (B \cap C)$ | (vi) $(A \cap C) \cup (B \cap C)$ |
| (vii) $(A \cup B) \cap C$ | (viii) $(A \cap B) \cup C$ | (ix) $(A \cup C) \cap (B \cup C)$ |

3.2.2.1 Properties of union and intersection

We now state the fundamental properties of union and intersection of two or three sets.

Properties

- | | |
|---|---|
| (i) $A \cup B = B \cup A$ | (Commutative property of Union) |
| (ii) $A \cap B = B \cap A$ | (Commutative property of Intersection) |
| (iii) $A \cup (B \cup C) = (A \cup B) \cup C$ | (Associative property of Union) |
| (iv) $A \cap (B \cap C) = (A \cap B) \cap C$ | (Associative property of Intersection) |
| (v) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ | (Distributivity of Union over intersection) |
| (vi) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ | (Distributivity of intersection over Union) |
| (vii) $(A \cup B)' = A' \cap B'$ | (De Morgan's Laws) |
| (viii) $(A \cap B)' = A' \cup B'$ | |

Verification of the Properties Using Sets

Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5\}$ and $C = \{3, 4, 5, 6, 7, 8\}$

- (i) $A \cup B = \{1, 2, 3\} \cup \{2, 3, 4, 5\}$; $B \cup A = \{2, 3, 4, 5\} \cup \{1, 2, 3\}$
 $= \{1, 2, 3, 4, 5\}$; $= \{1, 2, 3, 4, 5\}$
 $\therefore A \cup B = B \cup A$
- (ii) $A \cap B = \{1, 2, 3\} \cap \{2, 3, 4, 5\}$; $B \cap A = \{2, 3, 4, 5\} \cap \{1, 2, 3\}$
 $= \{2, 3\}$; $= \{2, 3\}$
 $\therefore A \cap B = B \cap A$

(iii) and (iv) Verify yourself.

- (v) $A \cup (B \cap C) = \{1, 2, 3\} \cup [\{2, 3, 4, 5\} \cap \{3, 4, 5, 6, 7, 8\}]$
 $= \{1, 2, 3\} \cup \{3, 4, 5\}$
 $= \{1, 2, 3, 4, 5\}$... (i)
- $(A \cup B) \cap (A \cup C) = [\{1, 2, 3\} \cup \{2, 3, 4, 5\}] \cap [\{1, 2, 3\} \cup \{3, 4, 5, 6, 7, 8\}]$
 $= \{1, 2, 3, 4, 5\} \cap \{1, 2, 3, 4, 5, 6, 7, 8\}$
 $= \{1, 2, 3, 4, 5\}$... (ii)

From (i) and (ii), $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(vi) Verify yourself.

- (vii) Let the universal set be $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A \cup B = \{1, 2, 3\} \cup \{2, 3, 4, 5\} = \{1, 2, 3, 4, 5\}$
 $(A \cup B)' = U - (A \cup B) = \{6, 7, 8, 9, 10\}$... (i)
 $A' = U - A = \{4, 5, 6, 7, 8, 9, 10\}$
 $B' = U - B = \{1, 6, 7, 8, 9, 10\}$
 $A' \cap B' = \{4, 5, 6, 7, 8, 9, 10\} \cap \{1, 6, 7, 8, 9, 10\}$
 $= \{6, 7, 8, 9, 10\}$... (ii)
 From (i) and (ii), $(A \cup B)' = A' \cap B'$

(viii) Verify yourself.

Verification of the properties with the help of Venn diagrams

(i) and (ii): Verification is very simple, therefore, do it by yourself.

(iii): In Fig. (1), set A is represented by a vertically lined region and $B \cup C$ is represented by a horizontally lined region. The set $A \cup (B \cup C)$ is represented by the region lined either in one way or both.

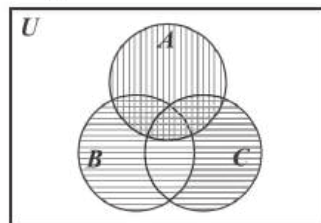


Fig. (1)

In Fig. (2) $A \cup B$ is represented by a horizontally lined region and C by a vertically lined region. $(A \cup B) \cup C$ is represented by the region lined in either one or both ways.

From Fig (1) and (2) we can see that

$$A \cup (B \cup C) = (A \cup B) \cup C$$

(iv) In Fig. (3), the doubly lined region represents

$$A \cap (B \cap C)$$

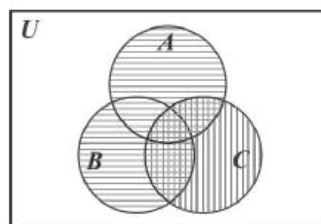


Fig. (2)

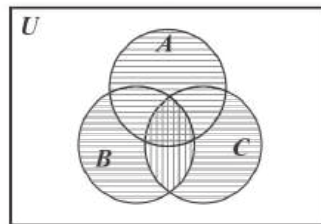


Fig. (3)

In Fig. (4), the doubly lined region represents $(A \cap B) \cap C$. Since in Fig. (3) and Fig. (4), these regions are the same, therefore, $A \cap (B \cap C) = (A \cap B) \cap C$.

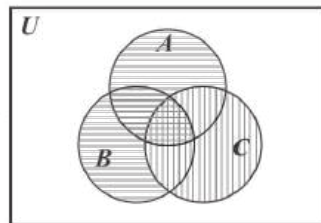


Fig. (4)

(v) In Fig. (5), $A \cup (B \cap C)$ is represented by the region which is lined horizontally or vertically or both ways.

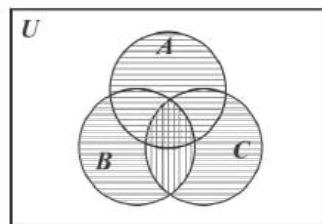


Fig. (5)

In Fig. (6), $(A \cup B) \cap (A \cup C)$ is represented by the doubly lined region.

Since the two regions in Fig (5) and (6) are the same, therefore,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

(vi) Verify yourselves.

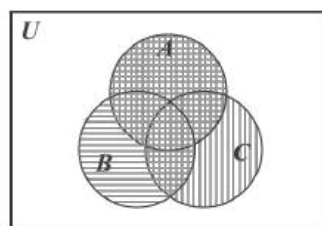


Fig. (6)

(vii) In Fig. (7), $(A \cup B)'$ is represented by a vertically lined region.

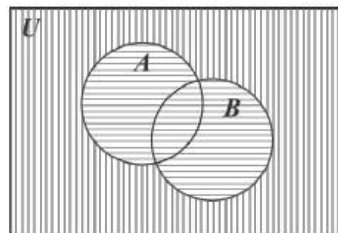


Fig. (7)

In Fig. (8), the doubly lined region represents $A' \cap B'$.

The two regions in Fig. (7) and (8) are the same, therefore,
 $(A \cup B)' = A' \cap B'$

(viii) Verify yourselves.

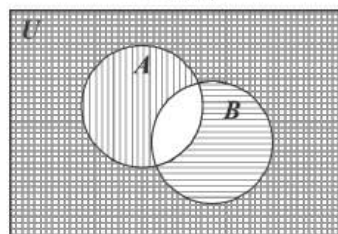


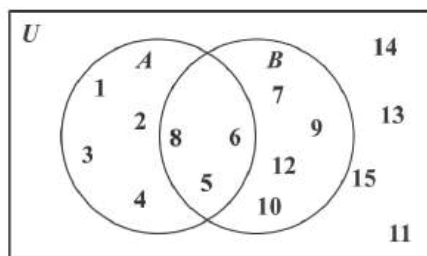
Fig. (8)

Note:

Only overlapping sets have been considered in the Venn diagrams above. Verification for other cases can be conducted similarly.

Example 1: Consider the adjacent Venn diagram illustrating two non-empty sets, A and B .

- Determine the number of elements common to sets A and B .
- Identify all the elements exclusively in set B and not in set A .
- Calculate the union of sets A and B .



Solution: From the information provided in the Venn diagram, we have:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$A = \{1, 2, 3, 4, 5, 6, 8\}$$

$$B = \{5, 6, 7, 8, 9, 10, 12\}$$

(a) The elements in both sets A and B are the intersection of the sets:

$$A \cap B = \{5, 6, 8\}$$

(b) The elements that are only in set B, not in set A, is the sets' differences.

$$B - A = \{7, 9, 10, 12\}$$

$$(c) \quad A \cup B = \{1, 2, 3, 4, 5, 6, 8\} \cup \{5, 6, 7, 8, 9, 10, 12\}$$

$$= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$$

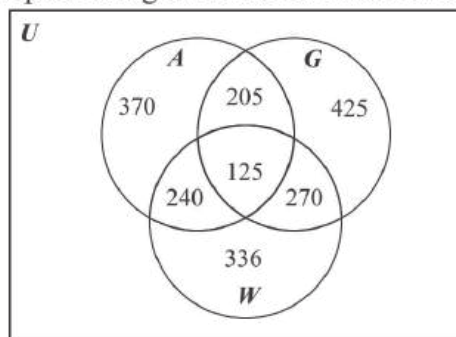
Example 2: Consider the adjacent Venn diagram representing the students enrolled in different courses in an IT institution.

U = Students enrolled in IT institutions

A = Students enrolled in an Applied Robotics

G = Students enrolled in a Game Development

W = Students enrolled in a Web Designing



(a) How many students enrolled in the applied Robotics course?

(b) Determine the total number of Students enrolled in a Game Development.

(c) How many students are enrolled in the Game development and Web designing course?

(d) Identify the students enrolled in Web development but not Applied Robotics.

(e) How many students are enrolled in IT institutions?

(f) How many students enrolled in all three courses?

Solution:

(a) Set A represents the total number of students enrolled in the Applied Robotics program.

$$\text{Total} = 370 + 205 + 125 + 240 = 940$$

So, the total number of students in the Applied Robotics course is 940.

(b) The total number of students enrolled in a Game Development is represented by the set G.

$$\text{Total} = 205 + 125 + 270 + 425 = 1025$$

Thus, the Students enrolled in a Game Development is 1025

- (c) Total students are enrolled in both the Game development and Web designing. The course is the intersection of G and W .

$$G \cap W = 125 + 270 = 395$$

Therefore, 395 students are enrolled in both the Game development and Web designing Course.

- (d) The students who are enrolled in Web development but not in Applied Robotics are the sum of values 336 and 270 in set W .

$$\text{Total} = 336 + 270 = 606$$

So, there are 606 students who enrolled in Web development courses but not in Applied Robotics.

- (e) The total number of students enrolled in IT institutions are represented by all the values inside the circles.

$$\text{Total} = 370 + 205 + 125 + 240 + 425 + 270 + 336 = 1971$$

There are a total of 1971 students enrolled in IT Institutions.

- (f) The students who enrolled in all three courses are the intersection of all the circles are represented by the value 125.

3.2.2 Real-World Applications

In this section, we will learn to apply concepts from set theory to real-world problems, such as solving problems on classification and cataloging using Venn diagrams. We will also explore some real-life situations, such as demographic classification and categorizing products in shopping malls.

For this purpose, we use the concept of cardinality of a set. The cardinality of a set is defined as the total number of elements of a set. The cardinality of a set is basically the size of the set. For a non-empty set A , the cardinality of a set is denoted by $n(A)$.

If $A = \{1, 3, 5, 7, 9, 11\}$, then $n(A)=6$. To find the cardinality of a set, we use the following rule called the **inclusion-exclusion principle** for two or three sets.

Principle of Inclusion and Exclusion for Two Sets

Let A and B be finite sets, then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

and $A \cup B$ and $A \cap B$ are also finite.

Principle of Inclusion and Exclusion for Three Sets

If A , B and C are finite sets, then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

and $A \cup B \cup C$, $A \cap B$, $A \cap C$, $B \cap C$ and $A \cap B \cap C$ are also finite.

Example 3: There are 98 secondary school students in a sports club. 58 students join the swimming club, and 50 join the tug-of-war club. How many students participated in both games?

Solution: Let $U = \{\text{total student in a sports club of school}\}$

$A = \{\text{students who participated in swimming club}\}$

$B = \{\text{students who participated in tug-of-war club}\}$

From the statement of problems, we have

$$n(U) = n(A \cup B) = 98, n(A) = 58, n(B) = 50.$$

We want to find the total number of students who participated in both clubs.

$$n(A \cap B) = ?$$

Using the principles of inclusion and exclusion for two sets:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

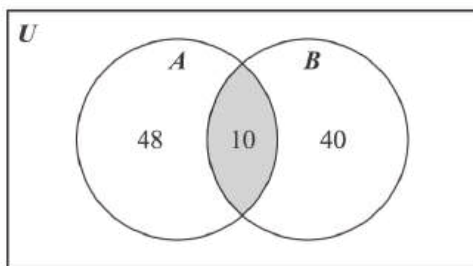
$$\Rightarrow n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= 58 + 50 - 98$$

$$= 10$$

Thus, 10 students participated in both clubs.

The adjacent Venn diagram shows the number of students in each sports club.



Example 4: Mr. Saleem, a school teacher, has a small library in his house containing 150 books. He has two main categories for these books: islamic and science. He categorized 70 books as islamic books and 90 books as science books. There are 15 books that neither belong to the islamic nor science books category. How many books are classified under both the islamic and science categories?

Solution: Let $U = \text{total number of books in library}$

$A = 70 \text{ books in Islamic category}$

$B = 90 \text{ books in Science category}$

$C = 15 \text{ books that do not belong to any category}$

$x = \text{number of books that belong to both the categories}$

The adjacent Venn diagram shows the number of books that are classified under both the islamic and science categories

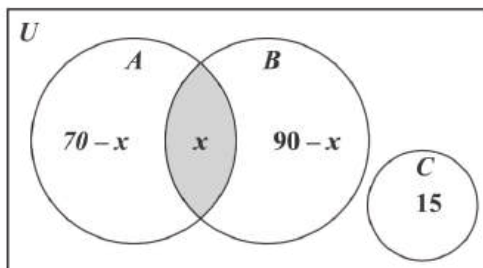
$$\text{As, } n(U) = 150$$

$$\text{So, } 70 - x + x + 90 - x + 15 = 150$$

$$\Rightarrow 175 - x = 150$$

$$\Rightarrow x = 25$$

Thus, 25 books are classified under both islamic and science categories.



Example 5: In a college, 45 teachers teach mathematics or physics or chemistry. Here is information about teachers who teach different subjects:

- 18 teach mathematics ▪ 12 teach physics
- 8 teach chemistry ▪ 6 teach both mathematics and physics
- 4 teach both physics and chemistry
- 2 teach both mathematics and chemistry.

How many teachers teach all three subjects?

Solution: Let U = total number of teachers in the college

M = teachers who teach mathematics

P = teachers who teach physics

C = teachers who teach chemistry

From the statement of problems, we have

$$n(M \cup P \cup C) = 45, n(M) = 18, n(P) = 12, n(C) = 8, n(M \cap P) = 6,$$

$$n(P \cap C) = 4, n(M \cap C) = 2$$

We want to find the total number of teachers who teach all the subjects.

$$n(M \cap P \cap C) = ?$$

Using the principle of inclusion and exclusion for three sets:

$$\begin{aligned} n(M \cup P \cup C) &= n(M) + n(P) + n(C) - n(M \cap P) - n(P \cap C) - n(M \cap C) \\ &\quad + n(M \cap P \cap C) \end{aligned}$$

$$\begin{aligned} \Rightarrow n(M \cap P \cap C) &= n(M \cup P \cup C) - n(M) - n(P) - n(C) + n(M \cap P) + n(P \cap C) \\ &\quad + n(M \cap C) \\ &= 45 - 18 - 12 - 8 + 6 + 4 + 2 \\ &= 19 \end{aligned}$$

Therefore 19 teachers teach all three subjects.

Example 6: A survey of 130 customers in a shopping mall was conducted in which they were asked about buying preferences.

The survey result showed the following statistics:

- 57 customers bought garments
 - 50 customers bought cosmetics
 - 46 customers bought electronics
 - 31 customers purchased both garments and cosmetics
 - 25 customers purchased both garments and electronics
 - 21 customers purchased both cosmetics and electronics
 - 12 customers purchased all three products i.e. garments, cosmetics, and electronics.
- (a) How many of the customers bought at least one of the products: garments, cosmetics or electronics?
- (b) How many of the customers bought only one of the products: garments, Cosmetics or electronics?
- (c) How many customers did not buy any of the three products?

Solution: Let U = total number of customers surveyed in the shopping mall

G = Customer who bought garments

C = Customer who bought cosmetics

E = Customer who bought electronics

From the statement of problems, we have

$$n(U) = 130, n(G) = 57, n(C) = 50, n(E) = 46, n(G \cap C) = 31,$$

$$n(G \cap E) = 25, n(C \cap E) = 21 \text{ and } n(G \cap C \cap E) = 12.$$

- (a) We want to find the total number of customers who have bought at least one of the products: garments, cosmetics, or electronics.

We are to find $n(G \cup C \cup E)$.

Using the principle of inclusion and exclusion for three sets:

$$\begin{aligned} n(G \cup C \cup E) &= n(G) + n(C) + n(E) - n(G \cap C) - n(G \cap E) - n(C \cap E) + n(G \cap C \cap E) \\ &= 57 + 50 + 46 - 31 - 25 - 21 + 12 = 88 \end{aligned}$$

Thus, 88 customers bought at least one of the products: garments, cosmetics, or electronics.

(b) Customers who bought only garments

$$\begin{aligned}
 &= n(G) - n(G \cap C) - n(G \cap E) + n(G \cap C \cap E) \\
 &= 57 - 31 - 25 + 12 \\
 &= 13
 \end{aligned}$$

Customers who bought only cosmetics

$$\begin{aligned}
 &= n(C) - n(G \cap C) - n(C \cap E) + n(G \cap C \cap E) \\
 &= 50 - 31 - 21 + 12 \\
 &= 10
 \end{aligned}$$

Customers who bought only electronics

$$\begin{aligned}
 &= n(E) - n(G \cap E) - n(C \cap E) + n(G \cap C \cap E) \\
 &= 46 - 25 - 21 + 12 = 12
 \end{aligned}$$

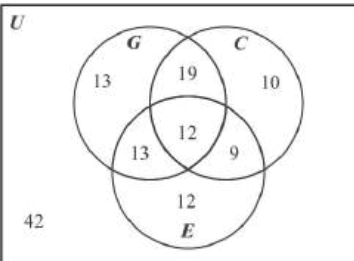
Therefore, the customers bought only one of the products: garments, cosmetics,

or electronics = $13 + 10 + 12 = 35$

(c) Since the total number of Customers surveyed was 130, and 88 customers bought at least one of the products: garments, cosmetics, or electronics. The customers who did not buy any of the three products can be calculated as:

$$\begin{aligned}
 n(G \cup C \cup E)' &= n(U) - n(G \cup C \cup E) \\
 &= 130 - 88 = 42
 \end{aligned}$$

So, 42 customers did not buy any of the three products.



Challenge!

The Venn diagram above illustrates the scenario presented in Example 6. Can you provide a justification for each value within the circles?

Exercise 3.2

- Consider the universal set $U = \{x \mid x \text{ is multiple of } 2 \text{ and } 0 < x \leq 30\}$,
 $A = \{x \mid x \text{ is a multiple of } 6\}$ and $B = \{x \mid x \text{ is a multiple of } 8\}$
 - List all elements of sets A and B in tabular form
 - Find $A \cap B$
 - Draw a Venn diagram
- Let $U = \{x \mid x \text{ is an integer and } 0 < x \leq 150\}$,
 $G = \{x \mid x = 2^m \text{ for integer } m\}$ and
 $H = \{x \mid x \text{ is a square}\}$
 - List all elements of sets G and H in tabular form
 - Find $G \cup H$
 - Find $G \cap H$
- Consider the sets $P = \{x \mid x \text{ is a prime number and } 0 < x \leq 20\}$ and
 $Q = \{x \mid x \text{ is a divisor of } 210 \text{ and } 0 < x \leq 20\}$
 - Find $P \cap Q$
 - Find $P \cup Q$

4. Verify the commutative properties of union and intersection for the following pairs of sets:
- (i) $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 6, 8, 10\}$ (ii) N, Z
- (iii) $A = \{x \mid x \in R \wedge x \geq 0\}$, $B = R$.
5. Let $U = \{a, b, c, d, e, f, g, h, i, j\}$
 $A = \{a, b, c, d, g, h\}$, $B = \{c, d, e, f, j\}$,
 Verify De Morgan's Laws for these sets. Draw Venn diagram
6. If $U = \{1, 2, 3, \dots, 20\}$ and $A = \{1, 3, 5, \dots, 19\}$, verify the following:
- (i) $A \cup A' = U$ (ii) $A \cap U = A$ (iii) $A \cap A' = \phi$
7. In a class of 55 students, 34 like to play cricket and 30 like to play hockey. Also each student likes to play at least one of the two games. How many students like to play both games?
8. In a group of 500 employees, 250 can speak Urdu, 150 can speak English, 50 can speak Punjabi, 40 can speak Urdu and English, 30 can speak both English and Punjabi, and 10 can speak Urdu and Punjabi. How many can speak all three languages?
9. In sports events, 19 people wear blue shirts, 15 wear green shirts, 3 wear blue and green shirts, 4 wear a cap and blue shirts, and 2 wear a cap and green shirts. The total number of people with either a blue or green shirt or cap is 34. How many people are wearing caps?
10. In a training session, 17 participants have laptops, 11 have tablets, 9 have laptops and tablets, 6 have laptops and books, and 4 have both tablets and books. Eight participants have all three items. The total number of participants with laptops, tablets, or books is 35. How many participants have books?
11. A shopping mall has 150 employees labelled 1 to 150, representing the Universal set U . The employees fall into the following categories:
- Set A: 40 employees with a salary range of 30k-45k, labelled from 50 to 89.
 - Set B: 50 employees with a salary range of 50k-80k, labelled from 101 to 150.
 - Set C: 60 employees with a salary range of 100k-150k, labelled from 1 to 49 and 90 to 100.
- (a) Find $(A' \cup B') \cap C$ (b) Find $n\{A \cap (B' \cap C')\}$

12. In a secondary school with 125 students participate in at least one of the following sports: cricket, football, or hockey.
- 60 students play cricket.
 - 70 students play football.
 - 40 students play hockey.
 - 25 students play both cricket and football.
 - 15 students play both football and hockey.
 - 10 students play both cricket and hockey.
- (a) How many students play all three sports?
- (b) Draw a Venn diagram showing the distribution of sports participation in all the games.
13. A survey was conducted in which 130 people were asked about their favourite foods. The survey results showed the following information:
- 40 people said they liked nihari
 - 65 people said they liked biryani
 - 50 people said they liked korma
 - 20 people said they liked nihari and biryani
 - 35 people said they liked biryani and korma
 - 27 people said they liked nihari and korma
 - 12 people said they liked all three foods nihari, biryani, and korma
- (a) At least how many people like nihari, biryani or korma?
- (b) How many people did not like nihari, biryani, or korma?
- (c) How many people like only one of the following foods: nihari, biryani, or korma?
- (d) Draw a Venn diagram.

3.3 Binary Relations

In everyday use, relation means an abstract type of connection between two persons or objects, for instance, (teacher, pupil), (mother, son), (husband, wife), (brother, sister), (friend, friend), (house, owner). In mathematics also some operations determine the relationship between two numbers, for example:

$> : (5, 4)$; square: $(25, 5)$; Square root: $(2, 4)$; Equal: $(2 \times 2, 4)$.

In the above examples $>$, square, square root and equal are examples of relations.

Mathematically, a relation is any set of ordered pairs. The relationship between the components of an ordered pair may or may not be mentioned.

- (i) Let A and B be two non-empty sets, then the Cartesian product is the set of all ordered pairs (x, y) such that $x \in A$ and $y \in B$ and is denoted by $A \times B$. Symbolically we can write it as $A \times B = \{(x, y) | x \in A \text{ and } y \in B\}$.
- (ii) Any subset of $A \times B$ is called a binary relation, or simply a relation, from A to B . Ordinarily a relation is denoted by the letter r .
- (iii) The set of the first elements of the ordered pairs forming a relation is called its domain. The domain of any relation r is denoted as $\text{Dom } r$.
- (iv) The set of the second elements of the ordered pairs forming a relation is called its range. The range of any relation r is denoted as $\text{Ran } r$.
- (v) If A is a non-empty set, any subset of $A \times A$ is called a relation in A .

Example 7: Let c_1, c_2, c_3 be three children and m_1, m_2 be two men such that the father of both c_1, c_2 is m_1 and father of c_3 is m_2 . Find the relation $\{(child, father)\}$

Solution: C = Set of children = $\{c_1, c_2, c_3\}$ and F = set of fathers = $\{m_1, m_2\}$

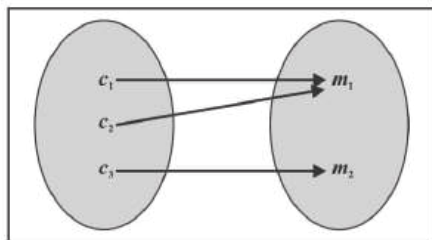
The Cartesian product of C and F :

$$C \times F = \{(c_1, m_1), (c_1, m_2), (c_2, m_1), (c_2, m_2), (c_3, m_1), (c_3, m_2)\}$$

r = set of ordered pairs (child, father).

$$= \{(c_1, m_1), (c_2, m_1), (c_3, m_2)\}$$

The relation is shown diagrammatically in adjacent figure.



Example 8: Let $A = \{1, 2, 3\}$. Determine the relation r such that $x r y$ iff $x < y$. Also find domain and range.

Solution: $A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

Clearly, required relation is:

$$r = \{(1, 2), (1, 3), (2, 3)\}, \text{ Dom } r = \{1, 2\}, \text{ Range } r = \{2, 3\}$$

3.3.1 Relation as Table, Ordered Pair and Graphs

We have learned that a relation in mathematics is any subset of the Cartesian product, which contains all ordered pairs. Each ordered pair consists of two coordinates, x and y . The x coordinate is called abscissa, and the y coordinate is ordinate, often representing an input and an output. Now, we describe the relation in three different ways.

Ordered Pairs: A relation can be represented by a set of ordered pairs. For example, consider a water tank that starts with 1 litre of water already inside. Each minute, 1 additional litre of water is added to the tank. The situation can be represented by the relation $r = \{ (x, y) \mid y = x + 1 \}$, where x is the number of minutes (time) that have passed since the filling started and y is the total amount of water (in litres) in the tank.

When $x = 0, y = 1$ and $x = 1, y = 2$

In order pair this relation is represented as:

$$\{(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$$

The above relation in table form can be represented as given below:

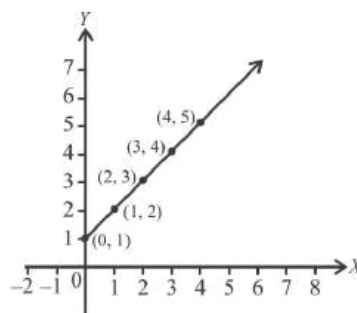
Table

x (time in minutes)	$y = x + 1$ (water in litres)
0	$y = 0 + 1 = 1$
1	$y = 1 + 1 = 2$
2	$y = 2 + 1 = 3$
3	$y = 3 + 1 = 4$
4	$y = 4 + 1 = 5$
5	$y = 5 + 1 = 6$

Graph: We can also represent the relations visually by drawing a graph. To draw the diagram, we use ordered pairs. Each ordered pair (x, y) is plotted as a point in the coordinate plane, where x is the first element and y is the second element of the ordered pair.

The relation is represented graphically by the line passing through the points,

$\{(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$ as shown in the adjacent Figure.



3.3.2 Function and its Domain and Range

Functions

A very important particular type of relation is a function defined as below:

Let A and B be two non-empty sets such that:

- (i) f is a relation from A to B , that is, f is a subset of $A \times B$
- (ii) Domain $f = A$

- (iii) First element of no two pairs of f are equal, then f is said to be a function from A to B .

The function f is also written as:

$$f : A \rightarrow B$$

Which is read as f is a function from A to B . The set of all first elements of each ordered pair represents the domain of f , and all second elements represent the range of f . Here, the domain of f is A , and the range of f is B .

If (x, y) is an element of f when regarded as a set of ordered pairs.

We write $y = f(x)$. y is called the value of f for x or the image of x under f .

Example 9: If $A = \{0, 1, 2, 3, 4\}$ and $B = \{3, 5, 7, 9, 11\}$, define a function $f: A \rightarrow B$, $f = \{(x, y) \mid y = 2x + 3, x \in A \text{ and } y \in B\}$, Find the value of function f , its domain, co-domain and range.

Solution: Given: $y = 2x + 3$; $x \in A$ and $y \in B$, then value of function,

$$f = \{(0, 3), (1, 5), (2, 7), (3, 9), (4, 11)\}$$

$$\text{Dom } f = \{0, 1, 2, 3, 4\} = A$$

$\Rightarrow$ Co-domain $f = B$ and

$\Rightarrow$ Range $f = \{3, 5, 7, 9, 11\} \subseteq B$

Types of functions

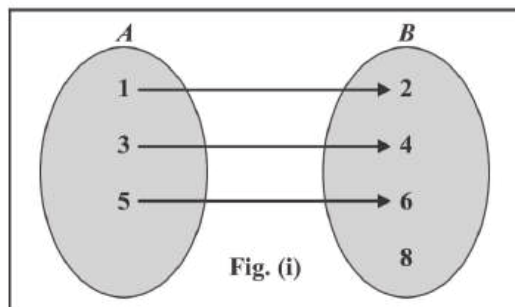
In this section we discuss different types of functions:

(i) Into Function

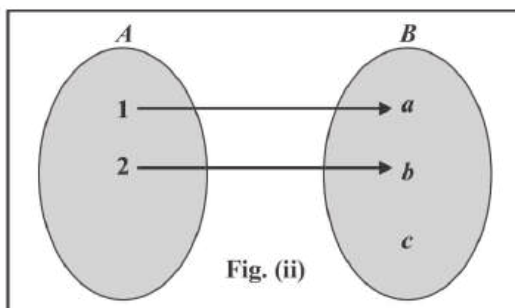
If a function $f: A \rightarrow B$ is such that $\text{Range } f \subset B$ i.e., $\text{Range } f \neq B$, then f is said to be a function from A into B . In Fig. (i), f is clearly a function. But $\text{Range } f \neq B$. Therefore, f is a function from A into B .

(ii) (One - One) Function (or Injective Function)

If a function f from A into B is such that second elements of no two of its ordered pairs are same, then it is



$$f = \{(1, 2), (3, 4), (5, 6)\}$$

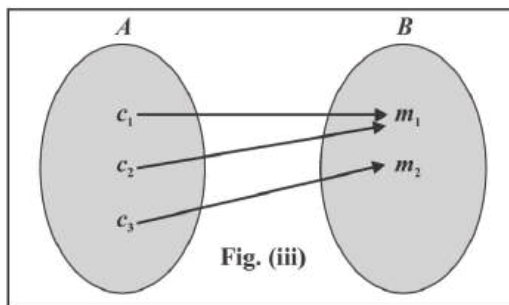


$$f = \{(1, a), (2, b)\}$$

called an injective function; the function shown in Fig. (ii) is such a function.

(iii) Onto Function (or Surjective function)

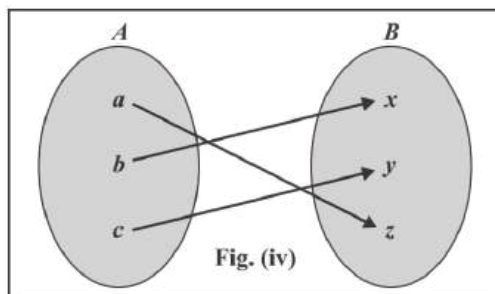
If a function $f: A \rightarrow B$ is such that $\text{Range } f = B$ i.e., every element of B is the image of some element of A , then f is called an **onto** function or a surjective function.



$$f = \{(c_1, m_1), (c_2, m_1), (c_3, m_2)\}$$

(iv) (One – One) and onto Function (or Bijective Function)

A function f from A to B is said to be a Bijective function if it is both one-one and onto. Such a function is also called (1 – 1) correspondence between the sets A and B .



$$f = \{(a, z), (b, x), (c, y)\}$$

(a, z) , (b, x) and (c, y) are the pairs of corresponding elements i.e., in this case $f = \{(a, z), (b, x), (c, y)\}$ which is a bijective function or (1 – 1) correspondence between the sets A and B .

3.3.3 Notation of Function

We know that set-builder notation is more suitable for infinite sets. So is the case with respect to a function comprising an infinite number of ordered pairs. Consider for instance, the function

$$f = \{(-1, 1), (0, 0), (1, 1), (2, 4), (3, 9), (4, 16), \dots\}$$

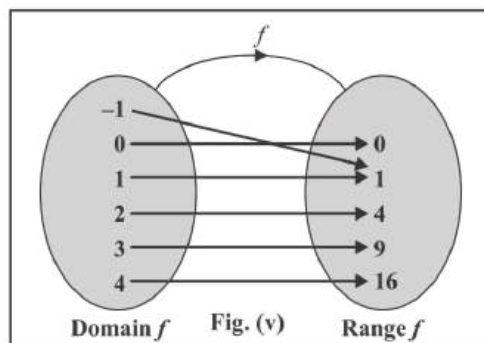
$$\text{Dom } f = \{-1, 0, 1, 2, 3, 4, \dots\} \text{ and}$$

$$\text{Range } f = \{0, 1, 4, 9, 16, \dots\}$$

This function may be written as:

$$f = \{(x, y) \mid y = x^2, x \in N\}$$

The mapping diagram for the function is shown in the Fig.(v).



3.3.4 Linear and Quadratic Functions

The function $\{(x, y) \mid y = mx + c\}$ is called a linear function because its graph (geometric representation) is a straight line. We know that an equation of the form $y = mx + c$ represents a straight line. The function $\{(x, y) \mid y = ax^2 + bx + c\}$ is called a quadratic function. We will study their geometric representation in Unit-10.

Example 10: If $f(x) = 2x - 1$ and $g(x) = x^2 - 3$, then find:

(i) $f(1)$ (ii) $f(-3)$ (iii) $f(7)$

(iv) $g(1)$ (v) $g(-3)$ (vi) $g(4)$

Solution: (i) $f(1) = 2 \times 1 - 1 = 1$ (ii) $f(-3) = 2 \times (-3) - 1 = -7$
 (iii) $f(7) = 2 \times 7 - 1 = 13$ (iv) $g(1) = (1)^2 - 3 = -3$
 (v) $g(-3) = (-3)^2 - 3 = 6$ (vi) $g(4) = (4)^2 - 3 = 13$

Example 11: Consider $f(x) = ax + b + 3$, where a and b are constant numbers. If $f(1) = 4$ and $f(5) = 9$, then find the value of a and b .

Solution: Given function $f(x) = ax + b + 3$

If $f(1) = 4$

Then $a \times 1 + b + 3 = 4$

$\Rightarrow a + b = 1$... (i)

Similarly, $f(5) = 9$

$\Rightarrow a \times 5 + b + 3 = 9$

$\Rightarrow 5a + b = 6$... (ii)

Subtract equation (i) from equation (ii), we get.

$$(5a + b) - (a + b) = 6 - 1$$

$$5a + b - a - b = 5$$

$$4a = 5 \Rightarrow a = \frac{5}{4}$$

Substitute $a = \frac{5}{4}$ in the equation (i)

$$\frac{5}{4} + b = 1$$

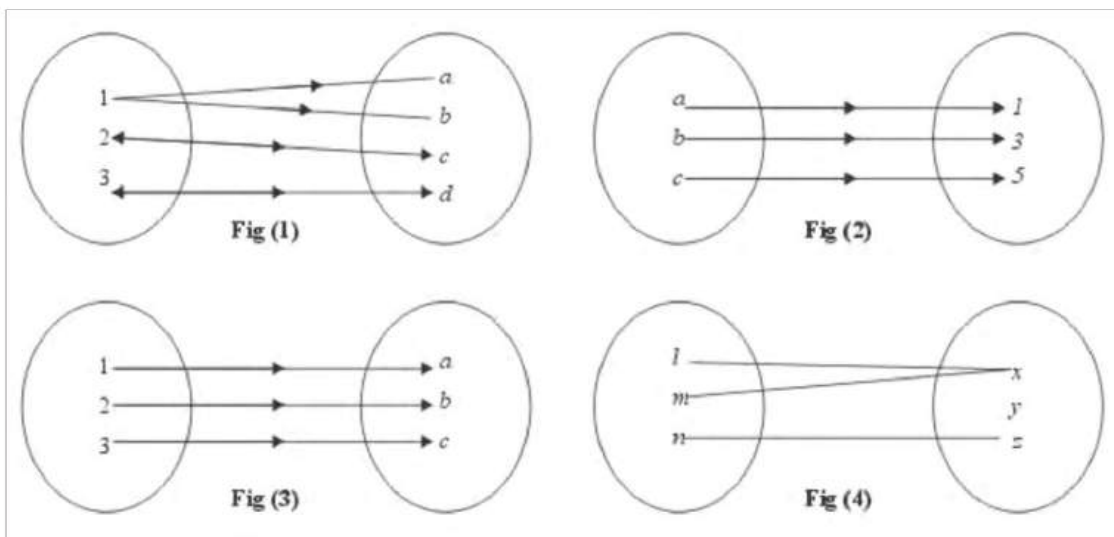
$$b = 1 - \frac{5}{4}$$

$$\Rightarrow b = -\frac{1}{4}$$

$$\text{Thus, } a = \frac{5}{4} \text{ and } b = -\frac{1}{4}$$

EXERCISE 3.3

- For $A = \{1, 2, 3, 4\}$, find the following relations in A . State the domain and range of each relation.
 - $\{(x, y) \mid y = x\}$
 - $\{(x, y) \mid y + x = 5\}$
 - $\{(x, y) \mid x + y < 5\}$
 - $\{(x, y) \mid x + y > 5\}$
- Which of the following diagrams represent functions and of which type?



- If $g(x) = 3x + 2$ and $h(x) = x^2 + 1$, then find:
 - $g(0)$
 - $g(-3)$
 - $g\left(\frac{2}{3}\right)$
 - $h(1)$
 - $h(-4)$
 - $h\left(-\frac{1}{2}\right)$
- Given that $f(x) = ax + b + 1$, where a and b are constant numbers. If $f(3) = 8$ and $f(6) = 14$, then find the values of a and b .
- Given that $g(x) = ax + b + 5$, where a and b are constant numbers. If $g(-1) = 0$ and $g(2) = 10$, find the values of a and b .
- Consider the function defined by $f(x) = 5x + 2$. If $f(x) = 32$, find the x value.
- Consider the function $f(x) = cx^2 + d$, where c and d are constant numbers. If $f(1) = 6$ and $f(-2) = 10$, then find the values of c and d .

REVIEW EXERCISE 3

1. Four options are given against each statement. Encircle the correct option.

(i) The set builder form of the set $\left\{1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots\right\}$ is:

(a) $\left\{x \mid x = \frac{1}{n}, n \in W\right\}$

(b) $\left\{x \mid x = \frac{1}{2n+1}, n \in W\right\}$

(c) $\left\{x \mid x = \frac{1}{n+1}, n \in W\right\}$

(d) $\{x \mid x = 2n+1, n \in W\}$

(ii) If $A = \{\}$, then $P(A)$ is:

(a) $\{\}$

(b) $\{1\}$

(c) $\{\{\}\}$

(d) $\emptyset$

(iii) If $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $U - (A \cap B)$ is:

(a) $\{1, 2, 4, 5\}$

(b) $\{2, 3\}$

(c) $\{1, 3, 4, 5\}$

(d) $\{1, 2, 3\}$

(iv) If A and B are overlapping sets, then $n(A - B)$ is equal to

(a) $n(A)$

(b) $n(B)$

(c) $A \cap B$

(d) $n(A) - n(A \cap B)$

(v) If $A \subseteq B$ and $B - A \neq \emptyset$, then $n(B - A)$ is equal to

(a) 0

(b) $n(B)$

(c) $n(A)$

(d) $n(B) - n(A)$

(vi) If $n(A \cup B) = 50$, $n(A) = 30$ and $n(B) = 35$, then $n(A \cap B) =$:

(a) 23

(b) 15

(c) 9

(d) 40

(vii) If $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$, then cartesian product of A and B contains exactly _____ elements.

(a) 13

(b) 12

(c) 10

(d) 6

(viii) If $f(x) = x^2 - 3x + 2$, then the value of $f(a + 1)$ is equal to:

(a) $a + 1$

(b) $a^2 + 1$

(c) $a^2 + 2a + 1$

(d) $a^2 - a$

(ix) Given that $f(x) = 3x + 1$, if $f(x) = 28$, then the value of x is:

(a) 9

(b) 27

(c) 3

(d) 18

(x) Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$ two non-empty sets and $f: A \rightarrow B$ be a function defined as $f = \{(1, a), (2, b), (3, b)\}$, then which of the following statement is true?

(a) f is injective (b) f is surjective (c) f is bijective (d) f is into only

2. Write each of the following sets in tabular forms:

(i) $\{x \mid x = 2n, n \in N\}$

(ii) $\{x \mid x = 2m + 1, m \in N\}$

(iii) $\{x | x = 11n, n \in W \wedge n < 11\}$ (iv) $\{x | x \in E \wedge 4 < x < 6\}$

(v) $\{x | x \in O \wedge 5 < x < 7\}$ (vi) $\{x | x \in Q \wedge x^2 = 2\}$

(vii) $\{x | x \in Q \wedge x = -x\}$ (viii) $\{x | x \in R \wedge x \notin Q'\}$

3. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5\}$ and $C = \{1, 3, 5, 7, 9\}$.

List the members of each of the following sets:

(i) A' (ii) B' (iii) $A \cup B$ (iv) $A - B$

(v) $A \cap C$ (vi) $A' \cup C'$ (vii) $A' \cup C$ (viii) U'

4. Using the Venn diagrams, if necessary, find the single sets equal to the following:

(i) A' (ii) $A \cap U$ (iii) $A \cup U$

(iv) $A \cup \emptyset$ (v) $\emptyset \cap \emptyset$

5. Use Venn diagrams to verify the following:

(i) $A - B = A \cap B'$ (ii) $(A - B)' \cap B = B$

6. Verify the properties for the sets A , B and C given below:

(i) Associativity of Union (ii) Associativity of intersection.

(iii) Distributivity of Union over intersection.

(iv) Distributivity of intersection over union.

(a) $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7, 8\}$, $C = \{5, 6, 7, 9, 10\}$

(b) $A = \emptyset$, $B = \{0\}$, $C = \{0, 1, 2\}$

(c) $A = N$, $B = Z$, $C = Q$

7. Verify De Morgan's Laws for the following sets:

$U = \{1, 2, 3, \dots, 20\}$, $A = \{2, 4, 6, \dots, 20\}$ and $B = \{1, 3, 5, \dots, 19\}$.

8. Consider the set $P = \{x | x = 5m, m \in N\}$ and $Q = \{x | x = 2m, m \in N\}$. Find $P \cap Q$

9. From suitable properties of union and intersection, deduce the following results:

(i) $A \cap (A \cup B) = A \cup (A \cap B)$ (ii) $A \cup (A \cap B) = A \cap (A \cup B)$

10. If $g(x) = 7x - 2$ and $s(x) = 8x^2 - 3$ find:

(i) $g(0)$ (ii) $g(-1)$ (iii) $g\left(-\frac{5}{3}\right)$ (iv) $s(1)$ (v) $s(-9)$ (vi) $s\left(\frac{7}{2}\right)$

11. Given that $f(x) = ax + b$, where a and b are constant numbers. If $f(-2) = 3$ and $f(4) = 10$, then find the values of a and b .

12. Consider the function defined by $k(x) = 7x - 5$. If $k(x) = 100$, find the value of x .

13. Consider the function $g(x) = mx^2 + n$, where m and n are constant numbers. If

$g(4) = 20$ and $g(0) = 5$, find the values of m and n .

14. A shopping mall has 100 products from various categories labeled 1 to 100, representing the universal set U . The products are categorized as follows:
- Set A : Electronics, consisting of 30 products labeled from 1 to 30.
 - Set B : Clothing comprises 25 products labeled from 31 to 55.
 - Set C : Beauty Products, comprising 25 products labeled from 76 to 100.
- Write each set in tabular form, and find the union of all three sets.
15. Out of the 180 students who appeared in the annual examination, 120 passed the math test, 90 passed the science test, and 60 passed both the math and science tests.
- How many passed either the math or science test?
 - How many did not pass either of the two tests?
 - How many passed the science test but not the math test?
 - How many failed the science test?
16. In a software house of a city with 300 software developers, a survey was conducted to determine which programming languages are liked more. The survey revealed the following statistics:
- 150 developers like Python.
 - 130 developers like Java.
 - 120 developers like PHP.
 - 70 developers like both Python and Java.
 - 60 developers like both Python and PHP.
 - 50 developers like both Java and PHP.
 - 40 developers like all three languages: Python, Java and PHP.
- How many developers use at least one of these languages?
 - How many developers use only one of these languages?
 - How many developers do not use any of these languages?
 - How many developers use only PHP?

Unit 4

Factorization and Algebraic Manipulation

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Identify common factors, trinomial factoring, concretely, pictorially and symbolically.
- Factorize quadratic and cubic algebraic expressions:

▪ $a^4 + a^2b^2 + b^4$ or $a^4 + b^4$	▪ $x^2 + px + q$
▪ $ax^2 + bx + c$	▪ $(ax^2 + bx + c)(ax^2 + bx + d) + k$
▪ $(x + a)(x + b)(x + c)(x + d) + k$	▪ $(x + a)(x + b)(x + c)(x + d) + kx^2$
▪ $a^3 + 3a^2b + 3ab^2 + b^3$	▪ $a^3 - 3a^2b + 3ab^2 - b^3$
▪ $a^3 \pm b^3$	
- Find highest common factor and least common multiple of algebraic expressions and know relationship of LCM and HCF.
- Find square root of algebraic expression by factorization and division.
- Apply the concepts of factorization of quadratic and cubic algebraic expression to real-world problems (such as engineering, physics, and finance.)

INTRODUCTION

Algebraic factorization is not just a mathematical technique limited to the classroom, it plays an important role in solving practical problems across various real-world scenarios. By breaking down complex algebraic expressions into simpler factors, we can make calculations more manageable and conceal important insights. Algebraic factorization has practical applications in finance, engineering science, business and daily life. This unit will explore the techniques of algebraic factorization and demonstrate how these methods can be applied to real-world situations, making math a valuable asset in various aspects of life.

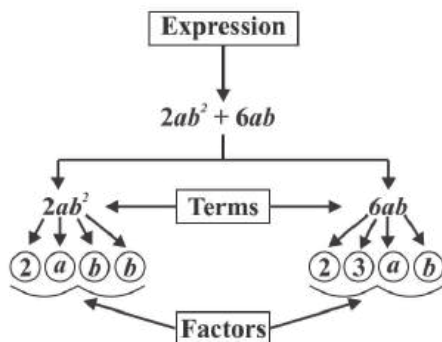
4.1 Identifying Common Factors and Trinomials Concretely, Pictorially and Symbolically

4.1.1 Common Factors

In algebra, a common factor is an expression that divides two or more expressions exactly. For example,

$$2x - 6 = 2(x - 3)$$

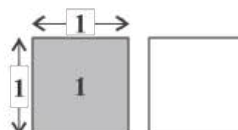
Here 2 is the common factor which exactly divides both terms $2x$ and 6 .



To represent trinomials concretely, we arrange unit tiles, rectangular tiles and the squared tiles into a rectangle. The factors of the trinomial are represented by the lengths of the sides of the rectangle.

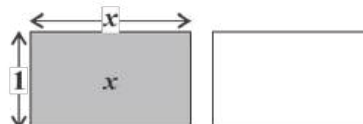
Unit Tiles

Here one grey unit tile represents 1 and one white unit tile represents -1 . Both grey and white unit tiles form a zero pair.



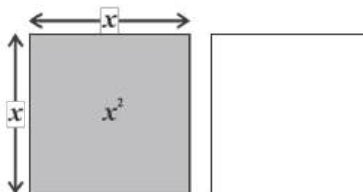
Rectangular Tiles

The grey rectangular tile represents x and the white rectangular tile represents $-x$. Both grey and white rectangular tiles also form a zero pair.



Squared Tiles

The grey squared tile measure x units on each side and it has an area of $x \times x = x^2$ units. This tile is labelled as x^2 tile. The white squared tile represents $-x^2$. Both grey and white squared tiles form a zero pair.



Example 1: Find common factor of $x^2 + 2x$ concretely, pictorially and symbolically

Solution: We arrange one x^2 tile and two x tiles into a rectangle.

Concretely	Pictorially	Symbolically
		$x^2 + 2x = x(x + 2)$

4.1.2 Trinomial Factoring

Trinomial factoring is converting trinomial expression as a product of two binomial expressions. A trinomial is an expression with three terms and binomial is an expression with two terms.

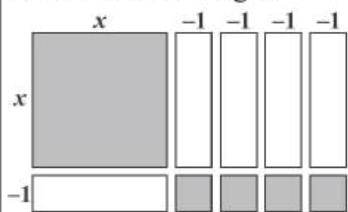
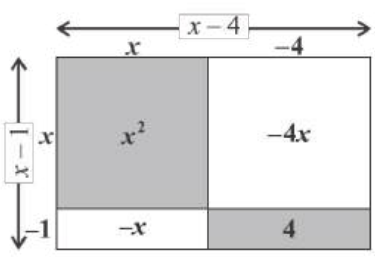
For example, $x^2 + 4x + 4$ and $3x^2 - x - 2$ are trinomials whereas $x + 2$ and $3x - 1$ are binomials.

Activity:

Divide students into groups. Divide some trinomial expressions and algebraic tiles of different sizes made from the cardboard among the students. Ask students to factorize the given expressions concretely.

Example 2: Factorize $x^2 - 5x + 4$ concretely, pictorially and symbolically.

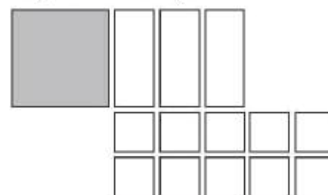
Solution:

Concretely	Pictorially	Symbolically
We arrange one x^2 tile, five $-x$ tiles and four unit tiles into a rectangle. 		$x^2 - 5x + 4$ $= (x - 1)(x - 4)$

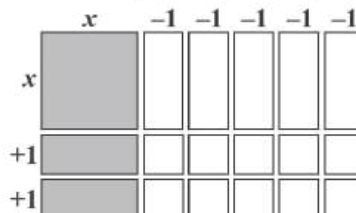
Example 3: Factorize $x^2 - 3x - 10$ concretely, pictorially and symbolically.

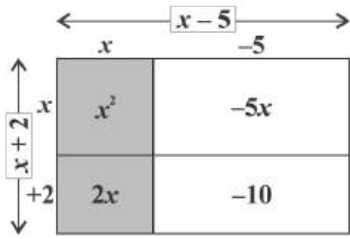
Solution:

Concretely we arrange one x^2 tile, three $-x$ tiles and ten -1 tiles into rectangle.



We see that there are not enough rectangular tiles to make a larger rectangle. To fix this issue, we add zero pair. Adding two x tiles and two $-x$ tiles does not change the given expression because $2x - 2x = 0$.



Pictorially	Symbolically
	$x^2 - 3x - 10 = (x + 2)(x - 5)$

4.1.3 Factorizing Quadratic and Cubic Algebraic Expressions

Type – I: Factorization of expression of the types $x^2 + px + q$ and $ax^2 + bx + c$

The procedure is explained in the following examples to factorize the above type of expressions:

Example 4: Factorize: $x^2 + 9x + 14$ **Solution:** Two numbers whose product is +14 and their sum is 9 are +2, +7.

$$\begin{aligned}
 \text{So, } & x^2 + 9x + 14 \\
 & \quad \quad \quad \uparrow \quad \quad \uparrow \\
 & = x^2 + 2x + 7x + 14 \\
 & = x(x + 2) + 7(x + 2) \\
 & = (x + 2)(x + 7)
 \end{aligned}$$

Product of factors	Sum of factors
$14 \times 1 = 14$	$14 + 1 = 15$
$7 \times 2 = 14$	$7 + 2 = 9$

Example 5: Factorize: $x^2 - 11x + 24$ **Solution:** Two numbers whose product is +24 and their sum is -11 are -8, -3.

$$\begin{aligned}
 \text{So, } & x^2 - 11x + 24 \\
 & = x^2 - 8x - 3x + 24 \\
 & = x(x - 8) - 3(x - 8) \\
 & = (x - 8)(x - 3)
 \end{aligned}$$

Product of factors	Sum of factors
$24 \times 1 = 24$	$24 + 1 = 25$
$8 \times 3 = 24$	$8 + 3 = 11$
$(-8) \times (-3) = 24$	$-8 - 3 = -11$
$6 \times 4 = 24$	$6 + 4 = 10$
$12 \times 2 = 24$	$12 + 2 = 14$

Example 6: Factorize: $p^2 + 11p + 18$ **Solution:** $p^2 + 11p + 18$

$$\begin{aligned}
 & = p^2 + 9p + 2p + 18 \\
 & = p(p + 9) + 2(p + 9) \\
 & = (p + 9)(p + 2)
 \end{aligned}$$

$$\therefore 9 + 2 = 11, 9 \times 2 = 18$$

In all quadratic trinomials factorized so far, the coefficient of x^2 was 1. We will now consider cases where the coefficient of x^2 is not 1.

Example 7: Factorize: $2x^2 + 17x + 26$ **Solution:****Step – I:** Multiply the coefficient of x^2 with constant term. i.e.,

$$2 \times 26 = 52$$

Step – II: List all the factors of 52:

$$\begin{array}{ll}
 1, 52 & -1, -52 \\
 2, 26 & -2, -26 \\
 4, 13 & -4, -13
 \end{array}$$

Remember!

An expression having degree 2 is called a quadratic expression.

Step – III: Sum of factors equals middle term (17)

$1 + 52 = 53$

$-1 - 52 = -53$

$2 + 26 = 28$

$-2 - 26 = -28$

$4 + 13 = 17$

$-4 - 13 = -17$

Try Yourself!

Factorize the following expressions:

(i) $x^2 + 7x - 18$

(ii) $t^2 - 5t - 24$

(iii) $6y^2 - y - 12$

Step – IV: Change the middle term in the given expression

$2x^2 + 17x + 26$

$= 2x^2 + 4x + 13x + 26$

Step – V: Take common from first two terms and last two terms

$= 2x(x + 2) + 13(x + 2)$

Step – VI: Again, take common from both terms

$= (x + 2)(2x + 13)$

Example 8: Factorize: $3x^2 - 4x - 4$ **Solution:** $3x^2 - 4x - 4$

$= 3x^2 + 2x - 6x - 4$

$\because 2 \times (-6) = -12, +2 - 6 = -4$

$= x(3x + 2) - 2(3x + 2)$

$= (3x + 2)(x - 2)$

EXERCISE 4.1

1. Factorize by identifying common factors.

(i) $6x + 12$

(ii) $15y^2 + 20y$

(iii) $-12x^2 - 3x$

(iv) $4a^2b + 8ab^2$

(v) $xy - 3x^2 + 2x$

(vi) $3a^2b - 9ab^2 + 15ab$

2. Factorize:

(i) $5x + 15$

(ii) $x^2 + 4x + 3$

(iii) $x^2 + 6x + 8$

(iv) $x^2 + 4x + 4$

3. Factorize:

(i) $x^2 + x - 12$

(ii) $x^2 + 7x + 10$

(iii) $x^2 - 6x + 8$

(iv) $x^2 - x - 56$

(v) $x^2 - 10x - 24$

(vi) $y^2 + 4y - 12$

(vii) $y^2 + 13y + 36$

(viii) $x^2 - x - 2$

4. Factorize:

- (i) $2x^2 + 7x + 3$ (ii) $2x^2 + 11x + 15$ (iii) $4x^2 + 13x + 3$
 (iv) $3x^2 + 5x + 2$ (v) $3y^2 - 11y + 6$ (vi) $2y^2 - 5y + 2$
 (vii) $4z^2 - 11z + 6$ (viii) $6 + 7x - 3x^2$

Type – II: Factorization of the expression of the types $a^4 + a^2b^2 + b^4$ or $a^4 + b^4$

Let's factorize the first expression

$$\begin{aligned}
 & a^4 + a^2b^2 + b^4 \\
 &= a^4 + b^4 + a^2b^2 \\
 &= (a^2)^2 + (b^2)^2 + a^2b^2 \\
 &= (a^2)^2 + (b^2)^2 + 2a^2b^2 - 2a^2b^2 + a^2b^2 \quad (\text{Adding and subtracting } 2a^2b^2) \\
 &= (a^2 + b^2)^2 - a^2b^2 \\
 &= (a^2 + b^2)^2 - (ab)^2 \\
 &= (a^2 + b^2 - ab)(a^2 + b^2 + ab) \\
 &= (a^2 - ab + b^2)(a^2 + ab + b^2)
 \end{aligned}$$

Remember!

$$\begin{aligned}
 a^2 - b^2 &= (a - b)(a + b) \\
 (a + b)^2 &= a^2 + 2ab + b^2 \\
 (a - b)^2 &= a^2 - 2ab + b^2
 \end{aligned}$$

Example 9: Factorize: $x^4 + x^2 + 25$

Solution:

$$\begin{aligned}
 & x^4 + x^2 + 25 \\
 &= x^4 + 25 + x^2 \\
 &= (x^2)^2 + (5)^2 + 2(x^2)(5) - 2(x^2)(5) + x^2 \quad (\text{Adding and subtracting } 2(x^2)(5)) \\
 &= (x^2 + 5)^2 - 10x^2 + x^2 \\
 &= (x^2 + 5)^2 - 9x^2 \\
 &= (x^2 + 5)^2 - (3x)^2 \\
 &= (x^2 + 5 - 3x)(x^2 + 5 + 3x) \\
 &= (x^2 - 3x + 5)(x^2 + 3x + 5)
 \end{aligned}$$

Activity

- Prepare cards by writing several expressions.
- Divide students in small groups.
- Each group will draw a card and factorize the expression.
- The group which completes the most correct factorizations in a set time will win.

Example 10: Factorize: $x^4 + y^4$

Solution:

$$\begin{aligned}
 & x^4 + y^4 \\
 &= (x^2)^2 + (y^2)^2 \\
 &= (x^2)^2 + (y^2)^2 + 2(x^2)(y^2) - 2(x^2)(y^2) \quad (\text{Adding and subtracting } 2x^2y^2) \\
 &= (x^2 + y^2)^2 - (\sqrt{2}xy)^2 \\
 &= (x^2 + y^2 - \sqrt{2}xy)(x^2 + y^2 + \sqrt{2}xy) \\
 &= (x^2 - \sqrt{2}xy + y^2)(x^2 + \sqrt{2}xy + y^2)
 \end{aligned}$$

Try Yourself!

- Factorize: (i) $64x^4y^4 + z^4$
 (ii) $81x^4 + \frac{1}{81x^4} - 11$

Example 11: Factorize: $a^4 + 64$

Solution:

$$\begin{aligned}
 & a^4 + 64 \\
 &= (a^2)^2 + (8)^2 \\
 &= (a^2)^2 + (8)^2 + 2(a^2)(8) - 2(a^2)(8) \quad (\text{Adding and subtracting } 2(a^2)(8)) \\
 &= (a^2 + 8)^2 - 16a^2 \\
 &= (a^2 + 8)^2 - (4a)^2 \\
 &= (a^2 + 8 - 4a)(a^2 + 8 + 4a) \\
 &= (a^2 - 4a + 8)(a^2 + 4a + 8)
 \end{aligned}$$

Type – III: Factorization of the expression of the types

- $(ax^2 + bx + c)(ax^2 + bx + d) + k$
- $(x + a)(x + b)(x + c)(x + d) + k$
- $(x + a)(x + b)(x + c)(x + d) + kx^2$

For explanation consider the following examples:

Example 12: Factorize: $(x^2 + 5x + 4)(x^2 + 5x + 6) - 3$

Solution:

$$\begin{aligned}
 & (x^2 + 5x + 4)(x^2 + 5x + 6) - 3 \\
 &= (y + 4)(y + 6) - 3 \quad (\text{Let } y = x^2 + 5x) \\
 &= y^2 + 6y + 4y + 24 - 3 \\
 &= y^2 + 10y + 21 \\
 &= y^2 + 7y + 3y + 21 \\
 &= y(y + 7) + 3(y + 7) \\
 &= (y + 7)(y + 3) \\
 &= (x^2 + 5x + 7)(x^2 + 5x + 3) \quad (\because y = x^2 + 5x)
 \end{aligned}$$

Example 13: Factorize: $(x + 2)(x + 3)(x + 4)(x + 5) - 15$

Solution: $(x + 2)(x + 3)(x + 4)(x + 5) - 15$

Re-arrange the given expression because $2 + 5 = 3 + 4$

$$\begin{aligned}
 & [(x + 2)(x + 5)][(x + 3)(x + 4)] - 15 \\
 &= (x^2 + 5x + 2x + 10)(x^2 + 4x + 3x + 12) - 15 \\
 &= (x^2 + 7x + 10)(x^2 + 7x + 12) - 15 \\
 &= (y + 10)(y + 12) - 15 \quad (\text{Let } y = x^2 + 7x)
 \end{aligned}$$

$$\begin{aligned}
 &= y^2 + 12y + 10y + 120 - 15 \\
 &= y^2 + 22y + 105 \\
 &= y^2 + 15y + 7y + 105 \\
 &= y(y + 15) + 7(y + 15) \\
 &= (y + 15)(y + 7) \\
 &= (x^2 + 7x + 15)(x^2 + 7x + 7) \quad (\because y = x^2 + 7x)
 \end{aligned}$$

Example 14: Factorize: $(x - 2)(x + 2)(x + 1)(x - 4) + 2x^2$

Solution: $(x - 2)(x + 2)(x + 1)(x - 4) + 2x^2$

$$\begin{aligned}
 &= [(x - 2)(x + 2)][(x + 1)(x - 4)] + 2x^2 \quad [\because (-2) \times 2 = 1 \times (-4)] \\
 &= (x^2 - 2^2)(x^2 - 4x + x - 4) + 2x^2 \\
 &= (x^2 - 4)(x^2 - 3x - 4) + 2x^2 \\
 &= y(y - 3x) + 2x^2 \quad (\text{Let } y = x^2 - 4) \\
 &= y^2 - 3xy + 2x^2 \\
 &= y^2 - 2xy - xy + 2x^2 \\
 &= y(y - 2x) - x(y - 2x) \\
 &= (y - 2x)(y - x) \\
 &= (x^2 - 4 - 2x)(x^2 - 4 - x) \quad (\because y = x^2 - 4) \\
 &= (x^2 - 2x - 4)(x^2 - x - 4)
 \end{aligned}$$

Type – IV: Factorization of the expression of the types

- $a^3 + 3a^2b + 3ab^2 + b^3$
- $a^3 - 3a^2b + 3ab^2 - b^3$

Factorization of such types of expressions is explained in the following examples:

Example 15: Factorize: $8x^3 + 60x^2 + 150x + 125$

Solution: $8x^3 + 60x^2 + 150x + 125$

$$\begin{aligned}
 &= (2x)^3 + 3(2x)^2(5) + 3(2x)(5)^2 + (5)^3 \\
 &= (2x + 5)^3 \\
 &= (2x + 5)(2x + 5)(2x + 5)
 \end{aligned}$$

Remember!

$$\begin{aligned}
 (a + b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
 (a - b)^3 &= a^3 - 3a^2b + 3ab^2 - b^3
 \end{aligned}$$

Example 16: Factorize: $x^3 - 18x^2 + 108x - 216$

Solution: $x^3 - 18x^2 + 108x - 216$

$$\begin{aligned}
 &= (x)^3 - 3(x)^2(6) + 3(x)(6)^2 - (6)^3 \\
 &= (x - 6)^3 \\
 &= (x - 6)(x - 6)(x - 6)
 \end{aligned}$$

Type – V: Factorization of the expression of the types $a^3 \pm b^3$

The expression $a^3 + b^3$ is a sum of cubes and it can be factorized using the following identity:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

The expression $a^3 - b^3$ is a difference of cubes and it can be factorized using the following identity:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example 17: Factorize: $8x^3 + 27$

Solution:

$$\begin{aligned} & 8x^3 + 27 \\ &= (2x)^3 + (3)^3 \\ &= (2x + 3)[(2x)^2 - (2x)(3) + (3)^2] \\ &= (2x + 3)(4x^2 - 6x + 9) \end{aligned}$$

Example 18: Factorize: $x^3 - 27y^3$

Solution:

$$\begin{aligned} & x^3 - 27y^3 \\ &= (x)^3 - (3y)^3 \\ &= (x - 3y)[(x)^2 + (x)(3y) + (3y)^2] \\ &= (x - 3y)(x^2 + 3xy + 9y^2) \end{aligned}$$

Do you know?

$$\begin{aligned} (a + b)^2 &\neq a^2 + b^2 \\ (a - b)^2 &\neq a^2 - b^2 \\ (a + b)^3 &\neq a^3 + b^3 \\ (a - b)^3 &\neq a^3 - b^3 \end{aligned}$$

EXERCISE 4.2

1. Factorize each of the following expressions:

(i) $4x^4 + 81y^4$ (ii) $a^4 + 64b^4$ (iii) $x^4 + 4x^2 + 16$
 (iv) $x^4 - 14x^2 + 1$ (v) $x^4 - 30x^2y^2 + 9y^4$ (vi) $x^4 - 7x^2y^2 + y^4$

2. Factorize each of the following expressions:

(i) $(x + 1)(x + 2)(x + 3)(x + 4) + 1$ (ii) $(x + 2)(x - 7)(x - 4)(x - 1) + 17$
 (iii) $(2x^2 + 7x + 3)(2x^2 + 7x + 5) + 1$ (iv) $(3x^2 + 5x + 3)(3x^2 + 5x + 5) - 3$
 (v) $(x + 1)(x + 2)(x + 3)(x + 6) - 3x^2$ (vi) $(x + 1)(x - 1)(x + 2)(x - 2) - 16x^2$

3. Factorize:

(i) $8x^3 + 12x^2 + 6x + 1$ (ii) $27a^3 + 108a^2b + 144ab^2 + 64b^3$
 (iii) $x^3 + 18x^2y + 108xy^2 + 216y^3$ (iv) $8x^3 - 125y^3 + 150xy^2 - 60x^2y$

4. Factorize:

(i) $125a^3 - 1$ (ii) $64x^3 + 125$ (iii) $x^6 - 27$
 (iv) $1000a^3 + 1$ (v) $343x^3 + 216$ (vi) $27 - 512y^3$

4.3 Highest Common Factor (HCF) and Least Common Multiple (LCM) of Algebraic Expressions

4.3.1 Highest Common Factor (HCF)

The HCF of two or more algebraic expressions refers to the greatest algebraic expression which divides them without leaving a remainder.

We can find HCF of given expressions by the following two methods:

- (a) By factorization (b) By division

(a) HCF by Factorization Method

Example 19: Find the HCF of $6x^2y$, $9xy^2$

Solution: $6x^2y = 2 \times 3 \times x \times x \times y$

$$9xy^2 = 3 \times 3 \times x \times y \times y$$

$$\therefore \text{HCF} = 3 \times x \times y \quad (\text{Product of common factors})$$

$$= 3xy$$

Example 20: Find the HCF by factorization method $x^2 - 27$, $x^2 + 6x - 27$, $x^2 - 9$

Solution: $x^3 - 27 = x^3 - 3^3$

$$= (x - 3)[(x)^2 + (3)(x) + (3)^2]$$

$$= (x - 3)(x^2 + 3x + 9)$$

$$x^2 + 6x - 27 = x^2 + 9x - 3x - 27$$

$$= x(x + 9) - 3(x + 9)$$

$$= (x + 9)(x - 3)$$

$$x^2 - 9 = x^2 - 3^2$$

$$= (x - 3)(x + 3)$$

Hence, HCF = $x - 3$

(b) HCF by Division Method

Example 21: Find HCF of $6x^3 - 17x^2 - 5x + 6$ and $6x^3 - 5x^2 - 3x + 2$ by using division method.

Solution:

$$6x^3 - 17x^2 - 5x + 6 \overline{) 6x^3 - 5x^2 - 3x + 2}$$

$$\underline{-6x^3 + 17x^2 + 5x + 6}$$

$$12x^2 + 2x - 4$$

$$\text{Here, } 12x^2 + 2x - 4 = 2(6x^2 + x - 2)$$

2 is not common in both the given polynomials, so we ignore it and consider only $6x^2 + x - 2$.

$$\begin{array}{r}
 \overline{) \begin{array}{r} 6x^3 - 17x^2 - 5x + 6 \\ -6x^3 + x^2 + 2x \\ \hline -18x^2 - 3x + 6 \\ +18x^2 + 3x + 6 \\ \hline 0 \end{array}} \\
 \overline{) \begin{array}{r} 6x^3 - 17x^2 - 5x + 6 \\ -6x^3 + x^2 + 2x \\ \hline -18x^2 - 3x + 6 \\ +18x^2 + 3x + 6 \\ \hline 0 \end{array}}
 \end{array}$$

Hence, HCF = $6x^2 + x - 2$

4.3.2 Least Common Multiple (LCM)

The LCM of two or more algebraic expressions is the smallest expression that is divisible by each of the given expressions.

To find the LCM by factorization, we use the formula.

$$\text{LCM} = \text{Common factors} \times \text{Non-common factors}$$

Example 22: Find the LCM of $4x^2y$, $8x^3y^2$.

Solution:

$$4x^2y = 2 \times 2 \times x \times x \times y$$

$$8x^3y^2 = 2 \times 2 \times 2 \times x \times x \times x \times y \times y$$

$$\text{Common factors} = 2 \times 2 \times x \times x \times y = 4x^2y$$

$$\text{Non-common factors} = 2 \times x \times y = 2xy$$

$$\text{LCM} = \text{Common factors} \times \text{Non-common factors}$$

$$= 4x^2y \times 2xy = 8x^3y^2$$

Example 23: Find the LCM of the polynomials $x^2 - 3x + 2$, $x^2 - 1$ and $x^2 - 5x + 4$.

Solution:

$$\text{As } x^2 - 3x + 2 = x^2 - 2x - x + 2$$

$$= x(x - 2) - 1(x - 2)$$

$$= (x - 2)(x - 1)$$

$$\text{And } x^2 - 1 = (x - 1)(x + 1)$$

$$x^2 - 5x + 4 = x^2 - 4x - x + 4$$

$$= x(x - 4) - 1(x - 4)$$

$$= (x - 4)(x - 1)$$

$$\text{Common factors} = x - 1$$

$$\text{Non-common factors} = (x + 1)(x - 2)(x - 4)$$

LCM = Common factors $\times$ Non-common factors

$$= (x - 1) \times (x + 1)(x - 2)(x - 4)$$

$$= (x - 1)(x + 1)(x - 2)(x - 4)$$

4.3.3 Relationship Between LCM and HCF

The relationship between LCM and HCF can be expressed as follows:

$$\text{LCM} \times \text{HCF} = p(x) \times q(x)$$

Where, $p(x) = 1^{\text{st}}$ polynomial

$q(x) = 2^{\text{nd}}$ polynomial

Example 24: LCM and HCF of two polynomials are $x^3 - 10x^2 + 11x + 70$ and $x - 7$. If one of the polynomials is $x^2 - 12x + 35$, find the other polynomial.

Solution: Given that: $\text{LCM} = x^3 - 10x^2 + 11x + 70$

$$\text{HCF} = x - 7$$

$$p(x) = x^2 - 12x + 35$$

As we know that: $q(x) = \frac{\text{LCM} \times \text{HCF}}{p(x)}$

$$= \frac{(x^3 - 10x^2 + 11x + 70)(x - 7)}{x^2 - 12x + 35}$$

$$\begin{array}{r}
 \overline{) x^3 - 10x^2 + 11x + 70} \\
 \underline{-x^3 + 12x^2 - 35x} \\
 2x^2 - 24x + 70 \\
 \underline{-2x^2 + 24x - 70} \\
 0
 \end{array}$$

$$\begin{aligned}
 \text{So, } q(x) &= (x + 2)(x - 7) \\
 &= x^2 - 7x + 2x - 14 \\
 &= x^2 - 5x - 14
 \end{aligned}$$

Example 25: The LCM of $x^2y + xy^2$ and $x^2 + xy$ is $xy(x + y)$. Find the HCF.

Solution: Given that: $\text{LCM} = xy(x + y)$

$$\text{HCF} = ?$$

$$1^{\text{st}} \text{ polynomial} = x^2y + xy^2$$

$$2^{\text{nd}} \text{ polynomial} = x^2 + xy$$

As we know that: $\text{LCM} \times \text{HCF} = \text{Product of two polynomials}$

$$\begin{aligned}\text{HCF} &= \frac{\text{Product of two polynomials}}{\text{LCM}} \\ &= \frac{(x^2y + xy^2)(x^2 + xy)}{xy(x + y)} \\ &= \frac{xy(x + y)x(x + y)}{xy(x + y)} \\ &= x(x + y)\end{aligned}$$

EXERCISE 4.3

- Find HCF by factorization method.
 - $21x^2y, 35xy^2$
 - $4x^2 - 9y^2, 2x^2 - 3xy$
 - $x^3 - 1, x^2 + x + 1$
 - $a^3 + 2a^2 - 3a, 2a^3 + 5a^2 - 3a$
 - $t^2 - 3t - 4, t^2 + 5t + 4, t^2 - 1$
 - $x^2 + 15x + 56, x^2 + 5x - 24, x^2 + 8x$
- Find HCF of the following expressions by using division method:
 - $27x^3 + 9x^2 - 3x - 10, 3x - 2$
 - $x^3 - 9x^2 + 23x - 15, x^2 - 4x + 3$
 - $2x^3 + 2x^2 + 2x + 2, 6x^3 + 12x^2 + 6x + 12$
 - $2x^3 - 4x^2 - 16x, x^3 - 4x, 3x^2 - 6x$
- Find LCM of the following expressions by using prime factorization method.
 - $2a^2b, 4ab^2, 6ab$
 - $x^2 + x, x^3 + x^2$
 - $a^2 - 4a + 4, a^2 - 2a$
 - $x^4 - 16, x^3 - 4x$
 - $16 - 4x^2, x^2 + x - 6, 4 - x^2$
- The HCF of two polynomials is $y - 7$ and their LCM is $y^3 - 10y^2 + 11y + 70$. If one of the polynomials is $y^2 - 5y - 14$, find the other.
- The LCM and HCF of two polynomial $p(x)$ and $q(x)$ are $36x^3(x + a)(x^3 - a^3)$ and $x^2(x - a)$ respectively. If $p(x) = 4x^2(x^2 - a^2)$, find $q(x)$.
- The HCF and LCM of two polynomials is $(x + a)$ and $12x^2(x + a)(x^2 - a^2)$ respectively. Find the product of the two polynomials.

4.4 Square Root of an Algebraic Expression

The square root of an algebraic expression refers to a value that, when multiplied by itself, gives the original expression. Just like finding the square root of a number, taking the square root of an algebraic expression involves determining what expression, when squared, results in the given expression.

For example, square root of $4a^2$ is $\pm 2a$ because $2a \times 2a = 4a^2$ and $(-2a) \times (-2a) = 4a^2$

There are following two methods for finding the square root of an algebraic expression:

(a) By factorization method

(b) By division method

(a) Square Root by Factorization Method

Example 26: Find the square root of the expression $36x^4 - 36x^2 + 9$

Solution:

$$\begin{aligned} & 36x^4 - 36x^2 + 9 \\ &= 9(4x^4 - 4x^2 + 1) \\ &= 9[(2x^2)^2 - 2(2x^2)(1) + (1)^2] \\ &= 3^2(2x^2 - 1)^2 \end{aligned}$$

Taking square root on both sides

$$\begin{aligned} \sqrt{36x^4 - 36x^2 + 9} &= \sqrt{3^2(2x^2 - 1)^2} \\ &= \sqrt{3^2} \cdot \sqrt{(2x^2 - 1)^2} \\ &= \pm 3(2x^2 - 1) \end{aligned}$$

(b) Square Root by Division Method

When the degree of the polynomial is higher, division method in finding the square root is very useful.

Example 27: Find the square root of the polynomial $x^4 - 12x^3 + 42x^2 - 36x + 9$.

Solution: Multiply x^2 by x^2 to get x^4

Multiply the quotient (x^2) by 2, so we get $2x^2$. By dividing $-12x^3$ by $2x^2$, we get $-6x$. By continuing in this way, we get the remainder.

Hence, square root of $x^4 - 12x^3 + 42x^2 - 36x + 9$ is $\pm (x^2 - 6x + 3)$

$$\begin{array}{r} x^2 - 6x + 3 \\ x^2 \overline{) x^4 - 12x^3 + 42x^2 - 36x + 9} \\ \underline{-x^4} \\ 2x^2 - 6x \\ \underline{-2x^2 + 6x} \\ 6x^2 - 36x + 9 \\ \underline{-6x^2 + 36x - 9} \\ 0 \end{array}$$

4.4.1 Real World Problems of Factorization

In this section, we will apply the concept of factorization of quadratic and cubic algebraic expressions to real world problems such as engineering, physics and finance.

Example 28: Cost function for producing a part is modeled by:

$$C(x) = 5x^2 - 25x + 30$$

Where x is the width of the component and $C(x)$ is the cost. Find the value of x where $C(x)$ is minimum.

Solution:

$$\begin{aligned}
 C(x) &= 5x^2 - 25x + 30 \\
 &= 5(x^2 - 5x + 6) \\
 &= 5(x^2 - 2x - 3x + 6) \\
 &= 5[x(x-2) - 3(x-2)] \\
 &= 5(x-2)(x-3)
 \end{aligned}$$

Thus, the minimum cost occurs when $x = 2$ or $x = 3$.

Example 29: The potential energy $U(x)$ of a particle moving in a cubic potential is expressed as:

$$U(x) = x^3 - 6x^2 + 12x - 8$$

Factorize the expression to find the points where the energy is minimized.

Solution:

$$\begin{aligned}
 U(x) &= x^3 - 6x^2 + 12x - 8 \\
 &= (x)^3 - 3(x)^2(2) + 3(x)(2)^2 - (2)^3 \\
 &= (x-2)^3 \\
 &= (x-2)(x-2)(x-2)
 \end{aligned}$$

The factorized form of the potential energy function shows that the energy is minimized at $x = 2$.

Example 30: A company's profit $P(x)$ is modeled by the quadratic equation:

$$P(x) = -5x^2 + 50x - 120$$

Where x represents the number of units produced and $P(x)$ represents the profit in dollars. Find how many units should be produced to maximize profit.

Solution:

$$\begin{aligned}
 P(x) &= -5x^2 + 50x - 120 \\
 &= -5(x^2 - 10x + 24) \\
 &= -5[x^2 - 4x - 6x + 24] \\
 &= -5[x(x-4) - 6(x-4)] \\
 &= -5(x-4)(x-6)
 \end{aligned}$$

We can see that profit will be 0 when $x = 4$ or $x = 6$. As coefficients of x^2 is negative, the maximum profit occurs at the midpoint between 4 and 6.

Which is:

$$x = \frac{4+6}{2} = \frac{10}{2} = 5$$

Thus, the company should produce 5 units to maximize profit.

EXERCISE 4.4

1. Find the square root of the following polynomials by factorization method:

(i) $x^2 - 8x + 16$

(ii) $9x^2 + 12x + 4$

(iii) $36a^2 + 84a + 49$

(iv) $64y^2 - 32y + 4$

(v) $200t^2 - 120t + 18$

(vi) $40x^2 + 120x + 90$

2. Find the square root of the following polynomials by division method:

(i) $4x^4 - 28x^3 + 37x^2 + 42x + 9$

(ii) $121x^4 - 198x^3 - 183x^2 + 216x + 144$

(iii) $x^4 - 10x^3y + 27x^2y^2 - 10xy^3 + y^4$

(iv) $4x^4 - 12x^3 + 37x^2 - 42x + 49$

3. An investor's return $R(x)$ in rupees after investing x thousand rupees is given by quadratic expression:

$$R(x) = -x^2 + 6x - 8$$

Factorize the expression and find the investment levels that result in zero return.

4. A company's profit $P(x)$ in rupees from selling x units of a product is modeled by the cubic expression:

$$P(x) = x^3 - 15x^2 + 75x - 125$$

Find the break-even point(s), where the profit is zero.

5. The potential energy $V(x)$ in an electric field varies as a cubic function of distance x , given by:

$$V(x) = 2x^3 - 6x^2 + 4x$$

Determine where the potential energy is zero.

6. In structural engineering, the deflection $Y(x)$ of a beam is given by:

$$Y(x) = 2x^2 - 8x + 6$$

This equation gives the vertical deflection at any point x along the beam. Find the points of zero deflection.

REVIEW EXERCISE 4

1. Four options are given against each statement. Encircle the correct option.

- i. The factorization of $12x + 36$ is:

(a) $12(x + 3)$ (b) $12(3x)$ (c) $12(3x + 1)$ (d) $x(12 + 36x)$

- ii. The factors of $4x^2 - 12x + 9$ are:

(a) $(2x + 3)^2$ (b) $(2x - 3)^2$
(c) $(2x - 3)(2x + 3)$ (d) $(2 + 3x)(2 - 3x)^2$

- iii. The HCF of a^3b^3 and ab^2 is:

(a) a^3b^3 (b) ab^2 (c) a^4b^5 (d) a^2b

- iv. The LCM of $16x^2$, $4x$ and $30xy$ is:
 (a) $480x^3y$ (b) $240xy$ (c) $240x^2y$ (d) $120x^4y$
- v. Product of LCM and HCF = _____ of two polynomials.
 (a) sum (b) difference (c) product (d) quotient
- vi. The square root of $x^2 - 6x + 9$ is:
 (a) $\pm(x-3)$ (b) $\pm(x+3)$ (c) $x-3$ (d) $x+3$
- vii. The LCM of $(a-b)^2$ and $(a-b)^4$ is:
 (a) $(a-b)^2$ (b) $(a-b)^3$ (c) $(a-b)^4$ (d) $(a-b)^6$
- viii. Factorization of $x^3 + 3x^2 + 3x + 1$ is:
 (a) $(x+1)^3$ (b) $(x-1)^3$
 (c) $(x+1)(x^2+x+1)$ (d) $(x-1)(x^2-x+1)$
- ix. Cubic polynomial has degree:
 (a) 1 (b) 2 (c) 3 (d) 4
- x. One of the factors of $x^3 - 27$ is:
 (a) $x-3$ (b) $x+3$ (c) x^2-3x+9 (d) Both a and c
2. Factorize the following expressions:
 (i) $4x^3 + 18x^2 - 12x$ (ii) $x^3 + 64y^3$
 (iii) $x^3y^3 - 8$ (iv) $-x^2 - 23x - 60$
 (v) $2x^2 + 7x + 3$ (vi) $x^4 + 64$
 (vii) $x^4 + 2x^2 + 9$ (viii) $(x+3)(x+4)(x+5)(x+6) - 360$
 (ix) $(x^2 + 6x + 3)(x^2 + 6x - 9) + 36$
3. Find LCM and HCF:
 (i) $4x^3 + 12x^2$, $8x^2 + 16x$ (ii) $x^3 + 3x^2 - 4x$, $x^2 - 4x + 3$
 (iii) $x^2 + 8x + 16$, $x^2 - 16$ (iv) $x^3 - 9x$, $x^2 - x - 6$
4. Find square root by factorization and division method of the expression $16x^4 + 8x^2 + 1$.
5. Huria is analyzing the total cost of her loan, modeled by the expression $C(x) = x^2 - 8x + 15$, where x represents the number of years. What is the optimal repayment period for Huria's loan?

Unit 5

Linear Equations and Inequalities

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Solve linear equations and inequalities with rational coefficients and represent the solution set on a real line.
- Solve two linear inequalities with two unknowns simultaneously.
- Interpret and identify regions in plane bounded by two linear inequalities in two unknowns.
- Find maximum and minimum values of a function using points in the feasible solution.

INTRODUCTION

Linear equations and inequalities are widely used in various fields to model and solve real-world problems. They help in understanding relationships between variables and making decisions. In this unit, our main goal will be to optimize (maximum or minimum) a quantity under consideration subject to certain constraint restrictions.

5.1 Linear Equation

An equation of the form $ax + b = 0$ where 'a' and 'b' are constants, $a \neq 0$ and 'x' is a variable, is called a linear equation in one variable. In linear equation, the highest power of the variable is always 1.

Remember!

$ax + b = 0$ and $a \neq 0$ is also called the general form of linear equation in one variable.

5.1.1 Solving a Linear Equation in One Variable

Solving a linear equation in one variable means finding the value of the variable that makes the equation true. To solve the equation, the goal is to isolate the variable on one side of the equation and determine its value.

Steps to Solve a Linear Equation in One Variable

Simplify Both Sides (if necessary)

- Combine like terms on each side of the equation.
- Simplify expressions, including distributing any multiplication over parentheses.

Isolate the Variable Term

- Move all terms containing the variable to one side of the equation and all

constant terms numbers to the other side. We can do this by adding or subtracting terms from both sides of the equation.

Solve for the Variable

- Once the variable term is isolated, solve for the variable by dividing or multiplying both sides of the equation by the co-efficient of the variable.

Check Your Solution

- Substitute the solution into the original equation to ensure that solution is correct.

Example 1: Solve the following equations and represent their solutions on real line:

(i) $3x - 5 = 7$

(ii) $\frac{x-2}{5} - \frac{x-4}{2} = 2$

Solution: (i) $3x - 5 = 7$

$$3x - 5 + 5 = 7 + 5$$

$$3x = 12$$

$$\frac{3x}{3} = \frac{12}{3}$$

$$x = 4$$

Remember!

A linear equation in one variable has only one solution.

Check: Substitute $x = 4$ into the original equation

$$3(4) - 5 = 7$$

$$12 - 5 = 7$$

$$7 = 7$$

So, $x = 4$ is a solution because it makes the original equation true.

Representation of the solution on a number line:



Fig. 5.1

(ii) $\frac{x-2}{5} - \frac{x-4}{2} = 2$

$$\frac{2(x-2) - 5(x-4)}{10} = 2$$

$$\frac{2x - 4 - 5x + 20}{10} = 2$$

$$\frac{-3x + 16}{10} = 2$$

Remember!

We check the solution after solving linear equation to ensure the accuracy of our work.

$$\begin{aligned}\frac{-3x+16}{10} \times 10 &= 2 \times 10 \\ -3x+16 &= 20 \\ -3x+16-16 &= 20-16 \\ -3x &= 4 \\ x &= -\frac{4}{3}\end{aligned}$$

Check: Substitute $x = -\frac{4}{3}$ into the original equation

$$\begin{aligned}\frac{-\frac{4}{3}-2}{5} - \frac{-\frac{4}{3}-4}{2} &= 2 \\ \Rightarrow \frac{-4-6}{5} - \frac{-4-12}{2} &= 2 \\ \Rightarrow \frac{-10}{5} - \frac{-16}{2} &= 2 \\ \Rightarrow -\frac{2}{3} + \frac{8}{3} &= 2 \\ \Rightarrow \frac{-2+8}{3} &= 2 \\ \Rightarrow \frac{6}{3} &= 2 \\ \Rightarrow 2 &= 2\end{aligned}$$

So, $x = -\frac{4}{3}$ is the solution of given equation.

Representation of the solution on a number line:

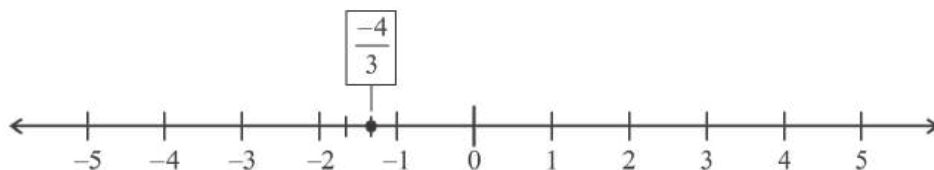


Fig. 5.2

5.2 Linear Inequalities

Inequalities are expressed by the following four symbols:

$>$ (greater than), $<$ (less than), $\geq$ (greater than or equal to), $\leq$ (less than or equal to)

For example,

(i) $ax < b$ (ii) $ax + b \geq c$ (iii) $ax + by > c$ (iv) $ax + by \leq c$

are inequalities. Inequalities (i) and (ii) are in one variable while inequalities (iii) and (iv) are in two variables. The following operations will not affect the order of inequality while changing it to simpler equivalent form:

- (i) Adding or subtracting a constant to each side of it.
- ii) Multiplying or dividing each side by a positive constant.

Do you know?

The order of an inequality is changed by multiplying or dividing each side by a negative constant.

Example 2: Find solution of $\frac{2}{3}x - 1 < 0$ and also represent it on a real line.

Solution:

$$\begin{aligned} \frac{2}{3}x - 1 &< 0 && \dots(i) \\ \Rightarrow \frac{2}{3}x &< 1 \\ \Rightarrow 2x &< 3 \\ \Rightarrow x &< \frac{3}{2} \end{aligned}$$

It means that all real numbers less than $\frac{3}{2}$ are in the solution of (i)

Thus, the interval $(-\infty, \frac{3}{2})$ or $-\infty < x < \frac{3}{2}$ is the solution of the given inequality which is shown in figure 5.3

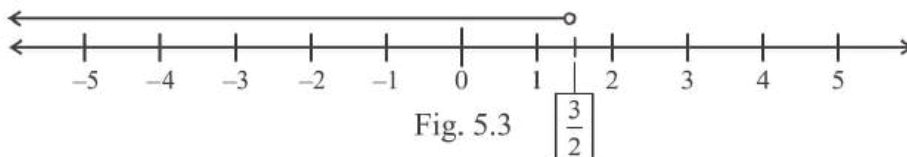
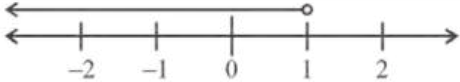
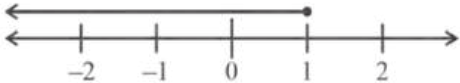


Fig. 5.3

We conclude that the solution of an inequality consists of all solutions of the inequality.

Following are the inequalities and their solutions on a real line:

Inequality	Solution	Representation on real line
$x > 1$	$(1, \infty)$ or $1 < x < \infty$	
$x < 1$	$(-\infty, 1)$ or $-\infty < x < 1$	
$x \geq 1$	$[1, \infty)$ or $1 \leq x < \infty$	
$x \leq 1$	$(-\infty, 1]$ or $-\infty < x \leq 1$	

5.2.1 Solution of a Linear Inequality in Two Variables

Generally, a linear inequality in two variables x and y can be one of the following forms:

$$ax + by < c; \quad ax + by > c; \quad ax + by \leq c; \quad ax + by \geq c$$

Where a, b, c are constants and a, b are not both zero.

We know that the graph of linear equation of the form $ax + by = c$ is a line which divides the plane into two disjoint regions as stated below:

- (i) The set of ordered pairs (x, y) such that $ax + by < c$
- (ii) The set of ordered pairs (x, y) such that $ax + by > c$

The regions (i) and (ii) are called **half planes** and the line $ax + by = c$ is called the boundary of each half plane.

Note that a **vertical line** divides the plane into **left and right half planes** while a **non-vertical line** divides the plane into **upper and lower half planes**.

Remember:

A solution of a linear inequality in x and y is an ordered pair of numbers which satisfies the inequality.

For example, the ordered pair $(1, 1)$ is a solution of the inequality $x + 2y < 6$ because $1 + 2(1) = 3 < 6$ which is true.

There are infinitely many ordered pairs that satisfy the inequality $x + 2y < 6$, so its graph will be a half plane.

Note that the linear equation $ax + by = c$ is called "**associated or corresponding equation**" of each of the above-mentioned inequalities.

Procedure for Graphing a linear Inequality in two Variables

- (i) The corresponding equation of the inequality is first graphed by using 'dashes' if the inequality involves the symbols $>$ or $<$ and a solid line is drawn if the inequality involves the symbols $\geq$ or $\leq$.
- (ii) A test point (not on the graph of the corresponding equation) is chosen which determines on which side of the boundary line the half plane line.

Example 3: Solve the inequality $x + 2y < 6$.

Solution: The associated equation of the inequality

$$x + 2y < 6 \quad \text{(i)}$$

$$\text{is } x + 2y = 6 \quad \text{(ii)}$$

The line (ii) intersects the x -axis and y -axis at $(6, 0)$ and $(0, 3)$ respectively. As no point of the line (ii) is a solution x of the inequality (i), so the graph of the line (ii) is shown by using dashes. We take $O(0, 0)$ as a test point because it is not on the line (ii).

Substituting $x = 0, y = 0$ in the expression $x + 2y$ gives $0 - 2(0) = 0 < 6$. So, the point $(0, 0)$ satisfies the inequality (i). Any other point below the line (ii) satisfies the inequality (i), that is all points in the half plane containing the point $(0, 0)$ satisfy the inequality (i).

Thus, the graph of the solution set of inequality (i) is a region on lies the origin-side of the line (ii), that is, the region below the line (ii). A portion of the open half plane below the line (ii) is shown as shaded region in figure 5.4(a)

Note:

All points above the dashed line satisfy the inequality $x + 2y > 6$ so on (iii).

A portion of the open half plane above the line (ii) is shown by shading in figure 5.4(b).

Do you know?

A test point is a point selected to determine which side of the boundary line represents the solution region for an inequality. Usually, we take origin $(0, 0)$ as a test point.

- If the inequality holds true with the test point, the region containing this point is part of the solution.
- If the inequality is false, the opposite region is the solution region.

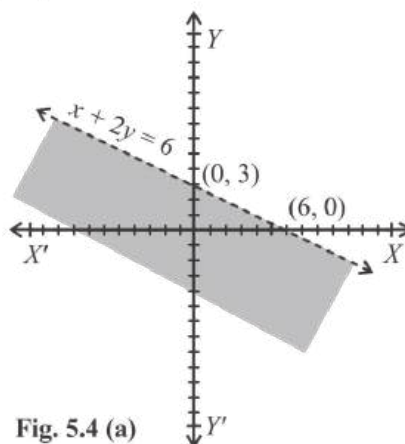


Fig. 5.4 (a)

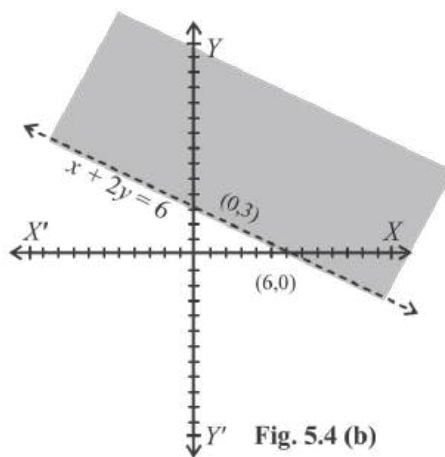


Fig. 5.4 (b)

Note: 1. The graph of the inequality $x + 2y \leq 6$... (iv) includes the graph of the line (ii). The open half-plane below the line (ii) including the graph of the line (ii) is the graph of the inequality (iv). A portion of the graph of the inequality (iv) is shown by shading in fig. 5.4 (c).

Note: 2 All points on the line (ii) and above the line (ii) satisfy the inequality $x + 2y \geq 6$... (v). This means that the solution set of the inequality (v) consists of all points above the line (ii) and all points on the lines (ii). The graph of the inequality (v) is partially shown as shaded region in fig. 5.4 (d).

Note: 3 The graphs of $x + 2y \leq 6$ and $x + 2y \geq 6$ are closed half planes.

Example 4: Solve the following linear inequalities in xy -plane:

(i) $2x \geq -3$

(ii) $y \leq 2$

Solution: (i) The inequality $2x \geq -3$ in xy -plane is considered as $2x + 0y \geq -3$ and its solution set consists of all point (x, y)

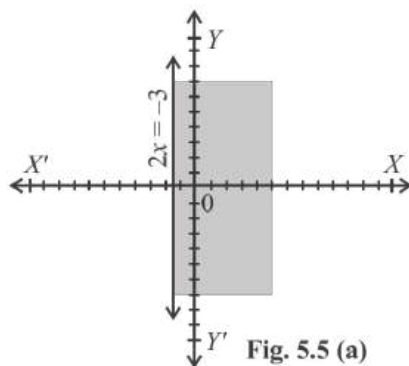
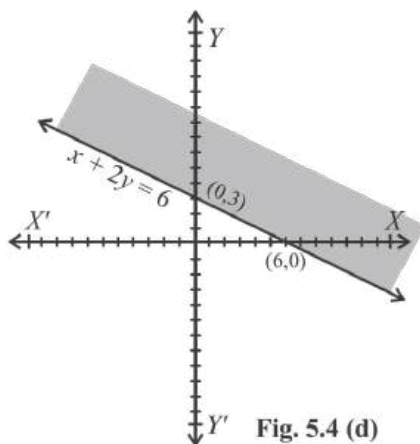
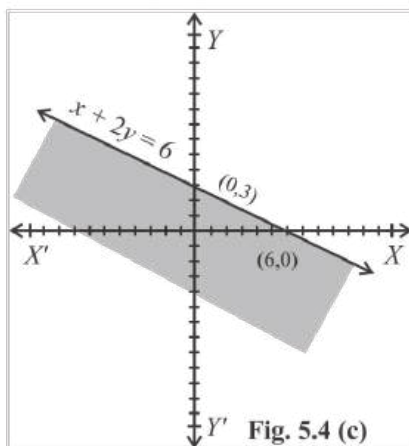
such that $x, y \in \mathbb{R}$ and $x \geq -\frac{3}{2}$

The corresponding equation of the given inequality is $2x = -3$... (a)

which is a vertical line (parallel to the y -axis) and its graph is drawn in figure 5.5(a).

Thus, the graph of $2x \geq -3$ consists of boundary line and the open half-plane to the right of the line (a).

(ii) The associated equation of the inequality $y \leq 2$ is $y = 2$



which is a horizontal line (parallel to the x -axis) and its graph is shown in figure 5.5 (b). Here the solution set of the inequality $y < 2$ is the open half plane below the boundary line $y = 2$. Thus, the graph of $y \leq 2$ consists of the boundary line and the closed half plane below it.

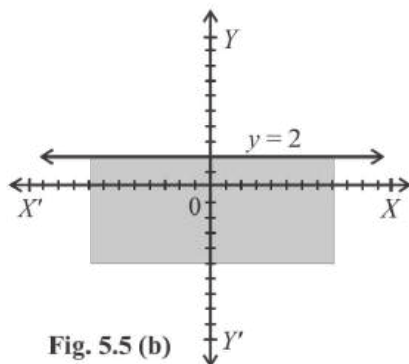


Fig. 5.5 (b)

5.2.2 Solution of Two Linear Inequalities in Two Variables

The graph of a system of linear inequalities consists of the set of all ordered pairs (x, y) in the xy -plane which simultaneously satisfies all the inequalities in the system. To find the graph of such a system, we draw the graph of each inequality in the system on the same coordinate axes and then take intersection of all the graphs. The common region so obtained is called the solution region for the system of inequalities.

Example 5: Find the solution region by drawing the graph of the system of inequalities

$$x - 2y \leq 6$$

$$2x + y \geq 2$$

Solution: $x - 2y \leq 6$... (i)

$$2x + y \geq 2$$
 ... (ii)

The associated equation of (i) is

$$x - 2y = 6$$
 ... (iii)

For x -intercept, put $y = 0$ in (iii), we get

$$x - 2(0) = 6$$

$$x - 0 = 6$$

$$\Rightarrow x = 6, \text{ so the point is } (6, 0)$$

For y -intercept, put $x = 0$ in (iii), we get

$$0 - 2y = 6$$

$$\Rightarrow -2y = 6$$

$$\Rightarrow y = \frac{6}{-2} = -3, \text{ so the point is } (0, -3)$$

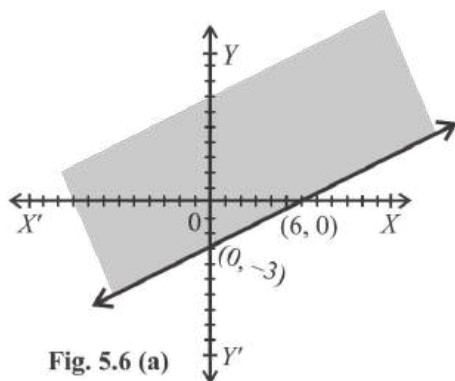


Fig. 5.6 (a)

The graph of the line $x - 2y = 6$ is drawn by joining the points $(6, 0)$ and $(0, -3)$. The point $(0, 0)$ satisfies the inequality $x - 2y < 6$ because $0 - 2(0) = 0 < 6$. Thus, the graph of $x - 2y \leq 6$ is the upper half-plane including the graph of the line $x - 2y = 6$. The closed half-plane is partially shown by shading in figure 5.6 (a).

The associated equation of (ii) is

$$2x + y = 2 \quad \dots(iv)$$

For x -intercept, put $y = 0$ in (iv), we get

$$2x + 0 = 2$$

$$\Rightarrow 2x = 2$$

$$\Rightarrow x = 1, \text{ so the point is } (1, 0)$$

For y -intercept, put $x = 0$ in (iv), we get

$$2(0) + y = 2$$

$$\Rightarrow y = 2, \text{ so the point is } (0, 2)$$

We draw the graph of the line $2x + y = 2$ joining the points $(1, 0)$ and $(0, 2)$. The point $(0, 0)$ does not satisfy the inequality $2x + y > 2$ because $2(0) + 0 = 0 \not> 2$. Thus, the graph of the inequality $2x + y \geq 2$ is the closed half-plane not on the origin-side of the line $2x + y = 2$ and partially shown by shading in figure 5.6 (b).

The solution region of the given system of inequalities is the intersection of the graphs indicated in figures 5.6 (a) and 5.6 (b) is shown as shaded region in figure 5.6 (c).

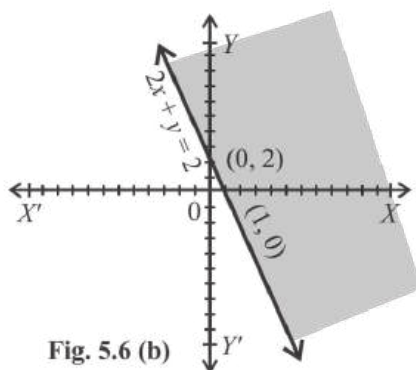


Fig. 5.6 (b)

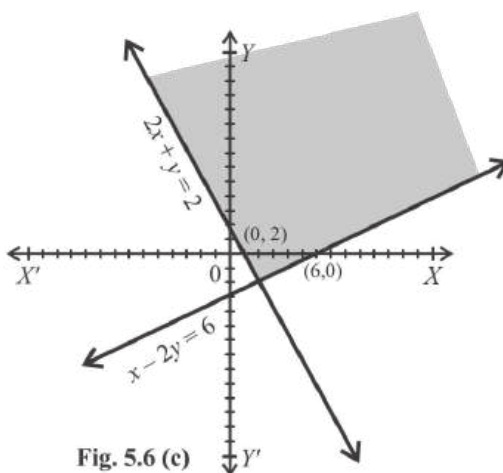


Fig. 5.6 (c)

EXERCISE 5.1

1. Solve and represent the solution on a real line.

(i) $12x + 30 = -6$

(ii) $\frac{x}{3} + 6 = -12$

(iii) $\frac{x}{2} - \frac{3x}{4} = \frac{1}{12}$

(iv) $2 = 7(2x + 4) + 12x$

(v) $\frac{2x-1}{3} - \frac{3x}{4} = \frac{5}{6}$

(vi) $\frac{-5x}{10} = 9 - \frac{10}{5}x$

2. Solve each inequality and represent the solution on a real line.

(i) $x - 6 \leq -2$

(ii) $-9 > -16 + x$

(iii) $3 + 2x \geq 3$

(iv) $6(x + 10) \leq 0$

(v) $\frac{5}{3}x - \frac{3}{4} < \frac{-1}{12}$

(vi) $\frac{1}{4}x - \frac{1}{2} \leq -1 + \frac{1}{2}x$

3. Shade the solution region for the following linear inequalities in xy -plane:
- (i) $2x + y \leq 6$ (ii) $3x + 7y \geq 21$ (iii) $3x - 2y \geq 6$
 (iv) $5x - 4y \leq 20$ (v) $2x + 1 \geq 0$ (vi) $3y - 4 \leq 0$
4. Indicate the solution region of the following linear inequalities by shading:
- (i) $2x - 3y \leq 6$ (ii) $x + y \geq 5$ (iii) $3x + 7y \geq 21$
 $2x + 3y \leq 12$ $-y + x \leq 1$ $x - y \leq 2$
 (iv) $4x - 3y \leq 12$ (v) $3x + 7y \geq 21$ (vi) $5x + 7y \leq 35$
 $x \geq -\frac{3}{2}$ $y \leq 4$ $x - 2y \leq 2$

5.3 Feasible Solution

While tackling a certain problem from everyday life each linear inequality concerning the problem is named as **problem constraint**. The system of linear inequalities involved in the problem concerned is called **problem constraints**. The variables used in the system of linear inequalities relating to the problems of everyday life are non-negative and are called **non-negative constraints**. These non-negative constraints play an important role for taking decision. So, these variables are called **decision variables**. A region which is restricted to the first quadrant is referred to as a **feasible region** for the set of given constraints. Each point of the feasible region is called a **feasible solution** of the system of linear inequalities (or for the set of a given constraints).

Example 6: Shade the feasible region and find the corner points for the following system of inequalities (or subject to the following constraints).

$$x - y \leq 3$$

$$x + 2y \leq 6, \quad x \geq 0, \quad y \geq 0$$

Solution: The associated equations for the inequalities

$$x - y \leq 3 \dots \text{(i)} \quad \text{and} \quad x + 2y \leq 6 \dots \text{(ii)}$$

$$\text{are } x - y = 3 \dots \text{(iii)} \quad \text{and} \quad x + 2y = 6 \dots \text{(iv)}$$

As the points $(3, 0)$ and $(0, -3)$ are on the line (iii), so the graph of $x - y = 3$ is drawn by joining the points $(3, 0)$ and $(0, -3)$ by solid line.

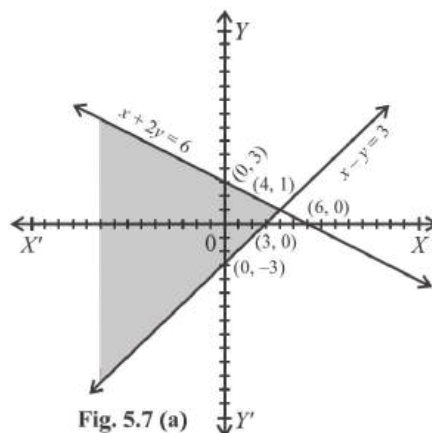


Fig. 5.7 (a)

Similarly, line (iv) is graphed by joining the points (6, 0) and (0, 3) by solid line.

For $x = 0$ and $y = 0$, we have;

$$0 - 0 = 0 < 3 \text{ and } 0 + 2(0) = 0 < 6$$

So, both the closed half-planes are on the origin sides of the lines (iii) and (iv). The intersection of these closed half-planes is partially displayed as shaded region in fig. 5.7(a).

The graph of $y \geq 0$, will be the closed upper half plane. The intersection of graph shown in figure 5.7(a) and closed upper half plane is partially displayed as shaded region in figure 5.7 (b).

The graph of $x \geq 0$ will be closed right half plane. The intersection of the graph shown in fig. 5.7(a) and closed right half plane is graphed in fig. 5.7 (c).

Finally, the graph of the given system of linear inequalities is displayed in figure 5.7 (d) which is the feasible region for the given system of linear inequalities. The points (0, 0), (3, 0), (4, 1) and (0, 3) are corner points of the feasible region.

Remember!

A point of a solution region where two of its boundary lines intersect, is called a **corner point** or **vertex** of the solution region.

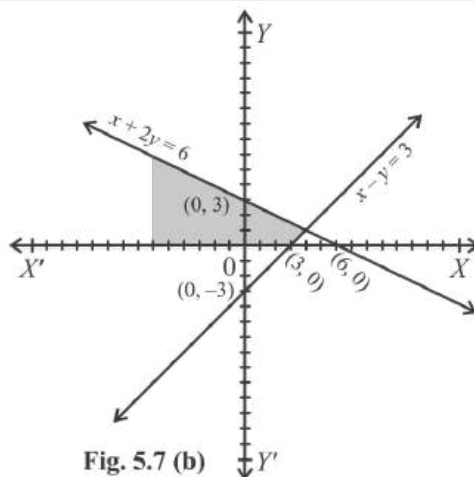


Fig. 5.7 (b)

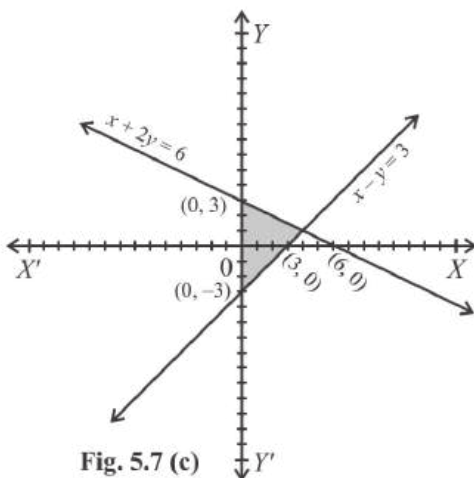


Fig. 5.7 (c)

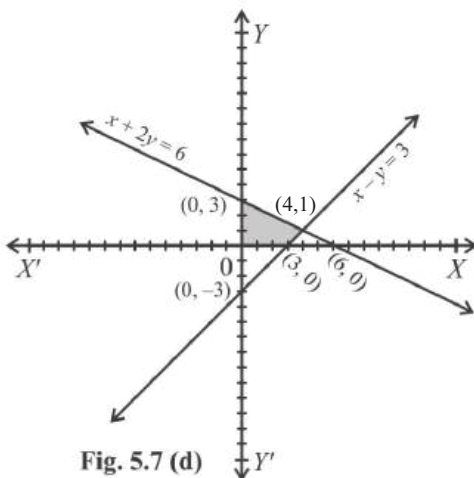


Fig. 5.7 (d)

5.3.2 Maximum and Minimum Values of a Function in the Feasible Solution

A function which is to be maximized or minimized is called an **objective function**. Note that there are infinitely many feasible solutions in the feasible region. The feasible solution which maximizes or minimizes the objective function is called the **optimal solution**.

Procedure for determining optimal solution

- Graph the solution set of linear inequality constraints to determine feasible region.
- Find the corner points of the feasible region.
- Evaluate the objective function at each corner point to find the optimal solution.

Example 7: Find the maximum and minimum values of the function defined as:

$$f(x, y) = 2x + 3y$$

subject to the constraints;

$$x - y \leq 2$$

$$x + y \leq 4$$

$$x \geq 0, y \geq 0$$

Solution: $x - y \leq 2$... (i)
 $x + y \leq 4$... (ii)

The associated equation of (i) is

$$x - y = 2$$

x -intercept and y -intercept of $x - y = 2$ are $(2, 0)$ and $(0, -2)$ respectively. The graph of the line $x - y = 2$ is drawn by joining the points $(2, 0)$ and $(0, -2)$. The point $(0, 0)$ satisfies the inequality $x - y \leq 2$ because $0 - 0 = 0 < 2$. Thus, the graph of $x - y \leq 2$ is the upper half-plane including the graph of the line $x - y = 2$. The closed half-plane is partially shown by shading in figure 5.8(a).

The associated equation of (ii) is $x + y = 4$

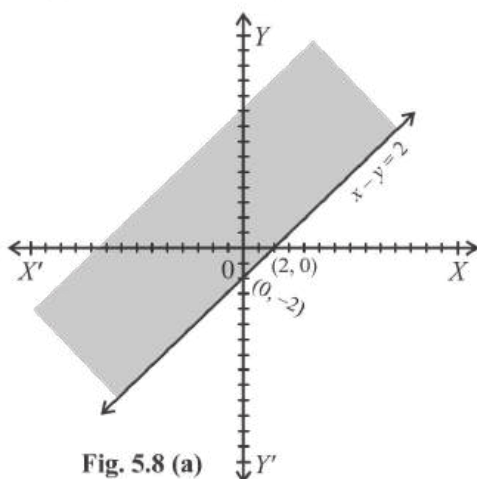


Fig. 5.8 (a)

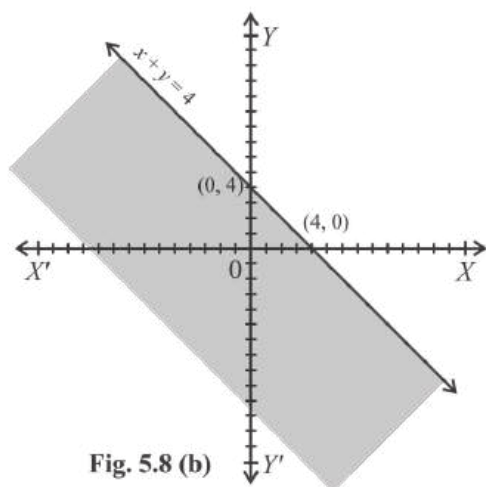


Fig. 5.8 (b)

x -intercept and y -intercept of $x + y = 4$ are $(4, 0)$ and $(0, 4)$ respectively. The graph of the line $x + y = 4$ is drawn by joining the points $(4,0)$ and $(0,4)$. The point $(0, 0)$ satisfies the inequality $x + y \leq 4$. The closed half-plane is partially shown by shading in figure 5.8 (b).

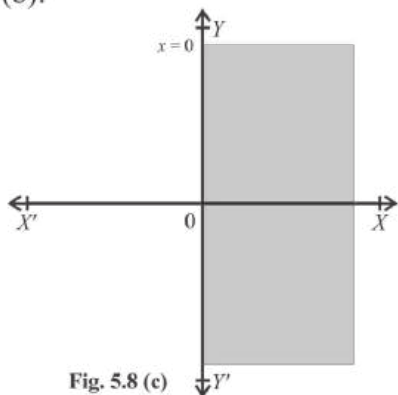


Fig. 5.8 (c)

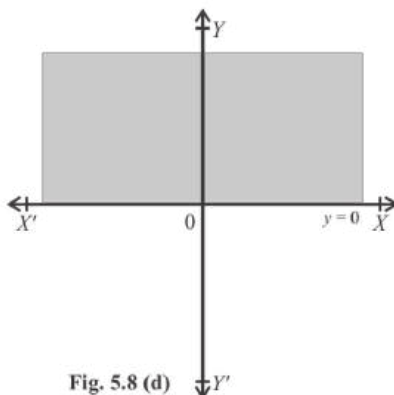


Fig. 5.8 (d)

The graph of $x \geq 0$ and $y \geq 0$ is shown by shading in figures 5.8 (c) and 5.8 (d) respectively.

The feasible region of the given system of inequalities is the intersection of the graphs indicated in figures 5.8 (a), 5.8 (b), 5.8 (c) and 5.8 (d) and is shown as shaded region $ABCD$ in figure 5.8 (e).

Corner points of the feasible region are $(0, 0)$, $(2, 0)$, $(3, 1)$ and $(0, 4)$. Now, we find values of $f(x, y) = 2x + 3y$ at the corner points.

$$f(0, 0) = 2(0) + 3(0) = 0$$

$$f(2, 0) = 2(2) + 3(0) = 4$$

$$f(3, 1) = 2(3) + 3(1) = 9$$

$$f(0, 4) = 2(0) + 3(4) = 12$$

Thus, the minimum value of f is 0 at the corner point $(0, 0)$ and maximum value of f is 12 at corner point $(0, 4)$.

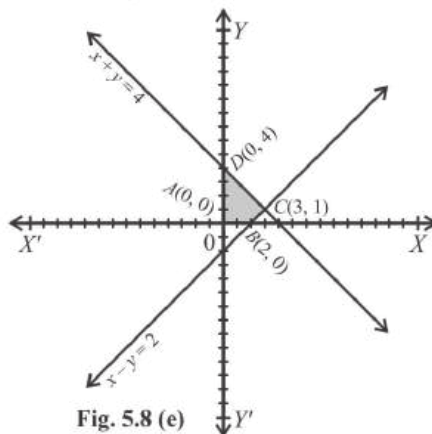


Fig. 5.8 (e)

EXERCISE 5.2

- Maximize $f(x, y) = 2x + 5y$; subject to the constraints
 $2y - x \leq 8$; $x - y \leq 4$; $x \geq 0$; $y \geq 0$
- Maximize $f(x, y) = x + 3y$; subject to the constraints
 $2x + 5y \leq 30$; $5x + 4y \leq 20$; $x \geq 0$; $y \geq 0$

3. Maximize $z = 2x + 3y$; subject to the constraints:

$$2x + y \leq 4 \quad ; \quad 4x - y \leq 2 \quad ; \quad x \geq 0; y \geq 0$$

4. Minimize $z = 2x + y$; subject to the constraints:

$$x + y \geq 3 \quad ; \quad 7x + 5y \leq 35 \quad ; \quad x \geq 0; y \geq 0$$

5. Maximize the function defined as; $f(x, y) = 2x + 3y$ subject to the constraints:

$$2x + y \leq 10 \quad ; \quad x + 2y \leq 14 \quad ; \quad x \geq 0; y \geq 0$$

6. Find minimum and maximum values of $z = 3x + y$; subject to the constraints:

$$3x + 5y \geq 15 \quad ; \quad x + 3y \leq 9 \quad ; \quad x \geq 0; y \geq 0$$

REVIEW EXERCISE 5

1. Four options are given against each statement. Encircle the correct one.

- i. In the following, linear equation is:

(a) $5x > 7$

(b) $4x - 2 < 1$

(c) $2x + 1 = 1$

(d) $4 = 1 + 3$

- ii. Solution of $5x - 10 = 10$ is:

(a) 0

(b) 50

(c) 4

(d) -4

- iii. If $7x + 4 < 6x + 6$, then x belongs to the interval

(a) $(2, \infty)$

(b) $[2, \infty)$

(c) $(-\infty, 2)$

(d) $(-\infty, 2]$

- iv. A vertical line divides the plane into

(a) left half plane

(b) right half plane

(c) full plane

(d) two half planes

- v. The equation formed from the linear inequality is called

(a) cubic equation

(b) associated equation

(c) quadratic equation

(d) feasible region

- vi. $3x + 4 < 0$ is:

(a) equation

(b) inequality

(c) not inequality

(d) identity

- vii. Corner point is also called:

(a) code

(b) vertex

(c) curve

(d) region

- viii. $(0,0)$ is solution of inequality:
- (a) $4x + 5y > 8$ (b) $3x + y > 6$
(c) $-2x + 3y < 0$ (d) $x + y > 4$
- ix. The solution region restricted to the first quadrant is called:
- (a) objective region (b) feasible region
(c) solution region (d) constraints region
- x. A function that is to be maximized or minimized is called:
- (a) solution function (b) objective function
(c) feasible function (d) none of these
2. Solve and represent their solutions on real line.
- (i) $\frac{x+5}{3} = 1 - x$ (ii) $\frac{2x+1}{3} + \frac{1}{2} = 1 - \frac{x-1}{3}$
(iii) $3x + 7 < 16$ (iv) $5(x-3) \geq 26x - (10x+4)$
3. Find the solution region of the following linear equalities:
- (i) $3x - 4y \leq 12$; $3x + 2y \geq 3$
(ii) $2x + y \leq 4$; $x + 2y \leq 6$
4. Find the maximum value of $g(x,y) = x + 4y$ subject to constraints $x + y \leq 4$, $x \geq 0$ and $y \geq 0$.
5. Find the minimum value of $f(x,y) = 3x + 5y$ subject to constraints $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.

Unit 6

Trigonometry

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Identify angles in standard positions expressed in degrees and radian.
- Apply Pythagoras theorem and the sine, the cosine and tangent ratios for acute angles of a right angle.
- Solve real life trigonometric problems in 2-D involving angles of elevation and depression
- Prove the trigonometric identities and apply them to draw different trigonometric relations.
- Solve real life problems involving trigonometric identities.

INTRODUCTION

Trigonometry is a branch of mathematics that deals with the relationships between the angles and sides of a triangle, especially right-angled triangle. It plays a vital role in various fields such as physics, engineering, architecture and astronomy. The trigonometric concepts can solve problems involving angles and distances that appear in real-life situations such as calculating the height of buildings, distance between objects and angle measurements in navigation.

6.1 Identifying Angles in Standard Position (Degrees and Radians)

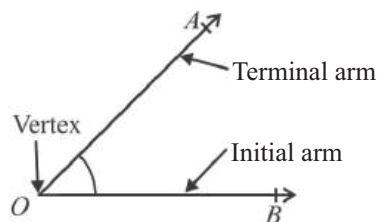
A plane figure which is formed by two rays sharing a common end point is called an angle. The two rays are known as the sides of the angle. The common end point is known as vertex. The amount of rotation or measure of opening between these rays is called an angle. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are rays and angle is AOB . Written as $\angle AOB$ or $\hat{AOB}$.

The angle is said to be in standard position if:

- Its vertex is located at the origin of the coordinate plane.
- One of its rays (the initial side) lies along the positive x -axis.

Brain teaser!

The plane geometry is the study of two dimensional figures. What is Euclidean geometry?



Types of angles are:

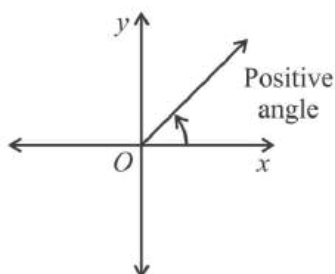
- Acute angle $0 < \theta < 90^\circ$
- Obtuse angle $90^\circ < \theta < 180^\circ$
- Right angle $\theta = 90^\circ$
- Straight angle $\theta = 180^\circ$
- Reflex angle $180^\circ < \theta < 360^\circ$
- Full rotation $\theta = 360^\circ$

(c) The other ray (the terminal side) determines the direction of the angle.

An angle is measured from the initial side to the terminal side. It is usually represented by Greek letters θ , α , β , γ etc.

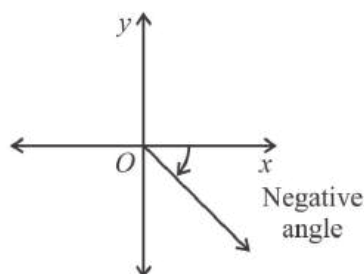
Positive angles

The angle will be positive if the terminal side is rotated counterclockwise from the initial side. The given angle is in 1st quadrant



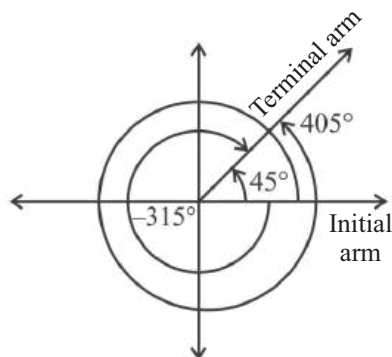
Negative angles

The angle will be negative if the terminal side is rotated clockwise from the initial side. The given angle is in 4th quadrant



Co-Terminal Angles

Co-terminal angles are angles that share the same initial side and terminal side in standard position, but they may have different measures. These angles differ by a multiple of 360° or 2π rad. For example, 45° , 405° and -315° are co-terminal angles because $405^\circ = 45^\circ + 360^\circ$ and $-315^\circ = 45^\circ - 360^\circ$.



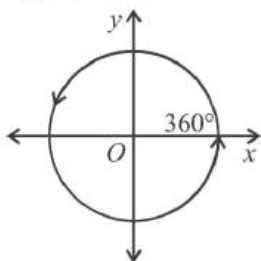
6.1.1 Degree Measurement

A degree ($^\circ$) is a unit of measurement of an angle. It represents $\left(\frac{1}{360}\right)^{\text{th}}$ of a full rotation around a point. In simpler terms, a degree is the measure of an angle, with a complete circle being 360° .

Why 360° Historically? The choice of 360° to divide a circle dates back to the **Babylonians**, who used a base-60 number system (sexagesimal system). They were among the first to formalize the concept of angle measurement, and 360 was chosen likely because it is a highly composite number (it can be divided by 2, 3, 4, 5, 6, 9, 10, 12, 15, and more), making calculations easier. This system persisted throughout ancient times and degrees became entrenched in various cultures and mathematical traditions.

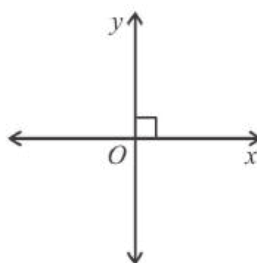
Full Circle

A full rotation around a central point forms an angle of 360° .



Right Angle

One-quarter of a full rotation, or a 90° angle, is called a right angle.



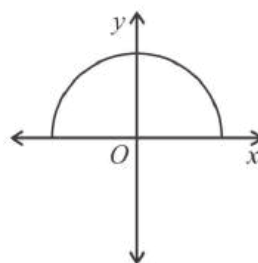
Half Circle

A straight angle, or half of a full rotation, measures 180° . The degree measure is further divided into minutes (') and seconds (").

$$1^\circ = 60' \text{ (60 minutes)}$$

$$1' = 60'' \text{ (60 seconds)}$$

$$1^\circ = 3600'' \text{ (60} \times \text{60 seconds)}$$



6.1.2 Converting Degrees to Minutes and Seconds

To convert decimal degrees to degrees, minutes and seconds (DMS), follow the steps:

- Separate the whole number part (degrees) of the decimal.
- Multiply the decimal part by 60 to get the minutes.
- The whole number part of the result is the minutes. Multiply the decimal part of the minutes by 60 to get the seconds.

Example 1: Convert 73.12° to degrees, minutes, and seconds.

Solution:

Degrees: The whole number part is 73° .

Minutes: Take the decimal part (0.12) and multiply by 60: $0.12 \times 60 = 7.2'$. The whole number part is 7, so it's 7 minutes.

Seconds: Now take the decimal part (0.2) and multiply by 60: $0.2 \times 60 = 12''$. So, it's 12 seconds.

Final result: $73^\circ 7' 12''$.

Example 2: Convert 109.42° to degrees, minutes, and seconds.

Solution:

Degrees: The whole number part is 109° .

Minutes: Take the decimal part (0.42) and multiply by 60: $0.42 \times 60 = 25.2'$. The whole number part is 25, so it's 25 minutes.

Seconds: Now take the decimal part (0.2) and multiply by 60: $0.2 \times 60 = 12''$. So, it's 12 seconds.

Final result: $109^\circ 25' 12''$.

6.1.3 Converting from Degrees, Minutes and Seconds to Decimal Degrees

To convert from degrees, minutes and seconds (DMS) to decimal degrees, follow the steps:

- Keep the degrees as they are.
- Convert minutes to decimal degrees: Divide the number of minutes by 60.
- Convert seconds to decimal degrees: Divide the number of seconds by 3600.
- Add all the values together.

Example 3: Convert $45^\circ 45' 45''$ to decimal degrees.

Solution: Degrees: Keep 45.

$$\text{Minutes to decimal: } \frac{45}{60} = 0.75; \quad \text{Seconds to decimal: } \frac{45}{3600} = 0.0125$$

$$\text{Add them together: } 45 + 0.75 + 0.0125 = 45.7625$$

$$\text{Final result: } 45.7625^\circ$$

Example 4: Convert $94^\circ 27' 54''$ to decimal degrees.

Solution: Degrees: Keep 94:

$$\text{Minutes to decimal: } \frac{27}{60} = 0.45; \quad \text{Seconds to decimal: } \frac{54}{3600} = 0.015$$

$$\text{Add them together: } 94 + 0.45 + 0.015 = 94.465$$

$$\text{Final result: } 94.465^\circ$$

6.1.4 Circular Measure (Radian)

There is another system of angular measurement called circular system.

The radian, denoted by the symbol “rad”, is the unit of angle in the International System of Units (SI) and is the standard unit of angular measure used in many areas of mathematics.

A radian is a unit of angular measure in mathematics, particularly in trigonometry. It is defined as, “the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle”. Unlike degrees, which are based on dividing a circle into 360 parts, the radian is inherently related to the circle's geometry and arc length.

Historical Background of the Radian

The concept of radian measure, was first formalized by mathematicians in the 18th century, but the principles behind it had been understood much earlier by Euclid and Archimedes.

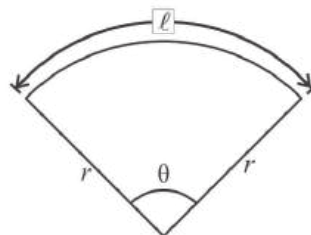
The word “radian” comes from the radius of a circle, as the radian is fundamentally related to the ratio between the arc length and the radius.

The first known use of the term radian in the context of angular measurement was by Scottish mathematician James Thomson in 1873. His brother, William Thomson, also known as Lord Kelvin, was made a prominent physicist and both were influential in establishing radians as a standard unit.

If a circle of radius r , has an arc length equal to the radius of the circle, then the angle θ subtended by that arc is 1 radian:

$$\theta = \frac{r}{r} = 1 \text{ radian} \quad \left(\therefore \theta = \frac{\text{Arc length}}{\text{Radius}} = \frac{\ell}{r} \right)$$

A complete circle has an arc length equal to the circumference ($2\pi r$), so the angle subtended by the entire circle (the full rotation) is 2π radians. This means:



- One full revolution of a circle is 2π radians, or 360° .
- Therefore, $1 \text{ radian} = \frac{360^\circ}{2\pi} \approx 57.2958^\circ$ and $1^\circ = \frac{2\pi}{360} = 0.01745 \text{ rad}$

Conversion between degrees and radians

Radians to degrees: $1 \text{ rad} = \frac{180}{\pi} \text{ degrees}$

Degrees to radians: $1^\circ = \frac{\pi}{180} \text{ rad}$

Example 5: Convert radians to degrees

- (i) $\frac{5\pi}{3} \text{ rad}$ (ii) $\frac{7\pi}{6} \text{ rad}$ (iii) $\frac{11\pi}{6} \text{ rad}$ (iv) 1.2 rad

Solution: (i) $\frac{5\pi}{3} \text{ rad} = \frac{5\pi}{3} \times \frac{180^\circ}{\pi} = 300^\circ$ ($1 \text{ rad} = \frac{180^\circ}{\pi}$)

(ii) $\frac{7\pi}{6} \text{ rad} = \frac{7\pi}{6} \times \frac{180^\circ}{\pi} = 210^\circ$

(iii) $\frac{11\pi}{6} \text{ rad} = \frac{11\pi}{6} \times \frac{180^\circ}{\pi} = 330^\circ$

(iv) $1.2 \text{ rad} = 1.2 \times \frac{180^\circ}{\pi} = 68.75^\circ$ ($\therefore \pi = 3.14159$)

Example 6: Convert degrees to radians

- (i) 15° (ii) 75° (iii) 315° (iv) $15^\circ 15'$

Solution: (i) $15^\circ = 15 \times \frac{\pi}{180} = \frac{\pi}{12} \text{ rad}$ or 0.262 rad

(ii) $75^\circ = 75 \times \frac{\pi}{180} = \frac{5\pi}{12} \text{ rad}$ or 1.309 rad

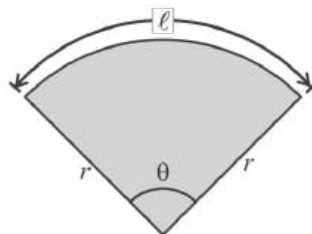
$$(iii) \quad 315^\circ = 315 \times \frac{\pi}{180} = \frac{7\pi}{4} \text{ rad} \quad \text{or} \quad 5.498 \text{ rad}$$

$$(iv) \quad 15^\circ 15' = 15^\circ + \left(\frac{15}{60}\right)^\circ = 15.25^\circ = 15.25 \times \frac{\pi}{180} \text{ rad} = 0.266 \text{ rad}$$

Turns	0 turn	$\frac{1}{12}$ turn	$\frac{1}{8}$ turn	$\frac{1}{6}$ turn	$\frac{1}{4}$ turn	$\frac{1}{2}$ turn	1 turn
Radians	0 rad	$\frac{\pi}{6}$ rad	$\frac{\pi}{4}$ rad	$\frac{\pi}{3}$ rad	$\frac{\pi}{2}$ rad	π rad	2π rad
Degrees	0°	30°	45°	60°	90°	180°	360°

Arc Length and Area of Sector

If r is radius and θ (rad) is the angle subtended by the arc of length ' ℓ ', then



$$\text{Arc length of sector} = \ell = r\theta$$

$$\text{and area of sector} = A = \frac{1}{2}r^2\theta$$

Proof: We know that:

$$\begin{aligned} \ell &= \frac{\theta}{360^\circ} \times 2\pi r \\ &= \frac{\theta}{2\pi} \times 2\pi r \quad (2\pi \text{ radians} = 360^\circ) \\ &= r\theta \end{aligned}$$

Proof: We know that

$$\begin{aligned} A &= \frac{\theta}{360^\circ} \times \pi r^2 \\ &= \frac{\theta}{2\pi} \times \pi r^2 \quad (2\pi \text{ radians} = 360^\circ) \\ &= \frac{1}{2}r^2\theta \end{aligned}$$

Hence arc length, $\ell = r\theta$ and area of sector, $A = \frac{1}{2}r^2\theta$

Example 7: Find the arc length of a sector with radius $r = 10$ cm and central angle $\theta = 60^\circ$.

Solution: Convert $\theta = 60^\circ$ to radians: $\theta = 60 \times \frac{\pi}{180} = \frac{\pi}{3}$ radians.

$$\ell = r\theta = 10 \times \frac{\pi}{3} \approx 10.47 \text{ cm}$$

The arc length is approximately 10.47 cm

Example 8: Find the area of a sector with radius $r = 8$ cm and central angle $\theta = 45^\circ$.

Solution: Convert $\theta = 45^\circ$ to radians: $\theta = 45 \times \frac{\pi}{180} = \frac{\pi}{4}$ radians.

$$A = \frac{1}{2} r^2 \theta = \frac{1}{2} \times 8^2 \times \frac{\pi}{4} = 8\pi \text{ cm}^2 \approx 25.12 \text{ cm}^2.$$

The area of the sector is approximately 25.12 cm^2 .

Example 9: If arc length of a sector of radius 5 cm is 11 cm, find the angle subtended by the arc in radians and degrees.

Solution: $r = 5$ cm ; $\ell = 11$ cm, ; $\theta = ?$

$$\therefore \ell = r \theta$$

$$11 = 5 \theta \quad \Rightarrow \theta = \frac{11}{5} = 2.2 \text{ rad}$$

$$\theta = 2.2 \times \frac{180^\circ}{\pi} \approx 126.1^\circ$$

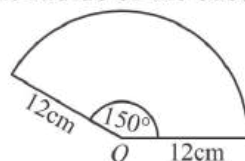
Thus, the angle subtended by the arc in radians is 2.2 rad and degrees is 126.1°

EXERCISE 6.1

- Find in which quadrant the following angles lie. Write a co-terminal angle for each:
(i) 65° (ii) 135° (iii) -40° (iv) 210° (v) -150°
- Convert the following into degrees, minutes, and seconds:
(i) 123.456° (ii) 58.7891° (iii) 90.5678°
- Convert the following into decimal degrees:
(i) $65^\circ 32' 15''$ (ii) $42^\circ 18' 45''$ (iii) $78^\circ 45' 36''$
- Convert the following into radians:
(i) 36° (ii) 22.5° (iii) 67.5°
- Convert the following into degrees:
(i) $\frac{\pi}{16}$ rad (ii) $\frac{11\pi}{5}$ rad (iii) $\frac{7\pi}{6}$ rad
- Find the arc length and area of a sector if:
(i) $r = 6$ cm and central angle $\theta = \frac{\pi}{3}$ radians.
(ii) $r = \frac{4.8}{\pi}$ cm and central angle $\theta = \frac{5\pi}{6}$ radians.

7. If the central angle of a sector is 60° and the radius of the circle is 12 cm, find the area of the sector and the percentage of the total area of the circle it represents.
8. Find the percentage of the area of sector subtending an angle $\frac{\pi}{8}$ radians.
9. A circular sector of radius $r = 12$ cm has an angle of 150° . This sector is cut out and then bent to form a cone. What is the slant height and the radius of the base of this cone?

Hint: Arc length of sector = circumference of cone.



6.2 Trigonometric Ratios

The functions that relate angles to side in a right-angled triangle are known as trigonometric functions (sine, cosine, tangent etc.) Their development is rooted in ancient geometry, blossomed through Indian and Islamic mathematics and became formalized in Europe during the Renaissance. Today, these functions are indispensable tools in both theoretical and applied sciences. Trigonometry has since been extensively used in various scientific disciplines such as physics (especially wave theory) engineering, and computer graphics.

History of Sine, Cosine and Tangent

Hipparchus of Nicaea (c. 190 - 120 BC) is considered the "father of trigonometry." He was the first to compile a trigonometric table for solving problems related to astronomy, using chord functions. Hipparchus divided a circle into 360 degrees and used this system for measuring angles.

In Islamic golden age, **Al-Battani** (c. 858 – 929 CE) was among the first to replace chord functions with the modern sine function and calculated tables of sines and tangents.

Al-Khwarizmi (c. 780–850 CE), known for his work in algebra, and Omar Khayyam (c. 1048–1131 CE) worked on spherical trigonometry, which has applications in astronomy.

Isaac Newton and **Gottfried Wilhelm Leibniz** (17th century) developed calculus, which further expanded the use of trigonometric functions beyond geometry into more abstract fields of mathematics.

Application of Trigonometric Ratios

When we make use of a ruler or measuring tape to measure the thickness of a book, the length of a pencil, the height of a chair or table or dimensions of a classroom, we are making direct measurements.

In some cases, it is not possible to obtain direct measurements, because these are difficult and dangerous. For example, it is difficult to climb upon a flag pole to measure its height. To measure the height of a cliff is also difficult and dangerous.

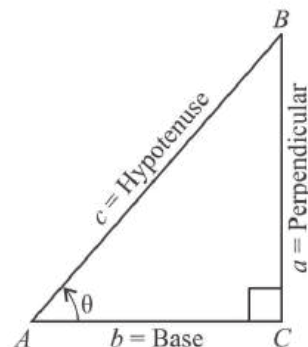
These problems can be solved by indirect measurement with the help of trigonometry. For indirect measurements of distance or height it is very much useful. It also plays an important role in the field of surveying, navigation, engineering and many other branches of physical sciences. We make use of these concepts of trigonometry to solve many of the problems in these fields.

6.2.1 Trigonometric Ratios of an Acute Angle

The trigonometric ratios are applied to acute angle in a right-angled triangle, but the concepts extend to angles greater than 90° and are widely used in many areas of mathematics and science.

Let us consider a right-angled triangle ACB with respect to an angle θ (theta) = $m\angle CAB$ with $m\angle ACB = 90^\circ$.

In the triangle ACB , the side BC is called perpendicular, which is opposite to an angle ' θ '.



The side AC is called the base and the side AB is called the hypotenuse. Let $m\overline{BC} = a$, $m\overline{AC} = b$ and $m\overline{AB} = c$.

For this right angled triangle ACB , the trigonometric ratios of an angle " θ " are defined as:

$$\begin{aligned}\sin \theta &= \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{a}{c} & : & \quad \text{cosec } \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{c}{a} \\ \cos \theta &= \frac{\text{Base}}{\text{Hypotenuse}} = \frac{b}{c} & : & \quad \text{sec } \theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{c}{b} \\ \tan \theta &= \frac{\text{Perpendicular}}{\text{Base}} = \frac{a}{b} & : & \quad \text{cot } \theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{b}{a}\end{aligned}$$

The six trigonometric ratios described with reference to a right-angled triangle ACB are: sine (sin), cosine(cos), tangent(tan), cosecant (cosec or csc), secant (sec) and cotangent (cot).

We note that: $\tan \theta = \frac{a}{b}$

$$= \frac{a/c}{b/c} \quad (\text{Dividing by } c)$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Similarly, $\cot \theta = \frac{\cos \theta}{\sin \theta}$

Note:

- (i) $\text{cosec } \theta = \frac{1}{\sin \theta}$
- (ii) $\text{sec } \theta = \frac{1}{\cos \theta}$
- (iii) $\text{cot } \theta = \frac{1}{\tan \theta}$

6.2.2 Trigonometric Ratios of Complementary Angles

We consider a right-angled triangle ACB , in which $m\angle A = \theta$, $m\angle C = 90^\circ$ then, $m\angle B = 90^\circ - \theta$. Using the trigonometric ratios of $\angle B$, we get

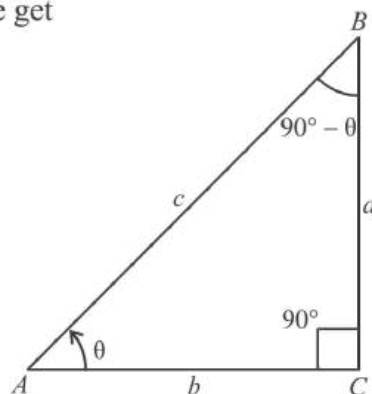
$$\sin m\angle B = \sin(90^\circ - \theta) = \frac{m\overline{AC}}{m\overline{AB}} = \frac{b}{c} \quad \dots(i)$$

Using ratios of $\angle A$, we get

$$\cos m\angle A = \cos \theta = \frac{m\overline{AC}}{m\overline{AB}} = \frac{b}{c} \quad \dots(ii)$$

Form (i) and (ii), we get,

$$\sin(90^\circ - \theta) = \cos \theta$$



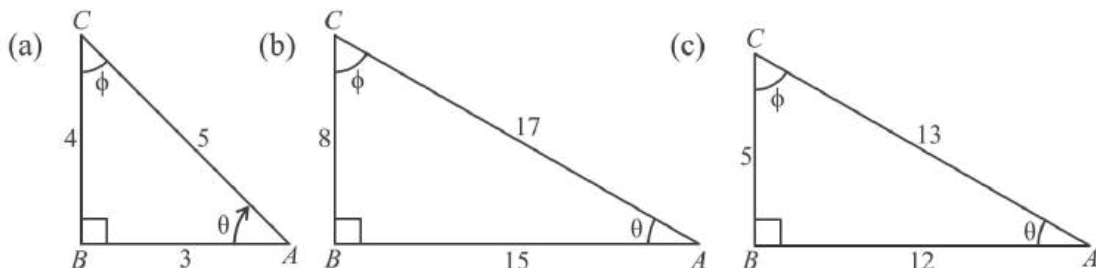
Similarly, we have

$$\begin{array}{lll} \cos(90^\circ - \theta) = \sin \theta & ; & \tan(90^\circ - \theta) = \cot \theta & ; & \cot(90^\circ - \theta) = \tan \theta \\ \sec(90^\circ - \theta) = \operatorname{cosec} \theta & ; & \operatorname{cosec}(90^\circ - \theta) = \sec \theta \end{array}$$

EXERCISE 6.2

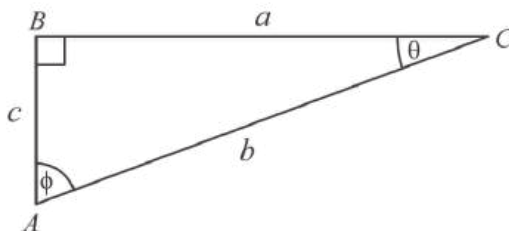
1. For each of the following right-angled triangles, find the trigonometric ratios:

- (i) $\sin \theta$ (ii) $\cos \theta$ (iii) $\tan \theta$ (iv) $\sec \theta$ (v) $\operatorname{cosec} \theta$
 (vi) $\cot \phi$ (vii) $\tan \phi$ (viii) $\operatorname{cosec} \phi$ (ix) $\sec \phi$ (x) $\cos \phi$



2. For the following right-angled triangle ABC find the trigonometric ratios for which $m\angle A = \phi$ and $m\angle C = \theta$

- (i) $\sin \theta$ (ii) $\cos \theta$
 (iii) $\tan \theta$ (iv) $\sin \phi$
 (v) $\cos \phi$ (vi) $\tan \phi$



3. Considering the adjoining triangle ABC , verify that:

(i) $\sin \theta \operatorname{cosec} \theta = 1$

(ii) $\cos \theta \sec \theta = 1$

(iii) $\tan \theta \cot \theta = 1$

4. Fill in the blanks.

(i) $\sin 30^\circ = \sin (90^\circ - 60^\circ) = \underline{\hspace{2cm}}$

(ii) $\cos 30^\circ = \cos (90^\circ - 60^\circ) = \underline{\hspace{2cm}}$

(iii) $\tan 30^\circ = \tan (90^\circ - 60^\circ) = \underline{\hspace{2cm}}$

(iv) $\tan 60^\circ = \tan (90^\circ - 30^\circ) = \underline{\hspace{2cm}}$

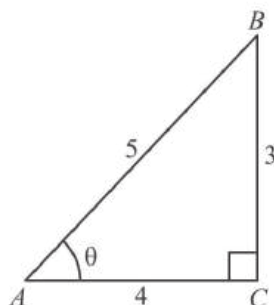
(v) $\sin 60^\circ = \sin (90^\circ - 30^\circ) = \underline{\hspace{2cm}}$

(vi) $\cos 60^\circ = \cos (90^\circ - 30^\circ) = \underline{\hspace{2cm}}$

(vii) $\sin 45^\circ = \sin (90^\circ - 45^\circ) = \underline{\hspace{2cm}}$

(viii) $\tan 45^\circ = \tan (90^\circ - 45^\circ) = \underline{\hspace{2cm}}$

(ix) $\cos 45^\circ = \cos (90^\circ - 45^\circ) = \underline{\hspace{2cm}}$



5. In a right angled triangle ABC , $m\angle B = 90^\circ$ and C is an acute angle of 60° . Also

$\sin m\angle A = \frac{a}{b}$, then find the following trigonometric ratios:

(i) $\frac{\overline{mBC}}{\overline{mAB}}$

(ii) $\cos 60^\circ$

(iii) $\tan 60^\circ$

(iv) $\operatorname{cosec} \frac{\pi}{3}$

(v) $\cot 60^\circ$

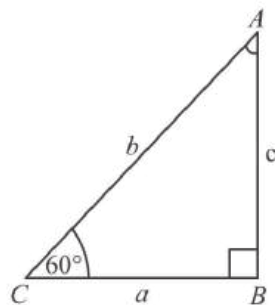
(vi) $\sin 30^\circ$

(vii) $\cos 30^\circ$

(viii) $\tan \frac{\pi}{6}$

(ix) $\sec 30^\circ$

(x) $\cot 30^\circ$



6.3 Trigonometric Identities

Fundamental Trigonometric Identities

We shall consider some of the fundamental identities used in trigonometry. The key to these basic identities is the Pythagoras theorem in geometry.

“The square of the length of the hypotenuse of a right triangle is equal to the sum of the squares of the lengths of the other two sides”.

$$c^2 = a^2 + b^2$$

$$5^2 = 3^2 + 4^2$$

$$25 = 9 + 16$$

In the given figure:

The perpendicular equals to the length 'a', base equals to the length 'b', and hypotenuse equals to the length 'c'.

By Pythagoras Theorem, we have

$$\boxed{a^2 + b^2 = c^2} \quad \dots(i)$$

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2} \quad (\text{Dividing by } c^2)$$

$$\boxed{\sin^2 \theta + \cos^2 \theta = 1} \quad \dots(ii)$$

$$a^2 + b^2 = c^2$$

$$\frac{a^2}{b^2} + \frac{b^2}{b^2} = \frac{c^2}{b^2}$$

Dividing equation (i) by b^2 , we have

$$\boxed{\tan^2 \theta + 1 = \sec^2 \theta} \quad \dots(iii)$$

$$\frac{a^2}{a^2} + \frac{b^2}{a^2} = \frac{c^2}{a^2}$$

Dividing equation (i) by a^2 , we have

$$\boxed{1 + \cot^2 \theta = \operatorname{cosec}^2 \theta} \quad \dots(iv)$$

The identities (ii), (iii) and (iv) are known as Pythagoras identities.

Example 10: Show that $(\sec^2 \theta - 1) \cos^2 \theta = \sin^2 \theta$

Solution: L.H.S = $(\sec^2 \theta - 1) \cos^2 \theta$

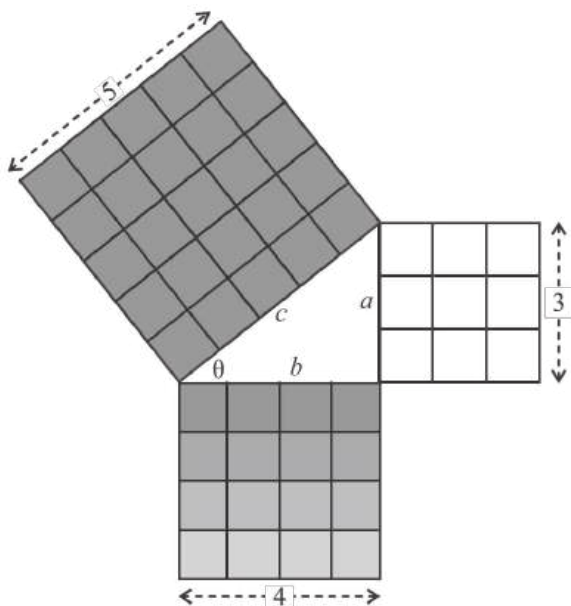
$$= \tan^2 \theta \cdot \cos^2 \theta \quad (\because 1 + \tan^2 \theta = \sec^2 \theta)$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta \quad \left(\because \tan \theta = \frac{\sin \theta}{\cos \theta} \right)$$

$$= \sin^2 \theta = \text{R.H.S}$$

$$\text{Hence, } (\sec^2 \theta - 1) \cos^2 \theta = \sin^2 \theta$$

Example 11: Show that $\tan \theta + \cos \theta = \sec \theta \operatorname{cosec} \theta$



Solution: L.H.S = $\tan \theta + \cot \theta$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin \theta \cdot \sin \theta + \cos \theta \cdot \cos \theta}{\cos \theta \cdot \sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} \quad (\because \sin^2 \theta + \cos^2 \theta = 1)$$

$$= \frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta}$$

$$= \sec \theta \cdot \operatorname{cosec} \theta = \text{R.H.S.}$$

Hence, $\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$ **Example 12:** Show that $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = \frac{1}{\sin \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta}$ **Solution:**

$$\text{L.H.S} = \frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta}$$

$$= \frac{1}{\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}} - \frac{1}{\sin \theta}$$

$$= \frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta}$$

$$= \frac{\sin \theta(1 + \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} - \frac{1}{\sin \theta}$$

$$= \frac{\sin \theta(1 + \cos \theta)}{1 - \cos^2 \theta} - \frac{1}{\sin \theta}$$

$$= \frac{\sin \theta(1 + \cos \theta)}{\sin^2 \theta} - \frac{1}{\sin \theta}$$

$$= \frac{1 + \cos \theta}{\sin \theta} - \frac{1}{\sin \theta}$$

$$= \frac{1 + \cos \theta - 1}{\sin \theta}$$

$$= \frac{\cos \theta}{\sin \theta} = \cot \theta$$

$$\text{R.H.S} = \frac{1}{\sin \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

$$= \frac{1}{\sin \theta} - \frac{1}{\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta}}$$

$$= \frac{1}{\sin \theta} - \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{1}{\sin \theta} - \frac{\sin \theta(1 - \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$$

$$= \frac{1}{\sin \theta} - \frac{\sin \theta(1 - \cos \theta)}{1 - \cos^2 \theta}$$

$$= \frac{1}{\sin \theta} - \frac{\sin \theta(1 - \cos \theta)}{\sin^2 \theta}$$

$$= \frac{1}{\sin \theta} - \frac{1 - \cos \theta}{\sin \theta}$$

$$= \frac{1 - 1 + \cos \theta}{\sin \theta}$$

$$= \frac{\cos \theta}{\sin \theta} = \cot \theta$$

Hence, L.H.S = R.H.S

Example 13: Show that $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$

Solution:

$$\begin{aligned} \text{L.H.S} &= \sin^6 \theta + \cos^6 \theta \\ &= (\sin^2 \theta)^3 + (\cos^2 \theta)^3 \\ &= (\sin^2 \theta + \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta) \\ &= (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta - \sin^2 \theta \cos^2 \theta \\ &= 1 - 3\sin^2 \theta \cos^2 \theta = \text{R.H.S} \end{aligned}$$

Hence, $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$

Example 14: If $\tan \theta = \frac{3}{4}$, find the remaining trigonometric ratios, when θ lies in first quadrant.

Solution: Given: $\tan \theta = \frac{3}{4} = \frac{a}{c}$,

Where, $a = 3, c = 4$

By Pythagoras theorem, we have

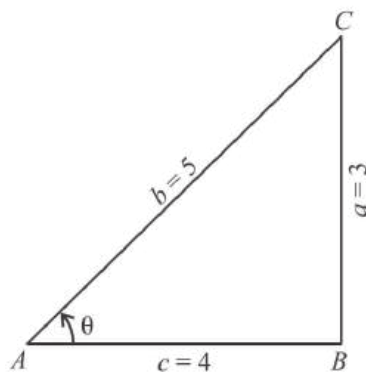
$$\begin{aligned} b^2 &= a^2 + c^2 \\ &= 9 + 16 = 25 \end{aligned}$$

$$\Rightarrow b = 5$$

Therefore, $\sin \theta = \frac{a}{b} = \frac{3}{5}$; $\cos \theta = \frac{c}{b} = \frac{4}{5}$

$\cos \theta = \frac{c}{b} = \frac{4}{5}$; $\sec \theta = \frac{b}{c} = \frac{5}{4}$

$\cot \theta = \frac{c}{a} = \frac{4}{3}$



EXERCISE 6.3

1. If θ lies in first quadrant, find the remaining trigonometric ratios of θ .

(i) $\sin \theta = \frac{2}{3}$ (ii) $\cos \theta = \frac{3}{4}$ (iii) $\tan \theta = \frac{1}{2}$

(iv) $\sec \theta = 3$ (v) $\cot \theta = \sqrt{\frac{3}{2}}$

Prove the following trigonometric identities:

2. $(\sin \theta + \cos \theta)^2 = 1 + 2 \sin \theta \cos \theta$
3. $\frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$
4. $\frac{\sin \theta}{\operatorname{cosec} \theta} + \frac{\cos \theta}{\sec \theta} = 1$
5. $\cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$
6. $\cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta$
7. $\frac{1 - \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$
8. $(\sec \theta - \tan \theta)^2 = \frac{1 - \sin \theta}{1 + \sin \theta}$
9. $(\tan \theta + \cot \theta)^2 = \sec^2 \theta \operatorname{cosec}^2 \theta$
10. $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \tan \theta + \sec \theta$
11. $\sin^3 \theta - \cos^3 \theta = (\sin \theta - \cos \theta)(1 + \sin \theta \cos \theta)$
12. $\sin^6 \theta - \cos^6 \theta = (\sin^2 \theta - \cos^2 \theta)(1 - \sin^2 \theta \cos^2 \theta)$

6.4 Values of Trigonometric Ratios of Special Angles

Trigonometric ratios of 45° $\left(\frac{\pi}{4} \text{ radian}\right)$:

Consider a square $ACBD$ of side length 1 unit.

We know that the diagonals bisect the angles.

So, in the triangle ABC

$$m\angle A = m\angle B = 45^\circ \text{ and } m\angle C = 90^\circ.$$

Using Pythagoras theorem in $\triangle ABC$,

$$c^2 = a^2 + b^2$$

$$c^2 = 1 + 1$$

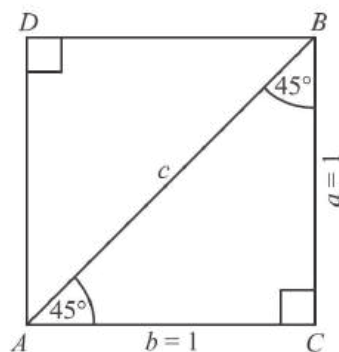
$$c^2 = 2 \Rightarrow c = \sqrt{2}$$

The trigonometric ratios are:

$$\sin 45^\circ = \frac{a}{c} = \frac{1}{\sqrt{2}} \quad ; \quad \operatorname{cosec} 45^\circ = \frac{c}{a} = \sqrt{2}$$

$$\cos 45^\circ = \frac{b}{c} = \frac{1}{\sqrt{2}} \quad ; \quad \sec 45^\circ = \frac{c}{b} = \sqrt{2}$$

$$\tan 45^\circ = \frac{a}{b} = 1 \quad ; \quad \cot 45^\circ = \frac{b}{a} = 1$$



Trigonometric Ratios of $30^\circ \left(\frac{\pi}{6} \text{radian} \right)$ and $60^\circ \left(\frac{\pi}{3} \text{radian} \right)$:

Consider an equilateral triangle ABD of side 2 units.

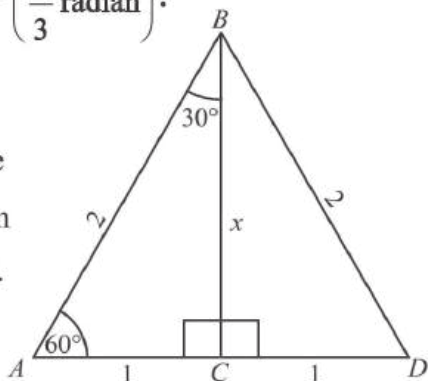
Draw a perpendicular bisector $\overline{BC}$ on $\overline{AD}$. The point C is the midpoint of $\overline{AD}$. So, $m\overline{AC} = m\overline{CD}$ in which $m\angle BAC = 60^\circ$, $m\angle ABC = 30^\circ$, $m\angle ACB = 90^\circ$.

Let $m\overline{BC} = x$ units.

Using Pythagoras theorem in the $\triangle ABC$.

$$2^2 = 1^2 + x^2$$

$$x^2 = 4 - 1 \Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3} \text{ (} m\overline{BC} = \sqrt{3} \text{ units)}$$



Trigonometric ratios of $30^\circ \left(\frac{\pi}{6} \text{radian} \right)$:

In the triangle ABC with $m\angle ABC = 30^\circ$

$$\sin 30^\circ = \frac{1}{2} \quad ; \quad \operatorname{cosec} 30^\circ = 2$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2} \quad ; \quad \sec 30^\circ = \frac{2}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}} \quad ; \quad \cot 30^\circ = \sqrt{3}$$

Trigonometric Ratios of $60^\circ \left(\frac{\pi}{3} \text{radian} \right)$:

In right angled triangle ABC , with $m\angle A = 60^\circ$

$$\sin 60^\circ = \frac{\sqrt{3}}{2} \quad ; \quad \cos 60^\circ = \frac{1}{2} \quad ; \quad \tan 60^\circ = \sqrt{3}$$

$$\operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}} \quad ; \quad \sec 60^\circ = 2 \quad ; \quad \cot 60^\circ = \frac{1}{\sqrt{3}}$$

These results in the form of a table can be written as:

θ	0°	$30^\circ = \frac{\pi}{6}$	$45^\circ = \frac{\pi}{4}$	$60^\circ = \frac{\pi}{3}$	$90^\circ = \frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

EXERCISE 6.4

1. Find the value of the following trigonometric ratios without using the calculator.

- (i) $\sin 30^\circ$ (ii) $\cos 30^\circ$ (iii) $\tan \frac{\pi}{6}$ (iv) $\tan 60^\circ$
 (v) $\sec 60^\circ$ (vi) $\cos \frac{\pi}{3}$ (vii) $\cot 60^\circ$ (viii) $\sin 60^\circ$
 (ix) $\sec 30^\circ$ (x) $\operatorname{cosec} 30^\circ$ (xi) $\sin 45^\circ$ (xii) $\cos \frac{\pi}{4}$

2. Evaluate:

- (i) $2 \sin 60^\circ \cos 60^\circ$ (ii) $2 \cos \frac{\pi}{6} \sin \frac{\pi}{6}$
 (iii) $2 \sin 45^\circ + 2 \cos 45^\circ$ (iv) $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$
 (v) $\cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$ (vi) $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$
 (vii) $\cos 60^\circ \cos 30^\circ + \sin 60^\circ \sin 30^\circ$ (viii) $\tan \frac{\pi}{6} \cot \frac{\pi}{6} + 1$

3. If $\sin \frac{\pi}{4}$ and $\cos \frac{\pi}{4}$ equal to $\frac{1}{\sqrt{2}}$ each, then find the value of the followings:

- (i) $2 \sin 45^\circ - 2 \cos 45^\circ$ (ii) $3 \cos 45^\circ + 4 \sin 45^\circ$
 (iii) $5 \cos 45^\circ - 3 \sin 45^\circ$

6.5 Solution of a Triangle

We know that there are three sides and three angles in a triangle. Out of these six elements, if we know three of them including at least one side, then we can find the

measures of the remaining elements. Finding the measures of the remaining elements is called the solution of a triangle. Here we learn the solution of a right angled triangle only.

Case I: When measures of one side and one angle are given.

Example 15: Solve triangle ABC , in which $m\angle B = 90^\circ$, $m\angle A = 30^\circ$, $a = 2$ cm

Solution:

We are required to find b , c and $m\angle C$.

$$\begin{aligned}\text{Now } m\angle C &= m\angle B - m\angle A \\ &= 90^\circ - 30^\circ \\ &= 60^\circ \quad \dots(i)\end{aligned}$$

$$\frac{a}{b} = \sin 30^\circ$$

$$\Rightarrow \frac{2}{b} = \sin 30^\circ \quad (\because a = 2)$$

$$\Rightarrow \frac{2}{b} = \frac{1}{2} \quad \left(\because \sin 30^\circ = \frac{1}{2} \right)$$

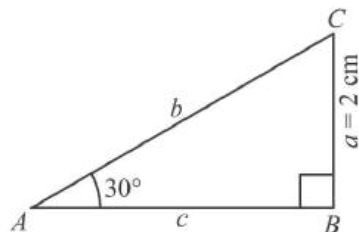
$$\Rightarrow b = 4 \text{ cm} \quad \dots(ii)$$

$$\text{and } \frac{a}{c} = \tan 30^\circ$$

$$\Rightarrow \frac{2}{c} = \frac{1}{\sqrt{3}} \quad \left(\because a = 2, \tan 30^\circ = \frac{1}{\sqrt{3}} \right)$$

$$\text{thus } c = 2\sqrt{3} \text{ cm} \quad \dots(iii)$$

(i), (ii) and (iii) are the required results.



Case II: When measure of the hypotenuse and an angle are given.

Example 16: Solve triangle ABC , when $m\angle A = 60^\circ$, $b = 5$ cm,

$$m\angle B = 90^\circ$$

Solution: We are required to find a , c and $m\angle C$

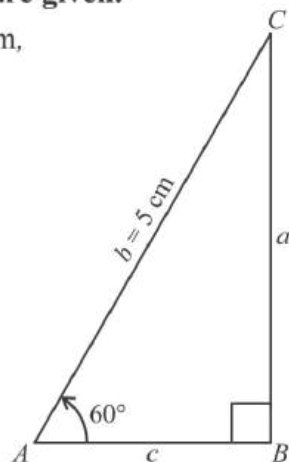
$$m\angle A = 60^\circ$$

$$m\angle B = 90^\circ$$

$$m\angle C = m\angle B - m\angle A$$

$$= 90^\circ - 60^\circ$$

$$= 30^\circ \quad \dots(i)$$



Now $\frac{a}{b} = \sin 60^\circ$

$$\frac{a}{5} = \frac{\sqrt{3}}{2} \quad \left(\because b = 5, \sin 60^\circ = \frac{\sqrt{3}}{2} \right)$$

$$\Rightarrow a = \frac{5\sqrt{3}}{2}$$

$$\Rightarrow a = 4.33 \text{ cm} \quad \dots(\text{ii})$$

and $\frac{c}{b} = \cos 60^\circ$

$$\frac{c}{5} = \frac{1}{2} \quad \left(\because b = 5, \cos 60^\circ = \frac{1}{2} \right)$$

$$\Rightarrow c = \frac{5}{2}$$

$$\Rightarrow c = 2.5 \text{ cm} \quad \dots(\text{iii})$$

(i), (ii) and (iii) are the required results.

Case III: When measure of two sides are given.

Example 17: Solve triangle ABC , when $a = \sqrt{2}$ cm,
 $c = 1$ cm and $m\angle B = 90^\circ$

Solution: We are required to find b , $m\angle A$, $m\angle C$.

By Pythagoras theorem, we have

$$b^2 = c^2 + a^2$$

$$\text{or } b^2 = (1)^2 + (\sqrt{2})^2$$

$$\text{or } b^2 = 1 + 2$$

$$\text{or } b^2 = 3$$

$$\text{or } b = \sqrt{3} \text{ cm} \quad \dots(\text{i})$$

$$\text{Now } \sin m\angle A = \frac{a}{b} = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow m\angle A = \sin^{-1} \frac{\sqrt{2}}{\sqrt{3}} = 54.7^\circ$$

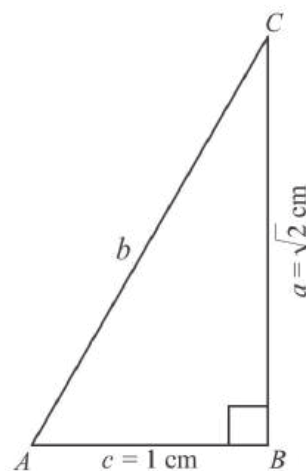
$$\Rightarrow m\angle A = 54.7^\circ \quad \dots(\text{ii})$$

and $m\angle C = m\angle B - m\angle A$

$$= 90^\circ - 54.7^\circ$$

$$= 35.3^\circ \quad \dots(\text{iii})$$

(i), (ii) and (iii) are the required results.



Case IV: When measure of one side and hypotenuse are given.

Example 18: Solve triangle ABC , when $a = 2\text{ cm}$, $b = 2\sqrt{2}\text{ cm}$ and $m\angle B = 90^\circ$

Solution: We are required to find $m\angle A$, $m\angle C$ and c .

By Pythagoras theorem, we have

$$b^2 = a^2 + c^2$$

$$\begin{aligned}\text{or } c^2 &= b^2 - a^2 \\ &= (2\sqrt{2})^2 - (2)^2 \\ &= 8 - 4 = 4\end{aligned}$$

$$\text{or } c = 2\text{ cm} \quad \dots(\text{i})$$

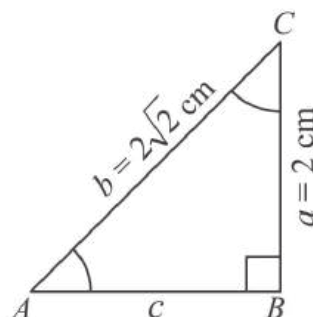
$$\text{Now } \frac{c}{b} = \cos m\angle A$$

$$\text{or } \frac{c}{b} = \cos m\angle A = \frac{1}{\sqrt{2}}$$

$$\Rightarrow m\angle A = 45^\circ \quad \dots(\text{ii})$$

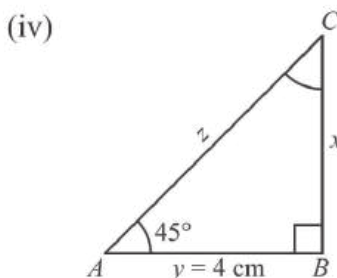
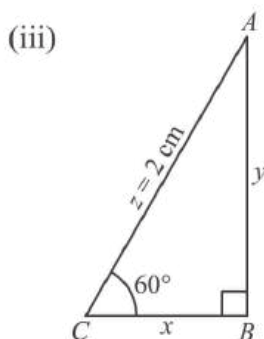
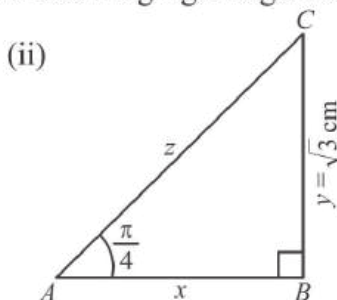
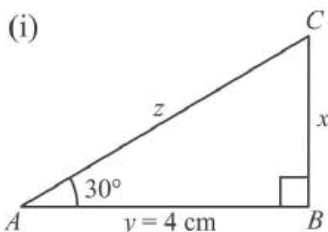
$$\begin{aligned}\text{Thus, } m\angle C &= m\angle B - m\angle A \\ &= 90^\circ - 45^\circ \\ &= 45^\circ \quad \dots(\text{iii})\end{aligned}$$

Hence (i), (ii) and (iii) are the required results.



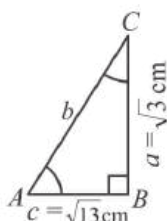
EXERCISE 6.5

1. Find the values of x , y and z from the following right angled triangles.

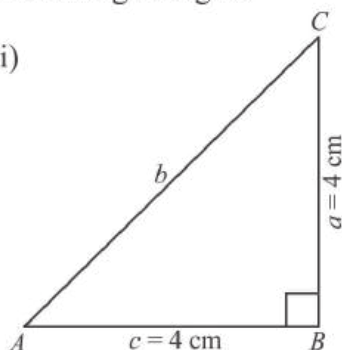


2. Find the unknown side and angles of the following triangles.

(i)



(ii)



3. Each side of a square field is 60 m long. Find the lengths of the diagonals of the field.

4. Solve the following triangles when $m\angle B = 90^\circ$:

(i) $m\angle C = 60^\circ$, $c = 3\sqrt{3}$ cm

(ii) $m\angle C = 45^\circ$, $a = 8$ cm

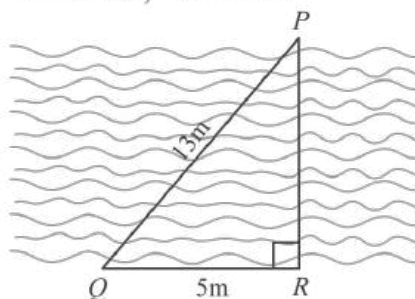
(iii) $a = 12$ cm, $c = 6$ cm

(iv) $m\angle A = 60^\circ$, $c = 4$ cm

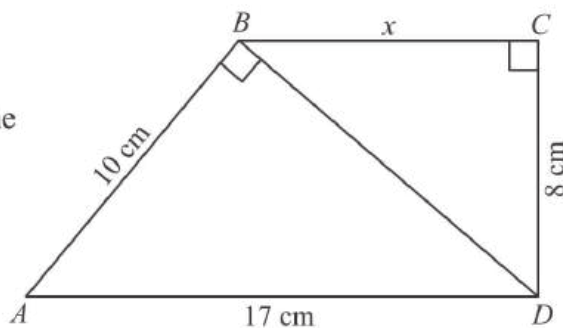
(v) $m\angle A = 30^\circ$, $c = 4$ cm

(vi) $b = 10$ cm, $a = 6$ cm

5. Let Q and R be the two points on the same bank of a canal. The point P is placed on the other bank straight to point R . Find the width of the canal and the angle PQR in radians.

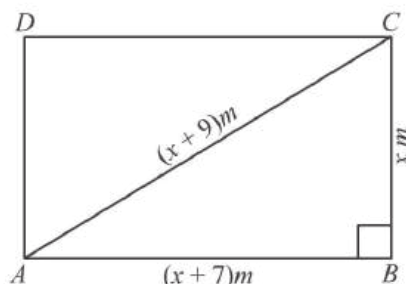


6. Calculate the length x in the adjoining figure.



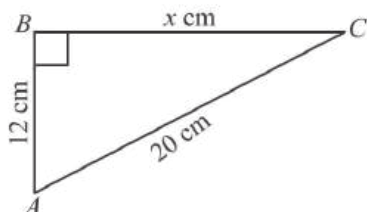
7. If the ladder is placed along the wall such that the foot of the ladder is 2 m away from the wall. If the length of the ladder is 8 m, find the height of the wall.

8. The diagonal of a rectangular field $ABCD$ is $(x + 9)m$ and the sides are $(x + 7)m$ and $x m$. Find the value of x .

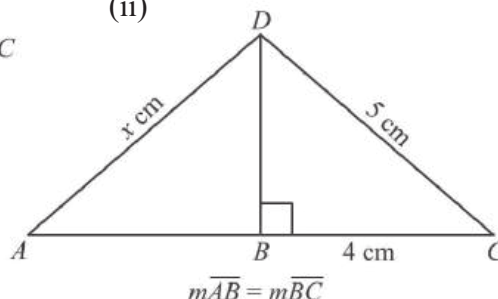


9. Calculate the value of 'x' in each case.

(i)

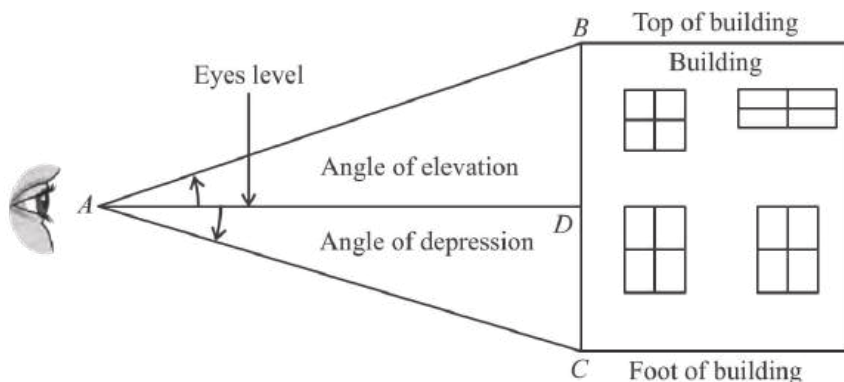


(ii)



6.6 The Angle of Elevation and the Angle of Depression

The angle between the horizontal line AD (eye level) and a line from the eye A to the top of building (B) is called an angle of elevation.



The angle between the horizontal line AD (eye level) and the line from the eye 'A' to the foot of the building (C) is called the angle of depression.

Example 19: The angle of elevation of the top of a pole 40 m high is 60° when seen from a point on the ground level. Find the distance of the point from the foot of the pole.

Solution: In the triangle ABC , we have

$$m\overline{BC} = 40 \text{ m}$$

$$m\angle A = 60^\circ$$

Let $m\overline{AB} = x$ (the point B is the foot of the pole BC)

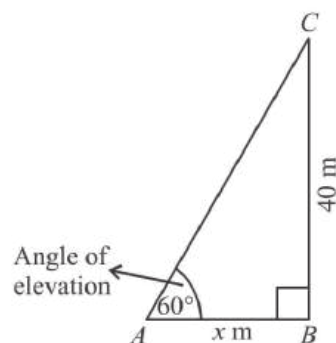
In right angled triangle ABC ,

$$\tan 60^\circ = \frac{m\overline{BC}}{m\overline{AB}}$$

$$\sqrt{3} = \frac{40}{x}$$

$$\Rightarrow x = \frac{40}{\sqrt{3}}$$

$$\Rightarrow x = 23.09 \text{ m}$$



Hence, distance of the point from the foot of the pole = 23.09 m

Example 20: From the top of a lookout tower, the angle of depression of a building has on the ground level of 45° . How far is a man on the ground from the tower, if the height of the tower is 30 m?

Solution: In the triangle ABC , AB is the tower and point C is the position of man. We have

$$m\overline{AB} = 30 \text{ m}$$

$$m\angle CAD = m\angle C = 45^\circ$$

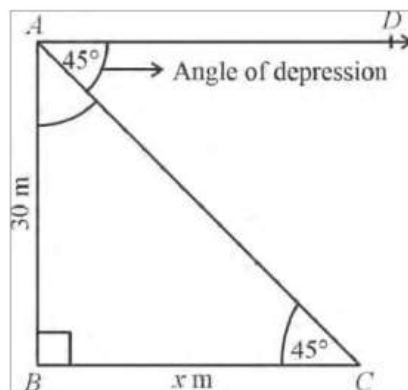
$$m\overline{BC} = x \text{ m} = ?$$

Let x be the base of right angled triangle ABC ,

$$\tan 45^\circ = \frac{m\overline{AB}}{m\overline{BC}}$$

$$\Rightarrow 1 = \frac{30}{x}$$

$$\Rightarrow x = 30 \text{ m}$$



Hence, man is 30 m far from the tower.

EXERCISE 6.6

1. The angle of elevation of the top of a flag post from a point on the ground level 40 m away from the flag post is 60° . Find the height of the post.
2. An isosceles triangle has a vertical angle of 120° and a base 10 cm long. Find the length of its altitude.
3. A tree is 72 m high. Find the angle of elevation of its top from a point 100 m away on the ground level.
4. A ladder makes an angle of 60° with the ground and reaches a height of 10m along the wall. Find the length of the ladder.
5. A light house tower is 150 m high from the sea level. The angle of depression from the top of the tower to a ship is 60° . Find the distance between the ship and the tower.
6. Measure of an angle of elevation of the top of a pole is 15° from a point on the ground, in walking 100 m towards the pole the measure of angle is found to be 30° . Find the height of the pole.
7. Find the measure of an angle of elevation of the Sun, if a tower 300 m high casts a shadow 450 m long.
8. Measure of angle of elevation of the top of a cliff is 25° , on walking 100 metres towards the cliff, measure of angle of elevation of the top is 45° . Find the height of the cliff.
9. From the top of a hill 300 m high, the measure of the angle of depression of a point on the nearer shore of the river is 70° and measure of the angle of depression of a point, directly across the river is 50° . Find the width of the river. How far is the river from the foot of the hill?
10. A kite has 120 m of string attached to it when at an angle of elevation of 50° . How far is it above the hand holding it? (Assume that the string is stretched.)

REVIEW EXERCISE 6

1. Four options are given against each statement. Encircle the correct one.
(i) The value of $\tan^{-1} 2$ in radians is:

(a) $\frac{\pi}{2}$

(b) $\frac{3\pi}{2}$

(c) 1.11π

(d) 1.11

- (ii) In a right triangle, the hypotenuse is 13 units and one of the angles is $\theta = 30^\circ$. The length of the opposite side is:

(a) 6.5 units (b) 7.5 units (c) 6 units (d) 5 units

- (iii) A person standing 50 m away from a building sees the top of the building at an angle of elevation of 45° . Height of the building is:

(a) 50 m (b) 25 m (c) 35 m (d) 70 m

- (iv) $\sec^2 \theta - \tan^2 \theta = \underline{\hspace{2cm}}$.

(a) $\sin^2 \theta$ (b) 1 (c) $\cos^2 \theta$ (d) $\cot^2 \theta$

- (v) If $\sin \theta = \frac{3}{5}$ and θ is an acute angle, $\cos^2 \theta = \underline{\hspace{2cm}}$.

(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) $\frac{16}{25}$ (d) $\frac{4}{25}$

- (vi) $\frac{5\pi}{24}$ rad = $\underline{\hspace{2cm}}$ degrees.

(a) 30° (b) 37.5° (c) 45° (d) 52.5°

- (vii) $292.5^\circ = \underline{\hspace{2cm}}$ rad.

(a) $\frac{17\pi}{6}$ (b) $\frac{17\pi}{4}$ (c) 1.6π (d) 1.625π

- (viii) Which of the following is a valid identity?

(a) $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$ (b) $\cos\left(\frac{\pi}{2} - \theta\right) = \cos \theta$
(c) $\cos\left(\frac{\pi}{2} - \theta\right) = \sec \theta$ (d) $\cos\left(\frac{\pi}{2} - \theta\right) = \operatorname{cosec} \theta$

- (ix) $\sin 60^\circ = \underline{\hspace{2cm}}$.

(a) 1 (b) $\frac{1}{2}$ (c) $\sqrt{(3)^2}$ (d) $\frac{\sqrt{3}}{2}$

(x) $\cos^2 100^\circ + \sin^2 100^\circ = \underline{\hspace{2cm}}$.

- (a) 1 (b) 2 (c) 3 (d) 4

2. Convert the given angles from:

(a) degrees to radians giving answer in terms of π .

- (i) 255° (ii) $75^\circ 45'$ (iii) 142.5°

(b) radians to degrees giving answer in degrees and minutes.

- (i) $\frac{17\pi}{24}$ (ii) $\frac{7\pi}{12}$ (iii) $\frac{11\pi}{16}$

3. Prove the following trigonometric identities:

(i) $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$

(ii) $\sin \theta (\operatorname{cosec} \theta - \sin \theta) = \frac{1}{\sec^2 \theta}$

(iii) $\frac{\operatorname{cosec} \theta - \sec \theta}{\operatorname{cosec} \theta + \sec \theta} = \frac{1 - \tan \theta}{1 + \tan \theta}$

(iv) $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$

(v) $\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} + \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \frac{2}{1 - 2 \sin^2 \theta}$

(vi) $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2$

4. If $\tan \theta = \frac{3}{\sqrt{2}}$ then find the remaining trigonometric ratios when θ lies in first quadrant.

5. From a point on the ground, the angle of elevation to the top of a 30 m high building is 28° . How far is the point from the base of the building?

6. A ladder leaning against a wall forms an angle of 65° with the ground. If the ladder is 10 m long, how high does it reach on the wall?

Unit 7

Coordinate Geometry

Students' Learning Outcomes

At the end of the unit, the students will be able to:

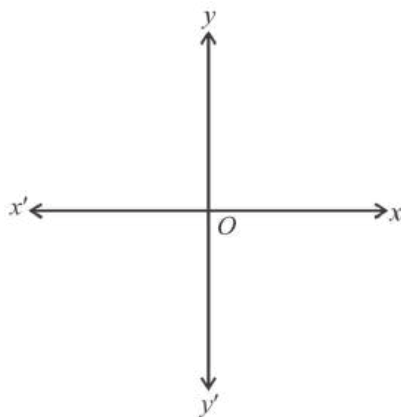
- Derive distance formula by locating the position of two points in coordinate plane.
- Calculate the midpoint of a line segment.
- Find the gradient of a straight line when coordinates of two points are given.
- Find the equation of a straight line in the form $y = mx + c$.
- Find the gradient of parallel and perpendicular lines.
- Apply distance and midpoint formulas to solve real-life situations such as physical measurements or distances between locations.
- Apply concepts from coordinate geometry to real world problems (such as, aviation and navigation, landscaping, map reading, longitude and latitude).
- Derive equation of a straight line in:
 - slope- intercept form
 - two-point form
 - symmetric form
 - point-slope form
 - intercepts form
 - normal form.
- Show that a linear equation in two variables represents a straight line and reduce the general form of the equation of a straight line to the other standard forms.

INTRODUCTION

Geometry is one of the most ancient branches of mathematics. The Greeks systematically studied it about four centuries B.C. Most of the geometry taught in schools is due to Euclid who expounded thirteen books on the subject (300 B.C.). A French philosopher and mathematician Rene Descartes (1596-1650 A.D.) introduced algebraic methods in geometry which gave birth to analytic geometry (or coordinate geometry). Our aim is to present fundamentals of the subject in this book.

7.1 Coordinate Plane

Draw in a plane two mutually perpendicular number lines $x'x$ and $y'y$, one horizontal and the other vertical. Let O be their point of intersection, called origin and the real number 0 of both the lines is represented by O . The two lines are called **coordinate axes**. The horizontal line $x'Ox$ is called the **x-axis** and the vertical line $y'Oy$ is called the **y-axis**.



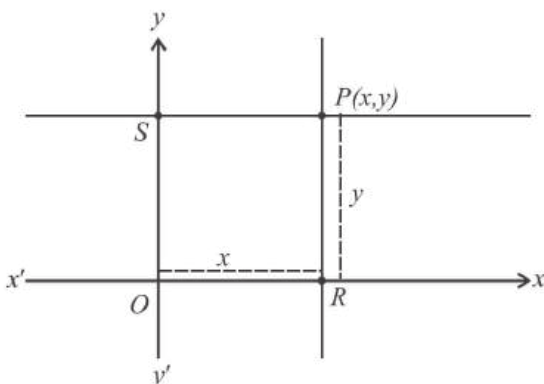
Important information:

The Cartesian coordinate system or the rectangular coordinate system was invented by French mathematician René Descartes, when he tried to describe the path of a fly crawling along criss-cross beams on the ceiling while he lay on his bed. The Cartesian coordinate system created a link between algebra and geometry. Geometric shapes could now be described algebraically using the coordinates of the points that make up the shapes.

The points lying on Ox are $+ve$ and on Ox' are $-ve$.

The points lying on Oy are $+ve$ and Oy' are $-ve$.

Suppose P is any point in the plane. Then P can be located by using an ordered pair of real numbers. Through P draw lines parallel to the coordinates axes meeting x -axis at R and y -axis at S .



Let the directed distance $\overline{OR} = x$ and the directed distance $\overline{OS} = y$.

The ordered pair (x, y) gives us enough information to locate the point P . Thus, P has coordinates (x, y) . The first component of the ordered pair (x, y) is called x -coordinate or **abscissa** and the second component is called y -coordinate or **ordinate** of P . The reverse of this technique also provides a method for associating exactly one point in the plane with any ordered pair (x, y) of real numbers. This method of pairing off in a one-to-one fashion the points in a plane with ordered pairs of real numbers is called the two dimensional rectangular (or Cartesian) coordinate system.

The coordinate axes divide the plane into four equal parts called quadrants. They are defined as follows:

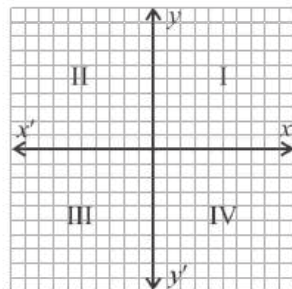
Quadrant I: All points (x, y) with $x > 0, y > 0$

Quadrant II: All points (x, y) with $x < 0, y > 0$

Quadrant III: All points (x, y) with $x < 0, y < 0$

Quadrant IV: All points (x, y) with $x > 0, y < 0$

The point P in the plane that corresponds to an ordered pair (x, y) is called the graph.



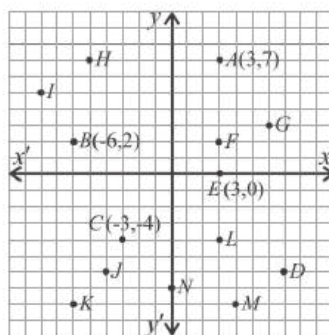
Thus, given a set of ordered pairs of real numbers, the graph of the set is the aggregate of all points in the plane that correspond to ordered pairs of the set.

Need to know!

The points on x -axis are of the form $(a, 0)$ and the points on y -axis are of the form $(0, b)$.

Challenges!

- Write down the coordinates of the points if not mentioned in the adjacent figure.
- Locate $(0, -1)$, $(2, 2)$, $(-4, 7)$ and $(-3, -3)$



7.1.1 The Distance Formula

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two points in the plane. To find the distance $d = |\overline{AB}|$, we draw a horizontal line from A to a point C lies directly below B , forming a right triangle ABC .

Note:

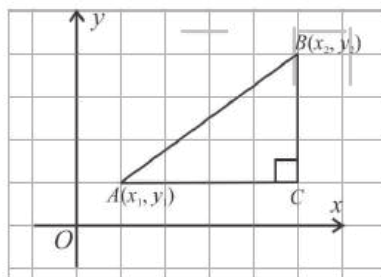
$|\overline{AB}|$ stands for $m\overline{AB}$

So that $|\overline{AC}| = |x_2 - x_1|$ and $|\overline{BC}| = |y_2 - y_1|$

By using Pythagoras Theorem, we have

$$\begin{aligned} d^2 &= |\overline{AB}|^2 = |\overline{AC}|^2 + |\overline{BC}|^2 \\ &= (x_2 - x_1)^2 + (y_2 - y_1)^2 \end{aligned}$$

or $d = |\overline{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \dots(i)$



The distance is always taken to be non-negative. It is not a directed distance from A to B .

If A and B lie on a line parallel to one of the coordinate axes, then by the formula(i), the distance $|\overline{AB}|$ is absolute value of the directed distance $\overline{AB}$.

The formula (i) shows that any of the two points can be taken as first point.

Example 1: Find the distance between the points:

- $A(5, 6)$, $B(5, -2)$
- $C(-4, -2)$, $D(0, 9)$

Solution: By the distance formula, we have

$$\begin{aligned} \text{(i)} \quad d &= |\overline{AB}| = \sqrt{(5-5)^2 + (-2-6)^2} & \text{(ii)} \quad d &= |\overline{CD}| = \sqrt{(0-(-4))^2 + (9-(-2))^2} \\ d &= |\overline{AB}| = \sqrt{(0)^2 + (-8)^2} & d &= |\overline{CD}| = \sqrt{(0+4)^2 + (9+2)^2} \\ d &= |\overline{AB}| = \sqrt{0+64} = 8 & d &= |\overline{CD}| = \sqrt{4^2 + 11^2} \\ & & d &= |\overline{CD}| = \sqrt{16+121} = \sqrt{137} \end{aligned}$$

Example 2: Show that the points $A(-1, 2)$, $B(7, 5)$ and $C(2, -6)$ are vertices of a right triangle.

Solution: Let a , b and c denote the lengths of the sides BC , CA and AB respectively.

By using the distance formula, we have

$$c = |AB| = \sqrt{(7 - (-1))^2 + (5 - 2)^2} = \sqrt{73}$$

$$a = |BC| = \sqrt{(2 - 7)^2 + (-6 - 5)^2} = \sqrt{146}$$

$$b = |CA| = \sqrt{(2 - (-1))^2 + (-6 - 2)^2} = \sqrt{73}$$

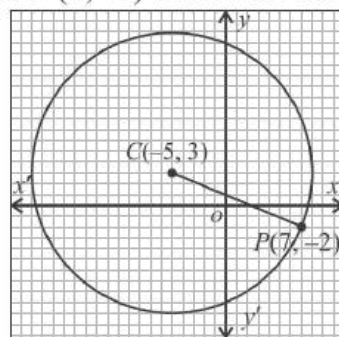
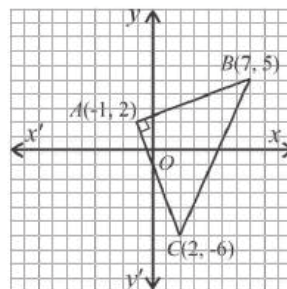
Clearly: $a^2 = b^2 + c^2$

Therefore, ABC is a right triangle with right angle at A .

Example 3: The point $C(-5, 3)$ is the centre of a circle and $P(7, -2)$ lies on the circle. What is the radius of the circle?

Solution: The radius of the circle is the distance from the points C to P . By the using distance formula, we have

$$\begin{aligned} \text{Radius} &= |CP| = \sqrt{(7 - (-5))^2 + (-2 - 3)^2} \\ &= \sqrt{144 + 25} = \sqrt{169} \\ &= 13 \text{ units} \end{aligned}$$



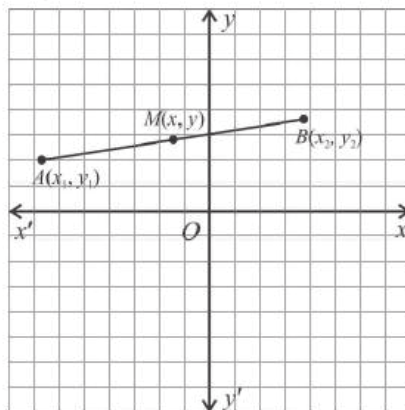
7.1.2 Mid Point Formula

The midpoint formula is used in geometry to find central point between two given points in a coordinate plane. This formula is particularly useful when you need to divide a line segment into two equal halves or parts.

Derivation of the Midpoint Formula

Consider two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on a two-dimensional plane. The line segment joining these two points has a midpoint $M(x, y)$, where x and y are the coordinates of the midpoint.

To derive the formula for $M(x, y)$ we need to find the average of the x -coordinates and y -coordinates of points A and B separately.



1. x-Coordinate of the Midpoint

The x-coordinate of the midpoint is the average of the x-coordinates of points A and B .

$$\text{i.e., } x = \frac{x_1 + x_2}{2}$$

2. y-Coordinate of the Midpoint

Similarly, the y-coordinate of the midpoint is the average of the y-coordinates of points A and B .

$$\text{i.e., } y = \frac{y_1 + y_2}{2}$$

Thus, the coordinates of the midpoint $M(x, y)$ are:

$$M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Example 4: Find the midpoint of the line segment joining the points $A(2, 3)$ and $B(8, 7)$.

Solution: Using the midpoint formula:

$$M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Substitute $x_1 = 2, y_1 = 3, x_2 = 8$ and $y_2 = 7$, into the midpoint formula

$$M(x, y) = \left(\frac{2 + 8}{2}, \frac{3 + 7}{2} \right)$$

$$M(x, y) = \left(\frac{10}{2}, \frac{10}{2} \right) = (5, 5)$$

EXERCISE 7.1

- Describe the location in the plane of the point $P(x, y)$, for which
 - $x > 0$
 - $x > 0$ and $y > 0$
 - $x = 0$
 - $y = 0$
 - $x > 0$ and $y \leq 0$
 - $y = 0, x = 0$
 - $x = y$
 - $x \geq 3$
 - $y > 0$
 - x and y have opposite signs.
- Find the distance between the points:
 - $A(6, 7), B(0, -2)$
 - $C(-5, -2), D(3, 2)$
 - $L(0, 3), M(-2, -4)$
 - $P(-8, -7), Q(0, 0)$
- Find in each of the following:
 - The distance between the two given points

(ii) Midpoint of the line segment joining the two points:

(a) $A(3, 1)$, $B(-2, -4)$ (b) $A(-8, 3)$, $B(2, -1)$

(c) $A\left(-\sqrt{5}, -\frac{1}{3}\right)$, $B(-3\sqrt{5}, 5)$

4. Which of the following points are at a distance of 15 units from the origin?

(i) $(\sqrt{176}, 7)$ (ii) $(10, -10)$ (iii) $(1, 15)$

5. Show that:

(i) the points $A(0, 2)$, $B(\sqrt{3}, 1)$ and $C(0, -2)$ are vertices of a right triangle.

(ii) the points $A(3, 1)$, $B(-2, -3)$ and $C(2, 2)$ are vertices of an isosceles triangle.

(iii) the points $A(5, 2)$, $B(-2, 3)$, $C(-3, -4)$ and $D(4, -5)$ are vertices of a parallelogram.

6. Find h such that the points $A(\sqrt{3}, -1)$, $B(0, 2)$ and $C(h, -2)$ are vertices of a right triangle with right angle at the vertex A .

7. Find h such that $A(-1, h)$, $B(3, 2)$ and $C(7, 3)$ are collinear.

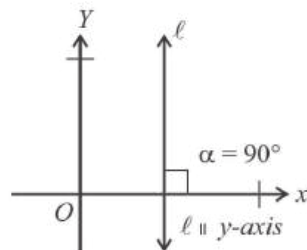
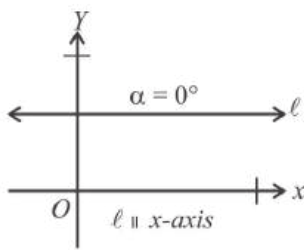
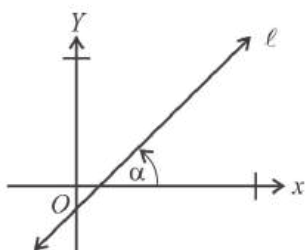
8. The points $A(-5, -2)$ and $B(5, -4)$ are ends of a diameter of a circle. Find the centre and radius of the circle.

9. Find h such that the points $A(h, 1)$, $B(2, 7)$ and $C(-6, -7)$ are vertices of a right triangle with right angle at the vertex A .

10. A quadrilateral has the points $A(9, 3)$, $B(-7, 7)$, $C(-3, -7)$ and $D(5, -5)$ as its vertices. Find the midpoints of its sides. Show that the figure formed by joining the midpoints consecutively is a parallelogram.

7.2 Equations of Straight Lines

Inclination of a Line: The angle α ($0^\circ < \alpha < 180^\circ$) is measured counterclockwise from positive x -axis to a non-horizontal straight line ℓ is called the inclination of ℓ .



Observe that the angle α in the different positions of the line ℓ is α , 0° and 90° respectively.

Slope or Gradient of a Line

When we walk on an inclined plane, we cover horizontal distance (**run**) as well as vertical distance (**rise**) at the same time.

It is harder to climb a steeper inclined plane. The measure of steepness (ratio of rise to the run) is termed as slope or gradient of the inclined path and is denoted by m .

$$m = \frac{\text{rise}}{\text{run}} = \frac{y}{x} = \tan \alpha$$

In analytical geometry, **slope or gradient** m of a non-vertical straight line with its inclination is defined by: $m = \tan \alpha$.

If ℓ is horizontal, its slope is zero and if ℓ is vertical then its slope is undefined.

If $0^\circ < \alpha < 90^\circ$, m is positive and if $90^\circ < \alpha < 180^\circ$, then m is negative.

7.2.1 Slope or Gradient of a Straight Line Joining Two Points

Theorem 1: If a non-vertical line ℓ with inclination α passes through two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, then the slope or gradient m of ℓ is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \alpha$$

Proof: Let m be the slope of the line ℓ .

Draw perpendiculars $\overline{PM}$ and $\overline{QM'}$ on x -axis and a perpendicular $\overline{PR}$ on $\overline{QM'}$.

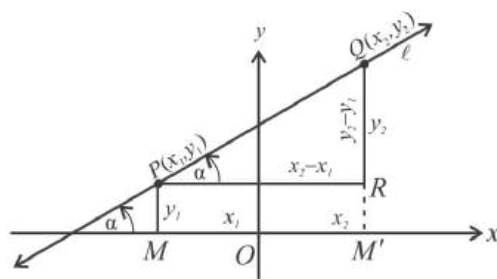
Then $m\angle RPQ = \alpha$, $m\overline{PR} = x_2 - x_1$ and $m\overline{QR} = y_2 - y_1$

The slope or gradient of ℓ is defined as:

$$m = \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

Note:

- (i) If ℓ is parallel to x -axis, then $\alpha = 0^\circ$
- (ii) If ℓ is parallel to y -axis, then $\alpha = 90^\circ$



Why are slopes important?

The concept of slope is widely used in engineering, architecture, and even sports like skiing, where understanding the steepness of a hill or ramp is essential.

Case (i). When $0 < \alpha < \frac{\pi}{2}$

In the right triangle PRQ , we have

$$m = \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

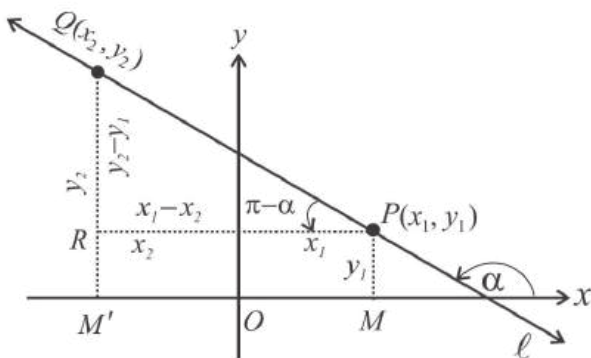
Case (ii). When $\frac{\pi}{2} < \alpha < \pi$

In the right triangle PRQ ,

$$\tan (\pi - \alpha) = \frac{y_2 - y_1}{x_1 - x_2}$$

$$\text{or } -\tan \alpha = \frac{y_2 - y_1}{x_1 - x_2}$$

$$\text{or } \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or } m = \frac{y_2 - y_1}{x_2 - x_1}$$



Thus if $P(x_1, y_1)$ and $Q(x_2, y_2)$ are two points on a line, then slope of $\overleftrightarrow{PQ}$ is given by:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

- Note:**
- (i) $m \neq \frac{y_2 - y_1}{x_1 - x_2}$ and $m \neq \frac{y_1 - y_2}{x_2 - x_1}$
 - (ii) ℓ is horizontal iff $m = 0$ ($\because \alpha = 0^\circ$)
 - (iii) ℓ is vertical iff m is not defined ($\because \alpha = 90^\circ$)
 - (iv) If slope of $\overline{AB}$ = slope of $\overline{BC}$, then the points A, B and C are collinear.

Theorem 2: The two lines ℓ_1 and ℓ_2 with slopes m_1 and m_2 respectively are:

- (i) parallel iff $m_1 = m_2$
- (ii) perpendicular iff $m_1 = \frac{-1}{m_2}$
or $m_1 m_2 = -1$

Remember !

The symbol:

- (i) $\parallel$ stands for “parallel”.
- (ii) $\nparallel$ stands for “not parallel”.
- (iii) $\perp$ stands for “perpendicular”

Example 5: Show that the points $A(-3, 6)$, $B(3, 2)$ and $C(6, 0)$ are collinear.

Solution: We know that the points A, B and C are collinear if the line AB and BC have the same slopes.

$$\text{Here slope of } \overline{AB} = \frac{2-6}{3-(-3)} = \frac{-4}{3+3} = \frac{-4}{6} = \frac{-2}{3} \quad \text{and slope of } \overline{BC} = \frac{0-2}{6-3} = \frac{-2}{3}$$

$$\therefore \text{ Slope of } AB = \text{Slope of } BC$$

Thus A, B and C are collinear.

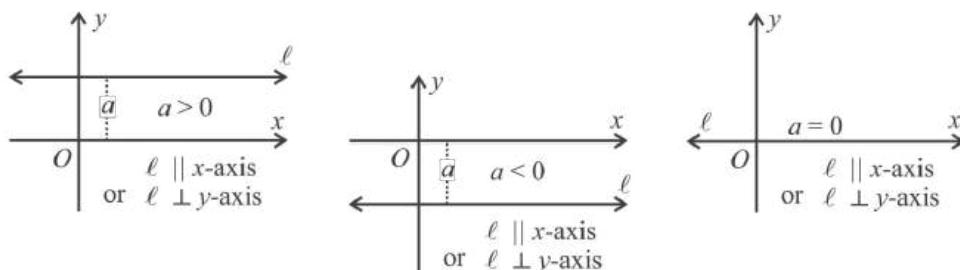
Example 6: Show that the triangle with vertices $A(1, 1)$, $B(4, 5)$ and $C(12, -1)$ is a right triangle.

Solution: Slope of $\overline{AB} = m_1 = \frac{5-1}{4-1} = \frac{4}{3}$ and slope of $\overline{BC} = m_2 = \frac{-1-5}{12-4} = \frac{-6}{8} = \frac{-3}{4}$

Since $m_1 \cdot m_2 = \left(\frac{4}{3}\right)\left(-\frac{3}{4}\right) = -1$, therefore, $\overline{AB} \perp \overline{BC}$

So $\triangle ABC$ is a right triangle.

7.2.2 Equation of a Straight Line Parallel to the x -axis (or perpendicular to the y -axis)



All the points on the line ℓ parallel to x -axis remain at a constant distance (say a) from x -axis. Therefore, each point on the line has its distance from x -axis equal to a , which is its y -coordinate (ordinate). So, all the points on this line satisfy the equation: $y = a$

Note:

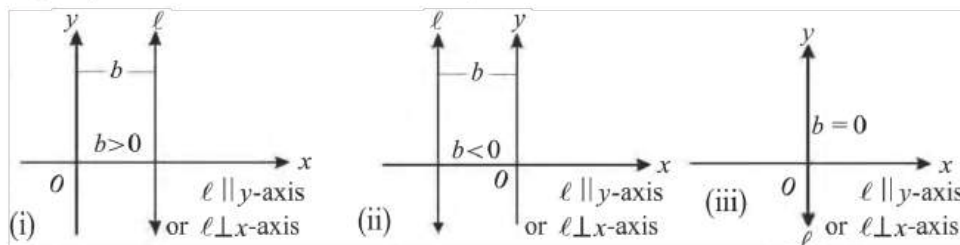
If $a > 0$, then the line ℓ is above the x -axis.

If $a < 0$, then the line ℓ is below the x -axis.

If $a = 0$, then the line ℓ becomes the x -axis.

Thus the equation of x -axis is $y = 0$

7.2.3 Equation of a straight Line Parallel to the y -axis (or perpendicular to the x -axis)

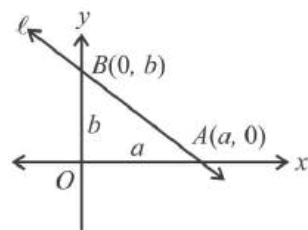


All the points on the line ℓ parallel to y -axis remain at a constant distance (say b) from y -axis. Each point on the line has its distance from y -axis equal to b , which is its x -coordinate (abscissa). So, all the points on this line satisfy the equation: $x = b$

7.2.4 Standard Forms of Equation of Straight Line:

Intercepts of a line

- If a line intersects x -axis at $(a, 0)$, then a is called **x -intercept** of the line.
- If a line intersects y -axis at $(0, b)$, then b is called **y -intercept** of the line.

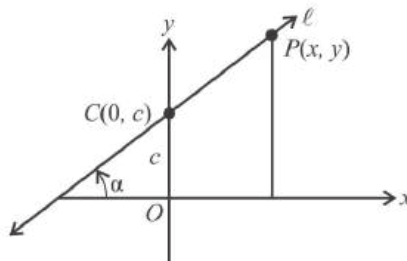


1. Slope-Intercept form of Equation of a Straight Line

Theorem 3: Equation of a non-vertical straight line with slope m and y -intercept c is given by:

$$y = mx + c$$

Proof: Let $P(x, y)$ be an arbitrary point of the straight line ℓ with slope m and y -intercept c . As $C(0, c)$ and $P(x, y)$ lie on the line, so the slope of the line is:



$$m = \frac{y-c}{x-0} \quad \text{or} \quad y-c = mx \quad \text{or} \quad y = mx + c \quad \text{is an equation of } \ell.$$

The equation of the line for which $c = 0$ is $y = mx$

In this case the line passes through the origin.

Example 7: Find an equation of the straight line if

- its slope is 2 and y -intercept is 5
- it is perpendicular to a line with slope -6 and its y -intercept is $\frac{4}{3}$

Solution: (a) The slope and y -intercept of the line are respectively:
 $m = 2$ and $c = 5$

Thus $y = 2x + 5$ (Slope-intercept form: $y = mx + c$) is the required equation.

- The slope of the given line is
 $m_1 = -6$

$\therefore$ The slope of the required line is: $m_2 = -\frac{1}{m_1} = \frac{1}{6}$

The slope and y -intercept of the required line are respectively:

$$m_2 = \frac{1}{6} \quad \text{and} \quad c = \frac{4}{3}$$

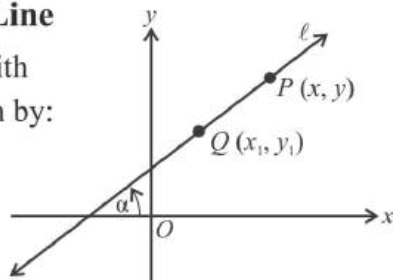
Thus, $y = \frac{1}{6}x + \frac{4}{3}$ or $6y = x + 8$ is the required equation.

2. Point-slope Form of Equation of a Straight Line

Theorem 4: Equation of a non-vertical straight line ℓ with slope m and passing through a point $Q(x_1, y_1)$ is given by:

$$y - y_1 = m(x - x_1)$$

Proof: Let $P(x, y)$ be an arbitrary point of the straight line with slope m and passing through $Q(x_1, y_1)$.



As $Q(x_1, y_1)$ and $P(x, y)$ both lie on the line, so the slope of the line is:

$$m = \frac{y - y_1}{x - x_1} \quad \text{or} \quad y - y_1 = m(x - x_1)$$

which is an equation of the straight line passing through $Q(x_1, y_1)$ with slope m .

3. Symmetric Form of Equation of a Straight Line

We have $m = \frac{y - y_1}{x - x_1} = \tan \alpha$, where α is the inclination of the line.

$$\frac{y - y_1}{x - x_1} = \frac{\sin \alpha}{\cos \alpha} \quad \text{or} \quad \boxed{\frac{x - x_1}{\cos \alpha} = \frac{y - y_1}{\sin \alpha} = r(\text{say})}$$

This is called **symmetric** form of equation of the line.

Example 8: Write down an equation of the straight line passing through $(5, 1)$ and parallel to a line passing through the points $(0, -1)$, $(7, -15)$.

Solution: Let m be the slope of the required straight line, then

$$\begin{aligned} m &= \frac{-15 - (-1)}{7 - 0} \quad (\because \text{Slopes of parallel lines are equal}) \\ &= -2 \end{aligned}$$

As the point $(5, 1)$ lies on the required line having slope -2 so, by point-slope form of equation of the straight line, we have

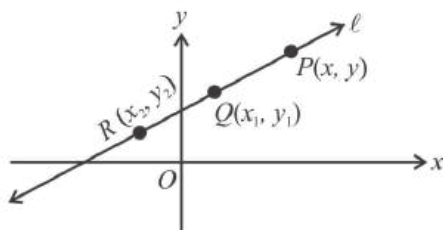
$$\begin{aligned} y - 1 &= -2(x - 5) \\ \text{or} \quad y &= -2x + 11 \\ \text{or} \quad 2x + y - 11 &= 0 \end{aligned}$$

is an equation of the required line.

4. Two-point Form of Equation of a Straight Line

Theorem 5: Equation of a non-vertical straight line passing through two points $Q(x_1, y_1)$ and

$$R(x_2, y_2) \text{ is: } \boxed{\begin{aligned} y - y_1 &= \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \\ \text{or} \\ y - y_2 &= \frac{y_2 - y_1}{x_2 - x_1} (x - x_2) \end{aligned}}$$



Proof: Let $P(x, y)$ be an arbitrary point of the line passing through $Q(x_1, y_1)$ and $R(x_2, y_2)$.

So, $\frac{y - y_1}{x - x_1} = \frac{y - y_2}{x - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$ (P, Q and R are collinear points)

We take

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

or $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$, the required equation of the line PQ .

or $(y_2 - y_1)x - (x_2 - x_1)y + (x_1y_2 - x_2y_1) = 0$

We may write this equation in determinant form as: $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

$y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_2)$ can be derived similarly.

Example 9: Find an equation of line through the points $(-2, 1)$ and $(6, -4)$.

Solution: Using two-points form of the equation of straight line, the required equation is:

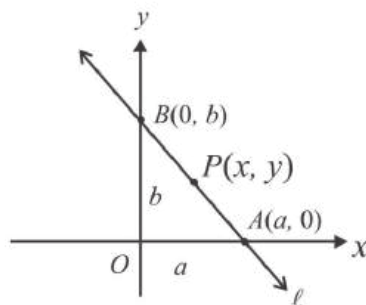
$$y - 1 = \frac{-4 - 1}{6 - (-2)} [x - (-2)] \quad \text{or} \quad y - 1 = \frac{-5}{8} (x + 2) \quad \text{or} \quad 5x + 8y + 2 = 0$$

5. Intercept Form of Equation of a Straight Line

Theorem 6: Equation of a line whose non-zero x and y -intercepts are a and b respectively is:

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

Proof: Let $P(x, y)$ be an arbitrary point of the line whose non-zero x and y -intercepts are a and b respectively. Obviously, the points $A(a, 0)$ and $B(0, b)$ lie on the



required line. So, by the two-point form of the equation of line, we have

$$\begin{aligned} y - 0 &= \frac{b-0}{0-a}(x-a) && (P, A \text{ and } B \text{ are collinear}) \\ \text{or} \quad -ay &= b(x-a) && \text{or} \quad bx + ay = ab \\ \text{or} \quad \frac{x}{a} + \frac{y}{b} &= 1 && (\text{dividing by } ab) \end{aligned}$$

Hence the result.

Example 10: Write down an equation of the line which cuts the x -axis at $(2, 0)$ and y -axis at $(0, -4)$.

Solution: As 2 and -4 are respectively x and y -intercepts of the required line, so by two-intercepts form of equation of a straight line, we have

$$\frac{x}{2} + \frac{y}{-4} = 1 \quad \text{or} \quad 2x - y - 4 = 0$$

Which is the required equation.

Example 11: Find an equation of the line through the point $P(2, 3)$ which forms an isosceles triangle with the coordinate axes in the first quadrant.

Solution: Let OAB be an isosceles triangle so that the line AB passes through $A(a, 0)$ and $B(0, a)$, where a is some positive real number.

Slope of $\overline{AB} = \frac{a-0}{0-a} = -1$. But $\overline{AB}$ passes through $P(2, 3)$.

Equation of the line through $P(2, 3)$ with slope -1 is

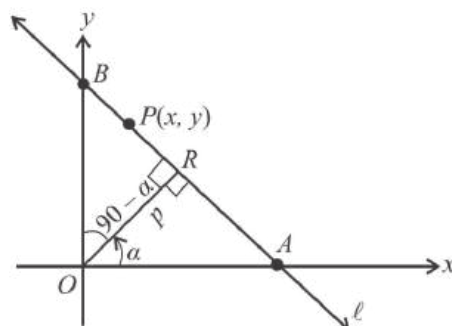
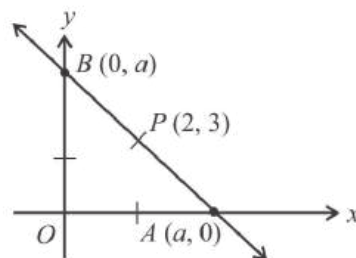
$$y - 3 = -1(x - 2) \quad \text{or} \quad x + y - 5 = 0$$

6. Normal Form of Equation of a Straight Line

Theorem 7: An equation of a non-vertical straight-line ℓ , such that length of the perpendicular from the origin to ℓ is p and α is the inclination of this perpendicular, is

$$\boxed{x \cos \alpha + y \sin \alpha = p}$$

Proof: Let the line ℓ meet the x -axis and y -axis at the points A and B respectively. Let $P(x, y)$ be an arbitrary point of line AB and let $\overline{OR}$ be perpendicular to the line ℓ . Then $|\overline{OR}| = p$



From the right triangles ORA and ORB , we have

$$\cos \alpha = \frac{p}{|OA|} \quad \text{or} \quad |OA| = \frac{p}{\cos \alpha}$$

$$\text{and } \cos(90^\circ - \alpha) = \frac{p}{|OB|} \quad \text{or} \quad |OB| = \frac{p}{\sin \alpha}$$

$$[\because \cos(90^\circ - \alpha) = \sin \alpha]$$

As $|OA|$ and $|OB|$ are the x and y -intercepts of the line AB , so equation of line AB is:

$$\frac{x}{p/\cos \alpha} + \frac{y}{p/\sin \alpha} = 1 \quad (\text{Two-intercept form})$$

That is $x \cos \alpha + y \sin \alpha = p$ is the required equation.

Example 12: The length of perpendicular from the origin to a line is 5 units and the inclination of this perpendicular is 120° . Find the slope and y -intercept of the line.

Solution. Here $p = 5$, $\alpha = 120^\circ$.

Equation of the line in normal form is

$$x \cos 120^\circ + y \sin 120^\circ = 5$$

$$\Rightarrow -\frac{1}{2}x + \frac{\sqrt{3}}{2}y = 5$$

$$\Rightarrow x - \sqrt{3}y + 10 = 0 \quad \dots(i)$$

To find the slope of the line, we re-write (i) as: $y = \frac{x}{\sqrt{3}} + \frac{10}{\sqrt{3}}$

which is slope-intercept form of the equation.

Here $m = \frac{1}{\sqrt{3}}$ and $c = \frac{10}{\sqrt{3}}$

7.2.5 A Linear Equation in two Variables Represents a Straight Line

Theorem 8:

A linear equation in two variables x and y is:

$$ax + by + c = 0 \quad \dots(i)$$

where a , b and c are constants and a and b are not simultaneously zero.

Proof: Here a and b cannot be both zero. So the following cases arise:

Remember!

The equation (i) represents a straight line and is called the **general equation of a straight line**.

Case I: $a \neq 0, b = 0$

In this case equation (i) takes the form:

$$ax + c = 0 \text{ or } x = -\frac{c}{a}$$

which is an equation of the straight line parallel to the y -axis at a directed distance $-\frac{c}{a}$ from the y -axis.

Case II: $a = 0, b \neq 0$

In this case equation (i) takes the form:

$$by + c = 0 \text{ or } y = -\frac{c}{b}$$

which is an equation of the straight line parallel to x -axis at a directed distance $-\frac{c}{b}$ from the x -axis.

Case III: $a \neq 0, b \neq 0$

In this case equation (i) takes the form:

$$by = -ax - c \text{ or } y = \frac{-a}{b}x - \frac{c}{b} = mx + c$$

which is the slope-intercept form of the straight line with slope $\frac{-a}{b}$ and y -intercept $-\frac{c}{b}$.

Thus, the equation $ax + by + c = 0$, always represents a straight line.

7.2.6 Transform the General Linear Equation to Standard Forms

Let's transform the equation $ax + by + c = 0$ into the standard forms:

i. Slope-Intercept Form

$$\text{We have: } by = -ax - c \text{ or } y = \frac{-a}{b}x - \frac{c}{b} = mx + c_1 \quad \dots(i)$$

$$\text{where } m = \frac{-a}{b}, c_1 = \frac{-c}{b}$$

ii. Point - Slope Form

We note from (i) above that slope of the line $ax + by + c = 0$ is $\frac{-a}{b}$. A point on the

line is $\left(\frac{-c}{a}, 0\right)$.

$$\text{Equation of the line becomes } y - 0 = -\frac{a}{b}\left(x + \frac{c}{a}\right)$$

which is in the point-slope form.

iii. Symmetric Form

$$m = \tan \alpha = \frac{-a}{b}, \sin \alpha = \frac{a}{\pm \sqrt{a^2 + b^2}}, \cos \alpha = \frac{b}{\pm \sqrt{a^2 + b^2}}$$

A point on $ax + by + c = 0$ is $\left(\frac{-c}{a}, 0\right)$

Equation of the line symmetric form becomes

$$\frac{x - \left(\frac{-c}{a}\right)}{b / \pm \sqrt{a^2 + b^2}} = \frac{y - 0}{a / \pm \sqrt{a^2 + b^2}} = r(\text{say})$$

is the required transformed equation. Sign of the radical to be chosen properly.

iv. Two-Point Form

We choose two arbitrary points on $ax + by + c = 0$. Two such points are

$\left(\frac{-c}{a}, 0\right)$ and $\left(0, \frac{-c}{b}\right)$. Equation of the line through these points is:

$$\frac{y - 0}{0 + \frac{c}{b}} = \frac{x + \frac{c}{a}}{-\frac{c}{a} - 0} \quad \text{i.e., } y - 0 = \frac{-a}{b} \left(x + \frac{c}{a}\right)$$

v. Intercept Form

$$ax + by = -c \quad \text{or} \quad \frac{ax}{-c} + \frac{by}{-c} = 1 \quad \text{i.e.,} \quad \frac{x}{-c/a} + \frac{y}{-c/b} = 1$$

which is an equation in two intercepts form.

vi. Normal Form

The equation: $ax + by + c = 0$... (i)

can be written in the normal form as:

$$\frac{ax + by}{\pm \sqrt{a^2 + b^2}} = \frac{-c}{\pm \sqrt{a^2 + b^2}} \quad \dots (ii)$$

The sign of the radical to be such that the right hand side of (ii) is positive.

Proof. We know that an equation of a line in normal form is

$$x \cos \alpha + y \sin \alpha = p \quad \dots (iii)$$

If (i) and (iii) are identical, we must have

$$\frac{a}{\cos \alpha} = \frac{b}{\sin \alpha} = \frac{-c}{p}$$

$$\text{i.e., } \frac{p}{-c} = \frac{\cos \alpha}{a} = \frac{\sin \alpha}{b} = \frac{\sqrt{\cos^2 \alpha + \sin^2 \alpha}}{\pm \sqrt{a^2 + b^2}} = \frac{1}{\pm \sqrt{a^2 + b^2}}$$

$$\text{Hence, } \cos\alpha = \frac{a}{\pm\sqrt{a^2+b^2}}, \quad \sin\alpha = \frac{b}{\pm\sqrt{a^2+b^2}} \quad \text{and} \quad p = \frac{-c}{\pm\sqrt{a^2+b^2}}$$

Substituting for $\cos\alpha$, $\sin\alpha$ and p into (iii), we have

$$\frac{ax+by}{\pm\sqrt{a^2+b^2}} = \frac{-c}{\pm\sqrt{a^2+b^2}}$$

Thus (i) can be reduced to the form (ii) by dividing it by $\pm\sqrt{a^2+b^2}$. The sign of the radical to be chosen so that the right hand side of (ii) is positive.

Example 13: Transform the equation $5x - 12y + 39 = 0$ into

- | | |
|--------------------------|-------------------------|
| (i) Slope intercept form | (ii) Two-intercept form |
| (iii) Normal form | (iv) Point-slope form |
| (v) Two-point form | (vi) Symmetric form |

Solution:

(i) We have $12y = 5x + 39$ or $y = \frac{5}{12}x + \frac{39}{12}$, $m = \frac{5}{12}$, y -intercept $c = \frac{39}{12}$

(ii) $5x - 12y = -39$ or $\frac{5x}{-39} + \frac{12y}{39} = 1$ or $\frac{x}{-39/5} + \frac{y}{39/12} = 1$ is the required equation.

(iii) $5x - 12y = -39$. Divide both sides by $\pm\sqrt{5^2+12^2} = \pm 13$. Since R.H.S is to be positive, we have to take negative sign.

Hence $\frac{5x}{-13} + \frac{12y}{13} = 3$ is the normal form of the equation.

(iv) A point on the line is $\left(\frac{-39}{5}, 0\right)$ and its slope is $\frac{5}{12}$.

Equation of the line can be written as: $y - 0 = \frac{5}{12}\left(x + \frac{39}{5}\right)$

(v) Another point on the line is $\left(0, \frac{39}{12}\right)$. Line through $\left(\frac{-39}{5}, 0\right)$ and $\left(0, \frac{39}{12}\right)$ is

$$\frac{y-0}{0-\frac{39}{12}} = \frac{x+\frac{39}{5}}{\frac{-39}{5}-0}$$

- (vi) We have $\tan \alpha = \frac{5}{12} = m$, so $\sin \alpha = \frac{5}{13}$, $\cos \alpha = \frac{12}{13}$. A point of the line is $\left(\frac{-39}{5}, 0\right)$

Equation of the line in symmetric form is

$$\frac{x + \frac{39}{5}}{\frac{12}{13}} = \frac{y - 0}{\frac{5}{13}} = r \text{ (say)}$$

EXERCISE 7.2

- Find the slope and inclination of the line joining the points:
 - $(-2, 4); (5, 11)$
 - $(3, -2); (2, 7)$
 - $(4, 6); (4, 8)$
- By means of slopes, show that the following points lie on the same line:
 - $A(-1, -3); B(1, 5); C(2, 9)$
 - $P(4, -5); Q(7, 5); R(10, 15)$
 - $L(-4, 6); M(3, 8); N(10, 10)$
 - $X(a, 2b); Y(c, a + b); Z(2c - a, 2a)$
- Find k so that the line joining $A(7, 3); B(k, -6)$ and the line joining $C(-4, 5); D(-6, 4)$ are:
 - parallel
 - perpendicular.
- Using slopes, show that the triangle with its vertices $A(6, 1), B(2, 7)$ and $C(-6, -7)$ is a right triangle.
- Two pairs of points are given. Find whether the two lines determined by these points are:
 - parallel
 - perpendicular
 - none.
 - $(1, -2), (2, 4)$ and $(4, 1), (-8, 2)$
 - $(-3, 4), (6, 2)$ and $(4, 5), (-2, -7)$
- Find an equation of:
 - the horizontal line through $(7, -9)$
 - the vertical line through $(-5, 3)$
 - through $A(-6, 5)$ having slope 7
 - through $(8, -3)$ having slope 0
 - through $(-8, 5)$ having slope undefined
 - through $(-5, -3)$ and $(9, -1)$
 - y -intercept: -7 and slope: -5
 - x -intercept: -3 and y -intercept: 4
 - x -intercept: -9 and slope: -4
- Find an equation of the perpendicular bisector of the segment joining the points $A(3, 5)$ and $B(9, 8)$.

8. Find an equation of the line through $(-4, -6)$ and perpendicular to a line having slope $\frac{-3}{2}$.
9. Find an equation of the line through $(11, -5)$ and parallel to a line with slope -24 .
10. Convert each of the following equations into slope intercept form, two intercept form and normal form:
 (a) $2x - 4y + 11 = 0$ (b) $4x + 7y - 2 = 0$ (c) $15y - 8x + 3 = 0$
11. In each of the following check whether the two lines are
 (i) parallel (ii) perpendicular (iii) neither parallel nor perpendicular
 (a) $2x + y - 3 = 0$; $4x + 2y + 5 = 0$
 (b) $3y = 2x + 5$; $3x + 2y - 8 = 0$
 (c) $4y + 2x - 1 = 0$; $x - 2y - 7 = 0$
12. Find an equation of the line through $(-4, 7)$ and parallel to the line $2x - 7y + 4 = 0$
13. Find an equation of the line through $(5, -8)$ and perpendicular to the join of $A(-15, -8), B(10, 7)$.

7.3 Applications of Coordinate Geometry in Real life Situations

Example 14: On a map, Town A is at coordinates $(2, 3)$ and Town B is at $(-4, -1)$. What is the distance between the two towns?

Solution: Use the distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Substitute the values:

$$d = \sqrt{(-4 - 2)^2 + (-1 - 3)^2} = \sqrt{(-6)^2 + (-4)^2} = \sqrt{36 + 16} = \sqrt{52} \approx 7.21 \text{ unit}$$

Thus, the distance between Town A and Town B is approximately 7.21 units.

Example 15: Suppose two cities, City A and City B, are represented by the coordinates $(3, 4)$ and $(7, 1)$ on a map. Find the straight-line distance between the two cities.

Solution: We apply the distance formula:

$$d = |AB| = \sqrt{(7 - 3)^2 + (1 - 4)^2}$$

$$d = |AB| = \sqrt{(4)^2 + (-3)^2}$$

$$d = |AB| = \sqrt{16 + 9} = \sqrt{25} = 5$$

Thus, the straight line distance between City A and City B is 5 units.

Example 16: An Engineer is building a bridge between two points on a riverbank. Suppose the coordinates of the two points where the bridge will start and end are (2, 5) and (8, 9). Find the coordinates of the midpoint, which will represent the centre of the bridge.

Solution: We apply the midpoint formula:

$$M = \left(\frac{2+8}{2}, \frac{5+9}{2} \right)$$

$$M = \left(\frac{10}{2}, \frac{14}{2} \right) = (5, 7)$$

Thus, the centre of the bridge is at the point (5, 7)

Example 17: A landscaper is designing a triangular garden with corners at points A(2, 3), B(5, 7), and C(6, 2). Calculate the lengths of the sides of the triangular garden.

Solution: Use the distance formula to find the length of each side:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$|AB| = \sqrt{(5-2)^2 + (7-3)^2}$$

$$|AB| = \sqrt{(3)^2 + (4)^2}$$

$$|AB| = \sqrt{9+16} = \sqrt{25} = 5 \text{ units}$$

$$|BC| = \sqrt{(6-5)^2 + (2-7)^2}$$

$$|BC| = \sqrt{(1)^2 + (-5)^2}$$

$$|BC| = \sqrt{1+25} = \sqrt{26} = 5.10 \text{ units}$$

$$|AC| = \sqrt{(6-2)^2 + (2-3)^2}$$

$$|AC| = \sqrt{(4)^2 + (-1)^2}$$

$$|AC| = \sqrt{16+1} = \sqrt{17} = 4.12 \text{ units}$$

Thus, the lengths of the sides of the triangular garden are:

$$m\overline{AB} = 5 \text{ units}, \quad m\overline{BC} \approx 5.10 \text{ units}, \quad m\overline{AC} \approx 4.12 \text{ units}$$

Example 18: A pilot needs to travel from city A(50, 60) to city B(120, 150). Determine the heading angle the plane should take relative to the east direction.

Solution: The heading angle can be calculated using the slope:

$$m = \frac{150 - 60}{120 - 50} = \frac{90}{70} = \frac{9}{7}$$

Let θ be the required angle, then

$$\tan \theta = m = \frac{9}{7}$$

$$\theta = \tan^{-1}\left(\frac{9}{7}\right)$$

$$\theta = \tan^{-1}(1.2857)$$

$$\theta \approx 52.13^\circ$$

Thus, the plane should take a heading angle of 52.13° north of east.

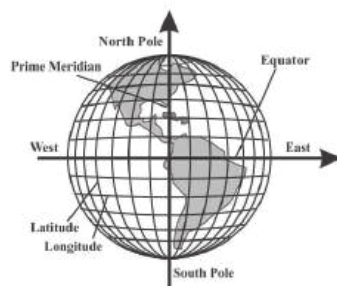
Do you know?

Aviation is the operation and flight of aircraft, including airplanes, helicopters and drones.

Navigation is the process of determining and controlling the route of a vehicle from one place to another.

Latitude measures how far a location is from the equator. It ranges from 0° at the equator to 90° north (at the North Pole) or 90° south (at the South Pole).

Longitude measures how far a location is from the Prime Meridian (which runs through Greenwich, London). It ranges from 0° at the Prime Meridian to 180° east and 180° west.



Example 19: Abdul Hadi is traveling from point A (Latitude 10° N, Longitude 50° E) to point B (Latitude 20° N, Longitude 60° E). Find the midpoint of his journey in terms of latitude and longitude.

Solution:

Given that

Point A (Latitude 10° N, Longitude 50° E)

Point B (Latitude 20° N, Longitude 60° E)

$$\text{Midpoint latitude} = \frac{10^\circ + 20^\circ}{2} = 15^\circ \text{ N}$$

$$\text{Midpoint longitude} = \frac{50^\circ + 60^\circ}{2} = 55^\circ \text{ E}$$

Thus, the midpoint of Abdul Hadi's journey would be at (Latitude 15° N, Longitude 55° E).

Example 20: A landscaper is designing a straight pathway from P(2, 3) to Q(8, 9). What is the length of the pathway?

Solution:

The length of the straight pathway can be found using the distance formula:

$$\begin{aligned}\text{Distance} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(8 - 2)^2 + (9 - 3)^2} \\ &= \sqrt{(6)^2 + (6)^2} \\ &= \sqrt{36 + 36} \\ &= 6\sqrt{2}\end{aligned}$$

So, the length of the pathway is approximately $6\sqrt{2}$ units.

Exercise 7.3

1. If the houses of two friends are represented by coordinates (2, 6) and (9, 12) on a grid. Find the straight line distance between their houses if the grid units represent kilometres?
2. Consider a straight trail (represented by coordinate plane) that starts at point (5, 7) and ends at point (15, 3). What are the coordinates of the midpoint?
3. An architect is designing a park with two buildings located at (10, 8) and (4, 3) on the grid. Calculate the straight-line distance between the buildings. Assume the coordinates are in metres.
4. A delivery driver needs to calculate the distance between two delivery locations. One location is at (7, 2) and the other is at (12, 10) on the city grid map, where each unit represents kilometres. What is the distance between the two locations?
5. The start and end points of a race track are given by coordinates (3, 9) and (9, 13). What is the midpoint of the track?
6. The coordinates of two points on a road are A(3, 4) and B(7, 10). Find the midpoint of the road.
7. A ship is navigating from port A located at (12° N, 65° W) to port B at (20° N, 45° W). If the ship travels along the shortest path on the surface of the Earth, calculate the straight line distance between the points.
8. Farah is fencing around a rectangular field with corners at (0,0), (0,5), (8, 5) and (8, 0). How much fencing material will she need to cover the entire perimeter of the field?

9. An airplane is flying from city X at $(40^\circ \text{ N}, 100^\circ \text{ W})$ to city Y at $(50^\circ \text{ N}, 80^\circ \text{ W})$. Use coordinate geometry, calculate the shortest distance between these two cities.
10. A land surveyor is marking out a rectangular plot of land with corners at $(3, 1)$, $(3, 6)$, $(8, 6)$, and $(8, 1)$. Calculate the perimeter.
11. A landscaper needs to install a fence around a rectangular garden. The garden has its corners at the coordinates: $A(0, 0)$, $B(5, 0)$, $C(5, 3)$, and $D(0, 3)$. How much fencing is required?

REVIEW EXERCISE 7

1. Four options are given against each statement. Encircle the correct option.
- (i) The equation of a straight line in the slope-intercept form is written as:
(a) $y = m(x + c)$ (b) $y - y_1 = m(x - x_1)$
(c) $y = c + mx$ (d) $ax + by + c = 0$
- (ii) The gradients of two parallel lines are:
(a) equal (b) zero
(c) negative reciprocals of each other (d) always undefined
- (iii) If the product of the gradients of two lines is -1 , then the lines are:
(a) Parallel (b) perpendicular
(c) Collinear (d) coincident
- (iv) Distance between two points $P(1, 2)$ and $Q(4, 6)$ is:
(a) 5 (b) 6 (c) $\sqrt{13}$ (d) 4
- (v) The midpoint of a line segment with endpoints $(-2, 4)$ and $(6, -2)$ is:
(a) $(4, 2)$ (b) $(2, 1)$ (c) $(1, 1)$ (d) $(0, 0)$
- (vi) A line passing through points $(1, 2)$ and $(4, 5)$ is:
(a) $y = x + 1$ (b) $y = 2x + 3$
(c) $y = 3x - 2$ (d) $y = x + 2$
- (vii) The equation of a line in point-slope form is:
(a) $y = m(x + c)$ (b) $y - y_1 = m(x - x_1)$
(c) $y = c + mx$ (d) $ax + by + c = 0$
- (viii) $2x + 3y - 6 = 0$ in the slope-intercept form is:
(a) $y = \frac{-2}{3}x + 2$ (b) $y = \frac{2}{3}x - 2$
(c) $y = \frac{2}{3}x + 1$ (d) $y = \frac{-2}{3}x - 2$

(ix) The equation of a line in symmetric form is:

(a) $\frac{x}{a} + \frac{y}{b} = 1$

(b) $\frac{x-x_1}{1} + \frac{y-y_1}{m} = \frac{z-z_1}{1}$

(c) $\frac{x-x_1}{\cos\alpha} = \frac{y-y_1}{\sin\alpha} = r$

(d) $y - y_1 = m(x - x_1)$

(x) The equation of a line in normal form is:

(a) $y = mx + c$

(b) $\frac{x}{a} + \frac{y}{b} = 1$

(c) $\frac{x-x_1}{\cos\alpha} = \frac{y-y_1}{\sin\alpha}$

(d) $x\cos\alpha + y\sin\alpha = p$

2. Find the distance between two points $A(2, 3)$ and $B(7, 8)$ on a coordinate plane.
3. Find the midpoint of the line segment joining the points $(4, -2)$ and $(-6, 3)$.
4. Calculate the gradient (slope) of the line passing through the points $(1, 2)$ and $(4, 6)$.
5. Find the equation of the line in the form $y = mx + c$ that passes through the points $(3, 7)$ and $(5, 11)$.
6. If two lines are parallel and one line has a gradient of $\frac{2}{3}$, what is the gradient of the other line?
7. An airplane needs to fly from city A at coordinates $(12, 5)$ to city B at coordinates $(8, -4)$. Calculate the straight-line distance between these two cities.
8. In a landscaping project, the path starts at $(2, 3)$ and ends at $(10, 7)$. Find the midpoint.
9. A drone is flying from point $(2, 3)$ to point $(10, 15)$ on the grid. Calculate the gradient of the line along which the drone is flying and the total distance travelled.
10. For a line with a gradient of -3 and a y -intercept of 2 , write the equation of the line in:
 - (a) Slope-intercept form
 - (b) Point-slope form using the point $(1, 2)$
 - (c) Two-point form using the points $(1, 2)$ and $(4, -7)$
 - (d) Intercepts form
 - (e) Symmetric form
 - (f) Normal form

Unit 8

Logic

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Understand a mathematical statement and its proof
- Differentiate between an axiom, conjecture and theorem.
- Formulate simple deductive proofs [algebraic proofs that require showing the LHS to be equal to the RHS. e.g., showing $(x - 3)^2 + 5 = x^2 - 6x + 14$]

INTRODUCTION

Logic is a systematic method of reasoning that enables one to interpret the meanings of statements, examine their truth, and deduce new information from existing facts. Logic plays a key role in problem-solving and decision-making.

We generally use logic in our daily life while engaging in mathematics. For example, we often draw general conclusions from a limited number of observations or experiences. A person gets a penicillin injection once or twice and experiences a reaction soon afterward. He generalises that he is allergic to penicillin. This way of drawing conclusions is called **induction**. Inductive reasoning is helpful in natural sciences, where we must depend upon repeated experiments or observations. In fact greater part of our knowledge is based on induction. On many occasions, we have to adopt the opposite course. We have to conclude from accepted or well-known facts. We often consult lawyers or doctors

History



The history of logic began with **Aristotle**, who is considered the father of formal logic. He developed a system of deductive reasoning known as syllogistic logic, which became the foundation of logical thought. The **Stoics** followed, contributing to propositional logic and exploring paradoxes such as the Liar Paradox. During the medieval period, scholars like **Peter Abelard** and **William of Ockham** expanded Aristotle's work, introducing theories of semantics and consequences. In the **19th century**, logic advanced through the works of **George Boole**, who developed Boolean algebra, and **Gottlob Frege**, who formalized modern predicate logic. **Bertrand Russell** and **Alfred North Whitehead** attempted to reduce mathematics to logic in their seminal work, *Principia Mathematica*. The **20th century** saw significant progress with **Kurt Gödel**, who introduced his incompleteness theorems, reshaping our understanding of mathematical logic(history-of-logic: <http://individual.utoronto.ca/pking/miscellaneous/history-of-logic.pdf>).

based on their good reputation. This way of reasoning i.e., drawing conclusions from premises believed to be true, is called **deduction**. One usual example of deduction is: All men are mortal. We are men. Therefore, we are also mortal. To study logic, we start with a statement.

8.1 Statement

A sentence or mathematical expression which may be true or false but not both is called a statement. This is correct so far as mathematics and other sciences are concerned. For instance, the statement $a = b$ can be either true or false. Similarly, any physical or chemical theory can be either true or false. However, in statistical or social sciences, it is sometimes impossible to divide all statements into two mutually exclusive classes. Some statements may be, for instance, undecided.

We can think of a mathematical statement as a unit of information that is either accurate or inaccurate.

Here, we discuss some examples of mathematical statements that are all true.

- (i) For a non-zero real number x and integers m and n , we have: $x^m \cdot x^n = x^{m+n}$
- (ii) The sum of the measures of the interior angles of a triangle is 180°
- (iii) The circumference of a circle with radius r is $2\pi r$
- (iv) $Q \subseteq R$ (The set of rational numbers is a subset of the set of real numbers)
- (v) $\frac{22}{7} \notin Q'$
- (vi) The sum of two odd integers is an even integer
- (vii) $x^2 - 5x + 6 = 0$, for $x = 2$ or $x = 3$

Further, we discuss some examples of mathematical statements that are all false.

- (i) $3 + 4 = 8$
- (ii) $Z \subseteq W$
- (iii) All isosceles triangle are equilateral triangle
- (iv) Between any two real numbers, there is no real number
- (v) $\{1, 2, 3, 4\} \cap \{-1, -2, -3, -4\} = \{1, 2, 3, 4\}$
- (vi) If a and b are the length and width of a rectangle, then the area of a rectangle is $\frac{1}{2}(a \times b)$.